Robust Replacement of Human Decision Makers under Distribution Shift

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Abstract

We study when a machine should replace a human decision maker before the deployment distribution is known. Historical decisions provide a posterior for human error rates, which are assumed stable across environments. A robust receiver operating characteristic frontier evaluates fixed machine thresholds over a label-preserving Wasserstein set. Replacement requires a specified posterior probability that the machine has no higher false-positive rate and no lower true-positive rate than the human throughout the set. We show that the estimated frontier converges uniformly and derive welfare bounds that separate replacement decisions from threshold choice. Even with a known frontier and increasingly informative human histories, threshold choice can leave a persistent welfare gap from a benchmark matching the human's false-positive rate at deployment. In simulations with a fixed score, some per-observation displacement bounds allow partial replacement with positive mean welfare gains over retention at evaluated distributions within the corresponding sets; average budgets of the same numerical sizes lead to complete retention. Among thresholds satisfying the same posterior requirement, welfare-based choice improves mean deployment welfare relative to probability maximization in these simulations.

Keywords: Distributionally robust optimization, Wasserstein distance, ROC curve, human-algorithm comparison, distribution shift.
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1 Introduction

A more accurate prediction is not enough to justify replacing a human decision maker. The choice depends on which errors matter, the machine threshold to be deployed, and how performance may change after deployment. We study a planner who must make this choice before the deployment distribution is known.

Recent work compares human and algorithmic decisions and develops methods for individual replacement decisions (Kleinberg et al., 2018; Mullainathan and Obermeyer, 2022; Feng et al., 2025). In Feng et al. (2025), a human is summarized by a false-positive rate (FPR) and a true-positive rate (TPR), and the machine by its receiver operating characteristic (ROC) curve. A point on the curve dominates the human if its FPR is no higher and its TPR no lower. Their Bayesian rule chooses the point with the largest posterior probability of dominance and replaces the human if that probability reaches a specified cutoff. The posterior accounts for uncertainty about the human's performance based on a finite history of decisions.

We extend their fixed-threshold replacement rule to an unknown deployment distribution. Historical decisions give a posterior for each human's performance, while the planner specifies a set of possible deployment distributions without assigning probabilities to them. We study settings in which human class-conditional error rates are stable but the predictors recorded for the machine may change. For example, clinical assessment may continue to use the same diagnostic information while a change in recording or measurement alters the inputs supplied to an algorithm.

For each threshold, we compute the largest FPR and the smallest TPR over a Wasserstein ambiguity set around the reference distribution. These rates define a robust ROC frontier. Replacement requires a fixed threshold with a sufficiently high posterior probability of no deterioration in either rate throughout the set. A second criterion determines which feasible threshold is deployed. The transport cost preserves label proportions and allows the class-conditional predictor distributions to change. This differs from classical covariate shift, which holds the conditional distribution of the label given the predictors fixed (Shimodaira, 2000; Sugiyama et al., 2007).

The welfare analysis separates whom to replace from which threshold to deploy. We bound harm relative to retaining the human in terms of the estimation errors in the worst-case rates and the concentration of the human posterior. We also compare the rule with a deployment matched-FPR benchmark, which knows the deployment ROC and uses a machine threshold matching each human's FPR. A margin condition controls losses from differences in the replacement sets, while a separate term accounts for threshold differences among humans replaced by both policies. An analytical example shows that this term can persist even with a known frontier and increasingly informative human histories: the posterior-maximizing threshold need not approach the matched-FPR threshold.

We establish uniform convergence of the estimated worst-case rates and frontier under function-class complexity and local FPR regularity conditions. The complexity condition is verified for logistic scores. Additional continuity conditions give a functional limit theorem for the estimated frontier.

The Wasserstein radius limits average displacement. An extension instead bounds every observation's displacement by a pointwise radius. For logistic scores, the local extrema are available in closed form, and the replacement and welfare results carry over under corresponding conditions. At a fixed threshold with continuous positive class-conditional index densities, small average and pointwise budgets affect the worst-case rates at square-root and linear orders, respectively. For tree ensembles, finite search introduces an optimization error that need not vanish as the validation sample grows.

In the simulations, the nominal rule, with radius zero, raises mean welfare near the reference distribution but lowers it after larger shifts. The average budgets $\varepsilon \in \{0.05, 0.1, 0.2, 0.3\}$ retain every doctor; the finer grid gives partial replacement at positive budgets of 0.003 or smaller. With the same fitted logistic score, a pointwise radius of 0.05 replaces 18.68% of doctors and gives a positive mean gain over retention at the evaluated set boundary. At equal radii the pointwise set is smaller, since it excludes concentrated displacements allowed by the average budget. Choosing thresholds by a welfare lower bound within the same posterior feasible set raises mean deployment welfare in this design, at a cost in posterior probability of dominance. An auxiliary income-data illustration uses observed predictors and synthetic human decisions.

The worst-case rate calculations build on optimal-transport risk bounds (Blanchet and Murthy, 2019; Gao and Kleywegt, 2023; Mohajerin Esfahani and Kuhn, 2018). In particular, Shafieezadeh-Abadeh et al. (2015, Section 3.3, Theorem 3) and Shafieezadeh-Abadeh et al. (2019, Section 5, Theorem 42) derive worst- and best-case misclassification bounds for a fixed linear classifier. The logistic case of Proposition 1 uses the same distance-to-half-space geometry within each label class, with the strict-threshold convention stated there. We combine these class-conditional bounds with each human's posterior to assess robust dominance and evaluate thresholds chosen before deployment.

More broadly, Kuhn et al. (2019) review Wasserstein distributionally robust optimization, and Duchi and Namkoong (2021) study uniform performance over f-divergence neighborhoods. Our margin condition is related to those used for plug-in classifiers (Mammen and Tsybakov, 1999; Audibert and Tsybakov, 2007), but applies to a fixed, finite group of decision makers. The analysis also complements work on binary decision problems and ROC inference (Elliott and Lieli, 2013; Feng et al., 2026). Importance weighting evaluates a given target distribution through a density ratio (Sugiyama and Kawanabe, 2012); here, the threshold must be chosen before the deployment distribution is known.

Section 2 presents the decision rule, and Section 3 gives the statistical and welfare

results. Section 4 studies pointwise budgets. Sections 5 and 6 report the simulations and income-data illustration, and Section 7 concludes. Appendix A contains proofs and additional inference results; Appendices B and C give numerical details.

2 The replacement decision under distribution shift

2.1 Information and human performance

We begin with the ROC comparison in Feng et al. (2025). There are J human decision makers, indexed by $j = 1, \ldots, J$, whom we call doctors, following the clinical example of the introduction. Let P_0 denote the reference law of (X, Y). Both the validation sample and the historical records $\mathcal{D}_j$ used to learn about doctor j's performance are drawn in this reference environment. On the basis of these data, the planner decides, for each doctor, whether to keep that doctor or to replace them with the machine operating at a single threshold. The threshold is chosen before the deployment law is known and is not adjusted afterwards.

The machine score m is fitted on a separate training sample. We condition on this sample throughout, so that m is a fixed function in the analysis of the estimators based on the validation sample, and the regularity conditions below are conditions on the fitted score. For observation i assigned to doctor j, $X_i \in \mathcal{X} \subseteq \mathbb{R}^d$ is the vector of covariates, $Y_i \in \{0, 1\}$ is the true label, $\hat{Y}_{i,j} \in \{0, 1\}$ is the doctor's decision, and $\hat{Y}_{i,M} \in \{0, 1\}$ is the machine's decision. Doctor j's false-positive and true-positive rates are

$$\alpha_{H,j} = \frac{\mathbb{E}[(1 - Y_i)\hat{Y}_{i,j}]}{1 - p}, \qquad \beta_{H,j} = \frac{\mathbb{E}[Y_i \hat{Y}_{i,j}]}{p}, \tag{2.1}$$

where $p = p_0 = P_0(Y = 1)$ is the prevalence in the reference environment and the expectations in (2.1) are taken under the doctor's decision process in that environment. We write $\theta_{H,j} = (\alpha_{H,j}, \beta_{H,j})$.

The machine produces a score $m(x) \in [0, 1]$ and decides by the strict threshold rule $\hat{Y}_{i,M} = \mathbb{I}(m(X_i) > c)$, where $\mathbb{I}(A)$ is the indicator of the event A. We define its ROC frontier through an FPR constraint:

$$g(\alpha) = \sup_{c \in [0,1]} \left\{ \frac{\mathbb{E}[Y \cdot \mathbb{I}(m(X) > c)]}{p} \;\middle|\; \frac{\mathbb{E}[(1 - Y) \cdot \mathbb{I}(m(X) > c)]}{1 - p} \leq \alpha \right\}. \tag{2.2}$$

The inequality in (2.2) accommodates FPR levels that are not attained when the score has atoms, without resorting to randomized thresholds. When the distribution function of the score in the negative class is continuous and strictly increasing over the relevant thresholds, the constraint binds and (2.2) is the usual matched-FPR ROC curve.[1]

[1] If $m(x) = \mathbb{P}(Y = 1 \mid X = x)$ and the class-conditional score distributions are continuous, thresholding

Feng et al. (2025) replace doctor j if and only if

$$\hat{q}_{\max,j} := \max_{\alpha\in[0,1]} \Pi_j\{\alpha_H \geq \alpha,\ \beta_H \leq g(\alpha) \mid \mathcal{D}_j\} \geq \tau, \tag{2.3}$$

where $\Pi_j(\cdot \mid \mathcal{D}_j)$ is the Dirichlet–multinomial posterior for doctor j's rate pair, $\theta_H = (\alpha_H, \beta_H)$ denotes this pair as a random element under the posterior, and $\theta_{H,j}$ is its true value. The cutoff $\tau \in (0,1)$ is the required posterior probability of dominance.

For a deployment law Q, let $\mathrm{FPR}_Q(c) = Q(m(X) > c \mid Y = 0)$ and $\mathrm{TPR}_Q(c) = Q(m(X) > c \mid Y = 1)$. We write $g(\alpha; P)$ for the frontier (2.2) under a law P when the law needs to be made explicit, and $g_Q(\alpha) := g(\alpha; Q)$.

Human performance across environments. A deployment environment consists of a law for (X, Y) and a decision process for each doctor. The ambiguity set introduced below restricts only the law of (X, Y). For the main comparison, we assume that in every environment considered the doctors' class-conditional error rates are the same as in the reference environment:

$$\mathbb{P}_Q(\hat{Y}_j = 1 \mid Y = 0) = \alpha_{H,j}, \qquad \mathbb{P}_Q(\hat{Y}_j = 1 \mid Y = 1) = \beta_{H,j}. \tag{2.4}$$

Here $\mathbb{P}_Q$ denotes probability in the environment whose (X, Y) marginal is Q, including the doctor's decision process. Under (2.4), the posterior computed from the historical decisions is a posterior for the rate pair that applies at deployment.

A simple sufficient condition is that doctor j decides according to $h_j(U_j)$, where U_j is a diagnostic signal, and that h_j and the law of U_j given Y are the same in all environments, while the law of the machine's predictors X given Y may change. The doctor's rates are then expectations of $h_j(U_j)$ under two conditional laws that do not vary. Condition (2.4) restricts the environments being compared and does not follow from the transport restriction on (X, Y) introduced below.

The historical decisions give a likelihood for the doctor's fixed rate pair, which is combined with the prior of Feng et al. (2025) to form the posterior. For the machine's inputs, the planner instead specifies a set of possible deployment distributions, without attaching probabilities to them, and requires the comparison to hold at every distribution in the set. If human performance were also allowed to change, the human decision process would have to be specified in each environment as well. Under (2.4), the historical posterior is informative about the common human rates, and the validation data about the machine frontier under the reference law.

gives the Neyman–Pearson ROC curve based on X.

2.2 Wasserstein ambiguity and the robust frontier

Let $\mathcal{Z} := \mathcal{X} \times \{0,1\}$ and let $\widehat{\mathbb{P}}_n = \frac{1}{n}\sum_{i=1}^n \delta_{(X_i,Y_i)}$ be the empirical distribution of n validation observations, where δ_z denotes a unit point mass at z. For a transport cost $\kappa : \mathcal{Z} \times \mathcal{Z} \to [0,\infty]$, define

$$\mathcal{W}_\kappa(P,Q) = \inf_{\pi \in \Pi(P,Q)} \int \kappa(z,z')\, d\pi(z,z'), \tag{2.5}$$

where $\Pi(P,Q)$ is the set of couplings with marginals P and Q.

We use a label-preserving transport cost:

$$\kappa\big((x,y),(x',y')\big) = \begin{cases} \|x - x'\|_2, & y = y', \\ +\infty, & y \neq y'. \end{cases} \tag{2.6}$$

Write $\mathcal{P}(\mathcal{Z})$ for the Borel probability laws on $\mathcal{Z}$. For a reference law P, the ambiguity set of radius ε is

$$\mathcal{B}_\varepsilon(P) = \{Q \in \mathcal{P}(\mathcal{Z}) : \mathcal{W}_\kappa(Q,P) \le \varepsilon\}. \tag{2.7}$$

If $\mathcal{W}_\kappa(P,Q)$ is finite, then $Q(Y=1) = P(Y=1)$. Let P^y and Q^y denote the conditional laws of X given $Y=y$, and let W_1 be the Wasserstein distance of order one with the Euclidean metric. With $p = P(Y=1) \in (0,1)$, the cost separates by class:

$$\mathcal{W}_\kappa(P,Q) = pW_1(P^1,Q^1) + (1-p)W_1(P^0,Q^0). \tag{2.8}$$

Hence the ambiguity set keeps the prevalence fixed but lets the class-conditional feature distributions move. Unlike classical covariate shift, it does not impose $Q(Y \mid X) = P(Y \mid X)$ (Sugiyama et al., 2007). The worst-case TPR and FPR defined below are each computed with the full budget ε, so the two need not be attained by the same law.

Definition 1 (Robust ROC frontier)**.** *Let P be a distribution on (X,Y) with $p := P(Y=1) \in (0,1)$. For each threshold $c \in [0,1]$, define*

$$\underline{\beta}_\varepsilon(c;P) := \inf_{Q \in \mathcal{B}_\varepsilon(P)} Q\big(m(X) > c \mid Y = 1\big),$$

$$\overline{\alpha}_\varepsilon(c;P) := \sup_{Q \in \mathcal{B}_\varepsilon(P)} Q\big(m(X) > c \mid Y = 0\big).$$

The robust ROC frontier is

$$g_\varepsilon(\alpha;P) := \sup\left\{\underline{\beta}_\varepsilon(c;P) : \overline{\alpha}_\varepsilon(c;P) \le \alpha\right\}, \qquad \alpha \in [0,1]. \tag{2.9}$$

We call $\underline{\beta}_\varepsilon(c;P)$ and $\overline{\alpha}_\varepsilon(c;P)$ the worst-case TPR and the worst-case FPR at threshold c, or jointly the worst-case rates. A threshold is feasible at level α if its worst-case FPR is

at most α, and $g_\varepsilon(\alpha; P)$ is the largest TPR that can be guaranteed by one such threshold chosen before deployment.

Proposition 1 (Exact dual representation of the worst-case rates at a fixed threshold)**.** *Let $\mathcal{X}$ be a closed subset of $\mathbb{R}^d$, let P have a finite first moment and $p \in (0,1)$, and let m be continuous. Fix $\varepsilon > 0$ and a threshold c for which $S_c := \{x \in \mathcal{X} : m(x) > c\}$ and its complement are both nonempty. Define distances within $\mathcal{X}$ by*

$$d_c^-(x) := \inf\{\|x - z\|_2 : m(z) \le c\}, \qquad d_c^+(x) := \inf\{\|x - z\|_2 : m(z) > c\}.$$

Then

$$\underline{\beta}_\varepsilon(c; P) = \sup_{\lambda \ge 0} \left\{ \frac{1}{p} \mathbb{E}_P\big[Y \min\{1, \lambda d_c^-(X)\}\big] - \frac{\lambda}{p}\,\varepsilon \right\}, \tag{2.10}$$

and

$$\overline{\alpha}_\varepsilon(c; P) = \inf_{\lambda \ge 0} \left\{ \frac{\lambda}{1-p}\,\varepsilon + \frac{1}{1-p} \mathbb{E}_P\big[(1-Y)(1 - \lambda d_c^+(X))_+\big] \right\}. \tag{2.11}$$

Both optimizations can be restricted to $0 \le \lambda \le 1/\varepsilon$. The same interval applies when P is replaced by an empirical law containing both classes. If S_c is empty, both rates are zero; if $S_c = \mathcal{X}$, both rates are one.

Here $(u)_+ = \max\{u, 0\}$. The proof conditions on the label and applies the duality result for closed events of Blanchet and Murthy (2019, Theorem 1 and Section 2.4). Since S_c is open, the proof also shows that at positive radius the supremum of the probability of S_c equals that of its closure, although a worst-case law for S_c need not exist.

At $\varepsilon = 0$ we use the nominal rates

$$\underline{\beta}_0(c; P) = P^1(S_c), \qquad \overline{\alpha}_0(c; P) = P^0(S_c).$$

We compute these directly rather than from the dual formulas, because at zero radius the dual formula for the FPR gives the probability of the closure of S_c. The two agree if and only if $P^0(\overline{S}_c \setminus S_c) = 0$, which need not hold at the thresholds used with an empirical law. Constant rules, for which one decision region is empty and the distance to it is infinite, are likewise evaluated directly, as in Proposition 1.

2.3 The robust replacement decision

Definition 2 (Robust dominance)**.** *A point $\theta_M = (\alpha_M, \beta_M)$ on the robust ROC frontier $g_\varepsilon(\cdot)$ robustly dominates a human FPR/TPR pair $\theta_H = (\alpha_H, \beta_H)$ if $\alpha_M \le \alpha_H$ and $\beta_M \ge \beta_H$.*

It follows that θ_H is dominated by some frontier point if and only if $\beta_H \le g_\varepsilon(\alpha_H)$. For

a fixed threshold c, the event

$$\overline{\alpha}_\varepsilon(c;P) \le \alpha_H, \qquad \underline{\beta}_\varepsilon(c;P) \ge \beta_H$$

means that, under (2.4), the single threshold c weakly dominates the human pair at every $Q \in \mathcal{B}_\varepsilon(P)$. Its posterior probability, taken over the human pair, is

$$q_j^\varepsilon(c;P) := \Pi_j\Big(\alpha_H \ge \overline{\alpha}_\varepsilon(c;P),\ \beta_H \le \underline{\beta}_\varepsilon(c;P) \mid \mathcal{D}_j\Big). \tag{2.12}$$

Definition 3 (Robust Bayesian replacement)**.** *Given $g_\varepsilon(\cdot;P)$ and the doctor posterior, define*

$$\hat{q}^\varepsilon_{\max,j} := \max_{\alpha\in[0,1]} \Pi_j(\alpha_H \ge \alpha,\ \beta_H \le g_\varepsilon(\alpha;P) \mid \mathcal{D}_j). \tag{2.13}$$

The rule is defined when the displayed maximum is attained. Replace doctor j if $\hat{q}^\varepsilon_{\max,j} \ge \tau$, using a threshold implementing a maximizing frontier point, and retain the doctor if $\hat{q}^\varepsilon_{\max,j} < \tau$.

The maximum in (2.13) exists, for instance, when the thresholds are restricted to a finite set of candidates. Lemma 2 in the Appendix shows that the frontier is attained at its smallest feasible threshold and that

$$\sup_{\alpha\in[0,1]} \Pi_j\{\alpha_H \ge \alpha,\ \beta_H \le g_\varepsilon(\alpha;P) \mid \mathcal{D}_j\} = \sup_{c\in[0,1]} q_j^\varepsilon(c;P).$$

Whenever the maximum exists, the threshold of Definition 3 can therefore be found by maximizing (2.12) over c. This remains true for the exact empirical worst-case rates, and for a finite grid of thresholds containing $c = 1$ if the frontier is computed over the same grid.

The replacement decision in this rule can be read as the solution of a simple decision problem. The planner would like to free up human capacity, but only if the posterior probability that the machine has no higher FPR and no lower TPR than the doctor is at least τ. With a replacement indicator $r_j \in \{0,1\}$, the first decision solves

$$\max_{r_j\in\{0,1\},\, c\in[0,1]} r_j \quad \text{subject to} \quad r_j\{1 - q_j^\varepsilon(c;P)\} \le 1-\tau. \tag{2.14}$$

Retention is always feasible, and replacement is feasible whenever some threshold satisfies the posterior requirement. Problem (2.14) does not say which of the feasible thresholds to deploy. Definition 3 takes the one with the largest posterior probability of dominance.

A planner who specifies a value $\phi \ge 0$ for true positives and a cost $\eta \ge 0$ for false positives can instead maximize a welfare lower bound within the same feasible set. To state the two threshold choices in terms of the estimated worst-case rates, let $\mathcal{C}_n$ be the

set of candidate thresholds and write $\widehat{q}_j^{\varepsilon}(c) = q_j^{\varepsilon}(c; \widehat{\mathbb{P}}_n)$. Define

$$\begin{aligned} \mathcal{C}_j^{\tau} &= \{c \in \mathcal{C}_n : \widehat{q}_j^{\varepsilon}(c) \geq \tau\}, \\ \widehat{V}_{\varepsilon}(c) &= \phi \underline{\beta}_{\varepsilon}(c; \widehat{\mathbb{P}}_n) - \eta \overline{\alpha}_{\varepsilon}(c; \widehat{\mathbb{P}}_n), \\ \widehat{c}_j^{q} &\in \arg\max_{c \in \mathcal{C}_n} \widehat{q}_j^{\varepsilon}(c), \qquad \widehat{c}_j^{W} \in \arg\max_{c \in \mathcal{C}_j^{\tau}} \widehat{V}_{\varepsilon}(c). \end{aligned} \tag{2.15}$$

If $\mathcal{C}_j^{\tau}$ is empty, the doctor is retained. Since a doctor is replaced under either choice exactly when $\mathcal{C}_j^{\tau}$ is nonempty, the two choices replace the same doctors and differ only in the threshold they deploy.[2] For each replaced doctor,

$$\widehat{q}_j^{\varepsilon}(\widehat{c}_j^{q}) \geq \widehat{q}_j^{\varepsilon}(\widehat{c}_j^{W}) \geq \tau, \qquad \widehat{V}_{\varepsilon}(\widehat{c}_j^{W}) \geq \widehat{V}_{\varepsilon}(\widehat{c}_j^{q}). \tag{2.16}$$

We take $\widehat{c}_j^{q}$ as the main rule because it does not require the planner to specify relative error costs. The welfare choice $\widehat{c}_j^{W}$ uses such costs, but only to choose among thresholds that already meet the posterior requirement. Under the average (Wasserstein) budget, $\widehat{V}_{\varepsilon}$ combines two worst-case rates that need not be attained by the same law and is therefore only a lower bound on welfare. Under the pointwise budget of Section 4.1, with the exact extrema of a logistic score, the corresponding quantity is the worst-case welfare over the set centered at the empirical distribution. Section 3.3 shows how estimation error enters the welfare comparison at the population level.

When P is the population reference law, (2.12) is exactly the posterior probability of the event above. In practice we replace P by the empirical law $\widehat{\mathbb{P}}_n$ of the validation sample and denote the selected threshold by $\hat{c}_j$. The resulting plug-in probability reflects the posterior uncertainty about the human pair but not the sampling error in the estimated worst-case rates. Section 3 gives the order of this error and shows how it enters the replacement and welfare bounds, and Appendix A.10 shows how simultaneous confidence bounds on the worst-case rates can be used when they are available.

We call the rule with $\varepsilon = 0$ the nominal rule and the rule with $\varepsilon > 0$ the robust rule. Figure 1 shows how the frontier falls as the radius increases. Since $g_{\varepsilon} \leq g$, the robust rule is more selective than the nominal rule at the same cutoff τ.

Proposition 2 (Conservativeness of robust replacement)**.** *For any $\varepsilon > 0$, the doctors replaced by the robust rule are a subset of those replaced by the nominal rule:*

$$\left\{j : \hat{q}^{\varepsilon}_{\max,j} \geq \tau\right\} \subseteq \left\{j : \hat{q}^{0}_{\max,j} \geq \tau\right\}.$$

The radius determines the deployment laws over which the dominance requirement has to hold. Distances between observed environments are informative about its scale only if they are computed with the same ground metric and relative to the same reference

[2] Maximizers are chosen measurably; on a finite grid, ties are broken in favor of the smallest threshold.

law as the ambiguity set. Under the label-preserving cost, for instance, two empirical laws with different class proportions are infinitely far apart. A distance between marginal feature distributions describes how the covariates differ, but neither its median nor its maximum across environments shows that an environment belongs to the ambiguity set for the joint law. For these reasons we report results over a grid of radii, which shows how the decision changes as the set is enlarged.

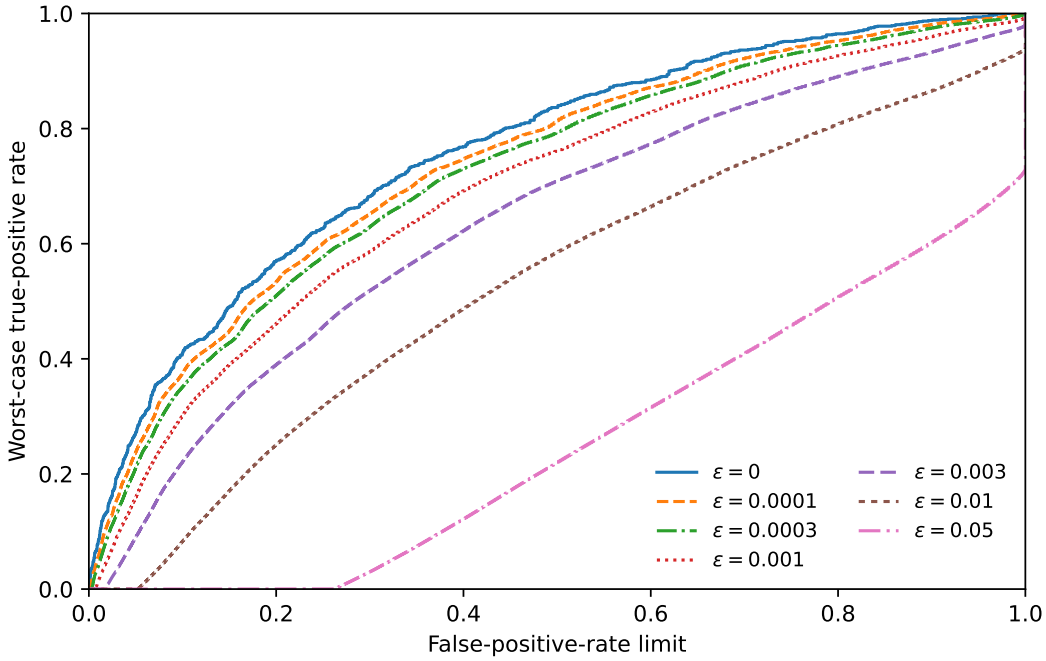


Figure 1: How the radius changes the robust ROC frontier.

Notes: Frontiers for the first replication in Section 5, using its fitted score, validation records and 1,001 candidate thresholds. Each curve gives the largest worst-case TPR subject to the limit on the worst-case FPR on the horizontal axis. Steps correspond to feasible candidate thresholds. All seven radii in the finer grid of Section 5.3 are shown.

2.4 Computation

Write $\widehat{\underline{\beta}}_{\varepsilon}(c) = \underline{\beta}_{\varepsilon}(c; \widehat{\mathbb{P}}_n)$ and $\widehat{\overline{\alpha}}_{\varepsilon}(c) = \overline{\alpha}_{\varepsilon}(c; \widehat{\mathbb{P}}_n)$ for the empirical worst-case rates. The empirical frontier is obtained from them by the same FPR constraint.

For doctor j, let $(\mathrm{TP}_j, \mathrm{FP}_j, \mathrm{FN}_j, \mathrm{TN}_j)$ count historical true positives, false positives, false negatives and true negatives. Under the symmetric Dirichlet prior with parameter $a_0 > 0$ on each cell, the posterior rates are independent and satisfy

$$\begin{aligned} \alpha_H \mid \mathcal{D}_j &\sim \mathrm{Beta}(a_0 + \mathrm{FP}_j, a_0 + \mathrm{TN}_j), \\ \beta_H \mid \mathcal{D}_j &\sim \mathrm{Beta}(a_0 + \mathrm{TP}_j, a_0 + \mathrm{FN}_j). \end{aligned} \tag{2.17}$$

This is the confusion-table posterior of Feng et al. (2025). Let $F_{\alpha,j}$ and $F_{\beta,j}$ denote the posterior cumulative distribution functions (CDFs) and $S_{\alpha,j} = 1 - F_{\alpha,j}$. With the

empirical worst-case rates, the posterior probability (2.12) is

$$\widehat{q}_j^{\varepsilon}(c) = S_{\alpha,j}\Big(\widehat{\overline{\alpha}}_{\varepsilon}(c)\Big)\, F_{\beta,j}\Big(\widehat{\underline{\beta}}_{\varepsilon}(c)\Big)\,. \tag{2.18}$$

Because the Beta laws are continuous, this expression is the same whether the dominance inequalities are weak or strict. Lemma 6 proves the factorization and the concentration properties of the posterior.

On a finite grid of candidate thresholds containing $c = 1$, we maximize (2.18), breaking ties by a rule fixed in advance, and replace the doctor if the maximum is at least τ, deploying the maximizing threshold. Because (2.18) is available in closed form, no posterior simulation is needed. The remaining errors come from estimating the worst-case rates, from computing them numerically and from searching over a grid.[3] For $\varepsilon > 0$, each threshold requires the distances $d_c^{\pm}(X_i)$ and a one-dimensional optimization over λ, in which each evaluation of a dual objective takes $O(n)$ operations.

Exact computation for the logistic score. For $m(x) = \{1 + \exp[-(b + x^\top \beta)]\}^{-1}$ on $\mathcal{X} = \mathbb{R}^d$, with $\beta \neq 0$, define $z(x) = b + x^\top \beta$ and $k_c = \log\{c/(1-c)\}$ for $0 < c < 1$. The distances used in Proposition 1 are

$$d_c^-(x) = \frac{(z(x) - k_c)_+}{\|\beta\|_2}, \qquad d_c^+(x) = \frac{(k_c - z(x))_+}{\|\beta\|_2}. \tag{2.19}$$

These are the Euclidean distances from x to the half-spaces $\{x' : z(x') \leq k_c\}$ and $\{x' : z(x') > k_c\}$, the geometry behind the fixed-classifier risk bounds of Shafieezadeh-Abadeh et al. (2015, 2019), applied here within each label class. If the covariates are restricted to a subset of $\mathbb{R}^d$, the distances must be computed within that subset.

At positive radius, the empirical dual is piecewise linear in λ, so it suffices to evaluate it at 0, at $1/\varepsilon$, and at those reciprocals of the positive distances that lie in $[0, 1/\varepsilon]$. Equivalently, one sorts the distances within each class and moves the closest observations across the boundary first, moving only a fraction of the last observation's mass if the budget runs out. Each observation has mass $1/n$ under the joint empirical law, so moving observation i entirely costs d_i/n, where d_i is its distance to the region into which it is moved. This cost is charged to the total budget ε, not to a separate conditional budget of size ε within each class. Lemma 3 justifies this procedure, and the resulting rates enter (2.18).

Remark 1 (The scale of the average budget)**.** *For the logistic score in* (2.19)*, let* $S = z(X)/\|\beta\|_2$ *and* $a_c = k_c/\|\beta\|_2$*. Write* H_y *for the conditional CDF of* $S \mid Y = y$*, and let* $\alpha_0(c) = \overline{\alpha}_0(c; P)$ *and* $\beta_0(c) = \underline{\beta}_0(c; P)$ *denote the nominal rates. At a fixed interior*

[3] Probabilities close to zero, and the complements of probabilities close to one, are compared on the logarithmic scale.

threshold where both CDFs have continuous positive densities, as $\varepsilon \downarrow 0$,

$$\overline{\alpha}_\varepsilon(c) - \alpha_0(c) = \sqrt{\frac{2\varepsilon H_0'(a_c)}{1-p}} + o(\sqrt{\varepsilon}), \tag{2.20}$$

$$\beta_0(c) - \underline{\beta}_\varepsilon(c) = \sqrt{\frac{2\varepsilon H_1'(a_c)}{p}} + o(\sqrt{\varepsilon}). \tag{2.21}$$

A rate is changed most cheaply by moving the mass closest to the boundary. Within its class, a strip of width w *next to the boundary has mass* $H_y'(a_c)w + o(w)$ *and is moved across at cost* $H_y'(a_c)w^2/2 + o(w^2)$. *Equating this cost to the conditional budget,* ε/p *or* $\varepsilon/(1-p)$, *gives the expansions. For the event* $\{m(X) > c\}$, *with strict inequality, moving mass just beyond the boundary gives the same supremum. Example 1 gives the exact calculation for uniform densities, in which* $s = 2\sqrt{\varepsilon}$.

Hence, for small ε, *an average budget can move substantially more probability across a threshold than a translation of every observation by* ε. *The expansions concern the worst-case rates under the population law at a fixed threshold. In a given sample, the worst-case rates are piecewise linear in* $\varepsilon > 0$. *The pointwise budget of Section 4.1, which bounds the displacement of each observation by a radius* ρ, *has an effect of order* ρ.

3 Statistical theory

We first bound the estimation error of the worst-case rates and of the frontier, and then turn to the implemented replacement rule and its welfare consequences.

3.1 Estimating the robust frontier

Recall that $p_0 = P_0(Y = 1)$, and let $\hat{p} := n^{-1}\sum_{i=1}^n \mathbb{I}(Y_i = 1)$ denote the empirical class prevalence. For $\varepsilon > 0$, on samples with $0 < \hat{p} < 1$ and at thresholds with two nonempty decision regions, Proposition 1 gives the estimated worst-case TPR

$$\widehat{\underline{\beta}}_\varepsilon(c) := \sup_{\lambda \geq 0} \left\{ \frac{1}{\hat{p}} \frac{1}{n} \sum_{i=1}^n Y_i \min\{1, \lambda d_c^-(X_i)\} - \frac{\lambda}{\hat{p}}\varepsilon \right\}, \tag{3.1}$$

and the estimated worst-case FPR

$$\widehat{\overline{\alpha}}_\varepsilon(c) := \inf_{\lambda \geq 0} \left\{ \frac{\lambda}{1-\hat{p}}\varepsilon + \frac{1}{1-\hat{p}} \frac{1}{n} \sum_{i=1}^n (1 - Y_i)\big(1 - \lambda d_c^+(X_i)\big)_+ \right\}. \tag{3.2}$$

Zero radius and constant rules are handled as in Section 2.2. The estimated frontier is

$$\widehat{g}_\varepsilon(\alpha) := \sup\left\{ \widehat{\underline{\beta}}_\varepsilon(c) : \widehat{\overline{\alpha}}_\varepsilon(c) \leq \alpha \right\}.$$

The corresponding dual functions are

$$\psi^{+}_{\lambda,c}(x,y) := \frac{y}{p_0}\min\{1, \lambda d^{-}_{c}(x)\},$$

$$\psi^{-}_{\lambda,c}(x,y) := \frac{1-y}{1-p_0}(1-\lambda d^{+}_{c}(x))_{+}.$$

We obtain uniform convergence when the classes of dual functions are not too rich and the map $c \mapsto \overline{\alpha}_\varepsilon(c; P_0)$ has a stable inverse. Assumption 1 states these requirements, with λ restricted to the interval $[0, 1/\varepsilon]$ of Proposition 1. Its covering bound is the usual polynomial bound for classes of Vapnik–Chervonenkis (VC) type.[4]

Assumption 1 (Complexity and regularity)**.** *Conditional on the independent training sample, validation observations are independent with common law P_0 satisfying the conditions of Proposition 1. Fix $\varepsilon > 0$ and a compact interval $\mathcal{C} = [c_l, c_u] \subset (0,1)$ with nonempty decision regions throughout. The following conditions hold:*

(i) population and empirical dual optimizations use $\Lambda = [0, 1/\varepsilon]$;

(ii) the classes $\mathcal{F}^{\pm} := \{\psi^{\pm}_{\lambda,c} : (\lambda, c) \in \Lambda \times \mathcal{C}\}$ are pointwise measurable and have bounded envelopes. For each class there are fixed finite constants K, v and a constant envelope M such that

$$\sup_H N(uM, \mathcal{F}^{\pm}, L_2(H)) \le (K/u)^v, \qquad 0 < u < 1,$$

where H ranges over finitely supported probability laws and $N(t, \mathcal{F}, L_2(H))$ is the minimum number of $L_2(H)$ balls of radius t covering $\mathcal{F}$;

(iii) $0 < p_{\min} \le p_0 \le p_{\max} < 1$;

(iv) on $\mathcal{C}$, the population worst-case rates are continuously differentiable, with

$$0 < \underline{L}_\alpha \le -\partial_c \overline{\alpha}_\varepsilon(c; P_0) \le \overline{L}_\alpha < \infty, \qquad |\partial_c \underline{\beta}_\varepsilon(c; P_0)| \le L_\beta < \infty.$$

For the interval $[\alpha_l, \alpha_u] \subset (0,1)$ under consideration, there is $b_C > 0$ such that

$$\overline{\alpha}_\varepsilon(c_l; P_0) > \alpha_u + b_C, \qquad \overline{\alpha}_\varepsilon(c_u; P_0) < \alpha_l - b_C.$$

That is, the FPR interval lies strictly inside the range of $\overline{\alpha}_\varepsilon(\cdot; P_0)$ on $\mathcal{C}$.

Remark 2 (A verifiable complexity condition for logistic scores)**.** *For the logistic score on $\mathbb{R}^d$ in (2.19), the exact dual functions are clipped affine functions of the fixed index $z(x)$. Lemma 3 shows that the classes $\mathcal{F}^{\pm}$ are then VC-subgraph and pointwise measurable,*

[4] A class is pointwise measurable if a countable subclass approximates every member pointwise.

which gives part (ii). Part (iv) requires, in addition, the stated derivative bounds and an FPR interval inside the range of the strictly decreasing worst-case FPR.

For $\alpha \in [\alpha_l, \alpha_u]$, write

$$c_\alpha = \min\{c \in \mathcal{C} : \overline{\alpha}_\varepsilon(c; P_0) \le \alpha\}, \qquad \mathcal{C}_\alpha = \{c_\alpha : \alpha \in [\alpha_l, \alpha_u]\}.$$

By the endpoint inequalities in Assumption 1(iv), the thresholds c_α and their sample analogues lie in $\mathcal{C}$ once the estimation error is small, even though the empirical worst-case FPR may jump.

Theorem 1 (Uniform convergence of the robust frontier)**.** *Under Assumption 1, for fixed $\varepsilon > 0$:*

(i)

$$\sup_{c\in\mathcal{C}} \left|\widehat{\underline{\beta}}_\varepsilon(c) - \underline{\beta}_\varepsilon(c; P_0)\right| + \sup_{c\in\mathcal{C}} \left|\widehat{\overline{\alpha}}_\varepsilon(c) - \overline{\alpha}_\varepsilon(c; P_0)\right| = O_p\big(n^{-1/2}\big)\,;$$

(ii) on the FPR interval $[\alpha_l, \alpha_u]$ of Assumption 1(iv),

$$\sup_{\alpha\in[\alpha_l,\alpha_u]} |\widehat{g}_\varepsilon(\alpha) - g_\varepsilon(\alpha; P_0)| = O_p\big(n^{-1/2}\big)\,. \tag{3.3}$$

The replacement and welfare results below rely on this uniform accuracy. Under additional continuity conditions, Appendix A.5 strengthens Theorem 1 to a functional limit theorem and verifies these conditions for a logistic example.

3.2 Replacement decisions and deployed thresholds

The probability of a correct replacement decision depends on both the estimated machine frontier and the doctor posterior. We first relate the errors in these two objects to the fixed-threshold rule, and then bound the performance of the selected threshold.

Let $\mathcal{C}_n \subseteq [0, 1]$ denote the set of candidate thresholds, which contains $c = 1$ and is either a finite grid or the full interval $[0, 1]$. In this subsection, $\widehat{g}_\varepsilon$ denotes the frontier computed over $\mathcal{C}_n$, and

$$\widehat{c}_j \in \arg\max_{c\in\mathcal{C}_n} \widehat{q}^{\,\varepsilon}_j(c), \qquad \widehat{q}^{\,\varepsilon}_{\max,j} = \widehat{q}^{\,\varepsilon}_j(\widehat{c}_j). \tag{3.4}$$

The maximizer in (3.4) is chosen measurably, which on a finite grid any fixed tie-breaking rule achieves. When $\mathcal{C}_n$ is the full interval, we also require the maximum to be attained, as in Definition 3. Under (2.17), this holds if both worst-case rates are continuous on the interval. Lemma 2 gives the equivalent representation in terms of the frontier. On a grid, the frontier is computed over the grid points, and its approximation error relative to the population frontier over all thresholds is part of the error terms below.

Assumption 2 (Separation for replacement)**.** *The number of doctors J is fixed. There are $\delta_0 > 0$ and $d_H > 0$ such that $\alpha_{H,j} \in [\alpha_l + d_H, \alpha_u - d_H]$ for every j, and either*

$$\beta_{H,j} \le g_\varepsilon(\alpha_{H,j}; P_0) - \delta_0 \quad \text{or} \quad \beta_{H,j} \ge g_\varepsilon(\alpha_{H,j}; P_0) + \delta_0.$$

Assumption 2 requires each doctor's TPR to differ by at least δ_0 from the frontier at that doctor's FPR. It is thus a condition on a fixed, finite group of doctors and differs from a margin condition on a distribution of doctor types.

Assumption 3 (Regularity of the frontier in FPR index)**.** *On $[\alpha_l, \alpha_u] \subset (0,1)$, the frontier $\alpha \mapsto g_\varepsilon(\alpha; P_0)$ is Lipschitz with constant $L_g \ge 0$.*

Define the random posterior tail

$$T_j(t) := \Pi_j(\|\theta_H - \theta_{H,j}\|_\infty > t \mid \mathcal{D}_j)\,. \tag{3.5}$$

The posterior tail is measured from the true pair $\theta_{H,j}$, and $T_j(t)$ is random because the posterior depends on the historical data.

Assumption 4 (Mean posterior concentration)**.** *For each doctor with historical sample size $n_j \ge 1$, there are constants $C_j \ge 1$ and $\zeta_j > 0$, independent of n_j and t, such that*

$$\mathbb{E}_{P_0}[T_j(t)] \le C_j \exp(-\zeta_j n_j t^2), \qquad 0 < t \le 1. \tag{3.6}$$

The expectation is over histories $\mathcal{D}_j$ drawn repeatedly in the reference environment, holding the doctor's type fixed. Write $C_0 = \max_j C_j$, $\zeta_0 = \min_j \zeta_j$, and $n_{\min} = \min_j n_j$.

Because (3.6) restricts only an expectation, an unusual history may still produce a posterior far from the truth. Lemma 6 verifies the condition for independent historical cases, prevalence bounded away from zero and one, and the fixed Dirichlet prior in (2.17). Since the results below combine the doctors through union bounds, they do not require independence across doctors or between the histories and the validation sample.

Theorem 2 (Consistency of the fixed-threshold replacement rule)**.** *Suppose that (2.4) holds, that thresholds are selected as in (3.4), and that Assumptions 2–4 hold. Let*

$$r_n := \sup_{\alpha \in [\alpha_l, \alpha_u]} |\widehat{g}_\varepsilon(\alpha) - g_\varepsilon(\alpha; P_0)|\,, \qquad 0 < t_0 \le \min\left\{\frac{d_H}{2}, \frac{\delta_0}{4(1+L_g)}\right\}.$$

On $\{r_n \le \delta_0/4\}$:

(i) if $\beta_{H,j} \le g_\varepsilon(\alpha_{H,j}; P_0) - \delta_0$, then $\widehat{q}^{\varepsilon}_{\max,j} \ge 1 - T_j(t_0)$;

(ii) if $\beta_{H,j} \ge g_\varepsilon(\alpha_{H,j}; P_0) + \delta_0$, then $\widehat{q}^{\varepsilon}_{\max,j} \le T_j(t_0)$.

Thus $\widehat{q}^{\varepsilon}_{\max,j} \to_p 1$ *or* 0*, respectively, when* $r_n \to_p 0$ *and* $n_{\min} \to \infty$*. For* $\bar{\tau} = \min\{\tau, 1-\tau\}$*, the probability that at least one doctor is assigned the wrong replacement decision obeys*

$$\mathbb{P}(\textit{at least one wrong decision}) \le \mathbb{P}(r_n > \delta_0/4) + \frac{JC_0}{\bar{\tau}} \exp(-\zeta_0 n_{\min} t_0^2). \tag{3.7}$$

Here a decision is wrong if it disagrees with the population robust-dominance classification of the doctor in Assumption 2.

The first term in (3.7) comes from the error in the estimated frontier and the second from the doctor posteriors. For a dominated doctor, the proof finds a single threshold whose rectangle of dominated rate pairs contains a neighborhood of the true pair. For a non-dominated doctor, no candidate rectangle meets that neighborhood. Because r_n includes any grid or numerical approximation error, consistency also requires the candidate grid to become fine enough, and a fixed coarse grid may not suffice.

Theorem 3 (Performance of the selected deployment threshold)**.** *Suppose (2.4) holds. Let* $\widehat{c}_j$ *be a selected candidate with* $\widehat{q}^{\varepsilon}_j(\widehat{c}_j) \ge \tau$*, and define*

$$\begin{aligned} e_{\alpha,n} &:= \sup_{c \in \mathcal{C}_n} \left| \widehat{\overline{\alpha}}_{\varepsilon}(c) - \overline{\alpha}_{\varepsilon}(c; P_0) \right|, \\ e_{\beta,n} &:= \sup_{c \in \mathcal{C}_n} \left| \widehat{\underline{\beta}}_{\varepsilon}(c) - \underline{\beta}_{\varepsilon}(c; P_0) \right|. \end{aligned} \tag{3.8}$$

For every $Q \in \mathcal{B}_{\varepsilon}(P_0)$*,*

$$\begin{aligned} \mathrm{FPR}_Q(\widehat{c}_j) &\le \widehat{\overline{\alpha}}_{\varepsilon}(\widehat{c}_j) + e_{\alpha,n}, \\ \mathrm{TPR}_Q(\widehat{c}_j) &\ge \widehat{\underline{\beta}}_{\varepsilon}(\widehat{c}_j) - e_{\beta,n}. \end{aligned} \tag{3.9}$$

Consequently, under the doctor posterior, the event

$$\mathrm{FPR}_Q(\widehat{c}_j) \le \alpha_H + e_{\alpha,n}, \qquad \mathrm{TPR}_Q(\widehat{c}_j) \ge \beta_H - e_{\beta,n} \quad \textit{for every } Q \in \mathcal{B}_{\varepsilon}(P_0)$$

has probability at least τ*.*

For any $t > 0$ *with* $T_j(t) < \tau$*, the true human pair satisfies*

$$\mathrm{FPR}_Q(\widehat{c}_j) \le \alpha_{H,j} + e_{\alpha,n} + t, \qquad \mathrm{TPR}_Q(\widehat{c}_j) \ge \beta_{H,j} - e_{\beta,n} - t \tag{3.10}$$

uniformly over the same ambiguity set. Under Assumption 4, for $\gamma \in (0,1)$ *put*

$$t_n(\gamma) := \min\left\{1, \sqrt{\frac{\log(2JC_0/(\gamma\tau))}{\zeta_0 n_{\min}}}\right\}. \tag{3.11}$$

With sampling probability at least $1-\gamma$*, (3.10) holds with* $t = t_n(\gamma)$ *for every replaced doctor and every* $Q \in \mathcal{B}_{\varepsilon}(P_0)$*.*

The first part of Theorem 3 is a statement under the posterior, with the dominance inequalities relaxed by the estimation errors $e_{\alpha,n}$ and $e_{\beta,n}$. The second holds under repeated sampling of the histories and, in addition, allows for the posterior radius $t_n(\gamma)$.

3.3 Welfare and regret

We compare the implemented rule with retaining all doctors and with the matched-FPR benchmark under the deployment law. The comparison with retention measures the harm that replacement can do. The comparison with the benchmark also counts forgone gains and differences between the thresholds of the two policies, and these differences remain even when both policies replace the same doctors.

For fixed weights $\phi, \eta \geq 0$, predictive welfare is the value of true positives less the cost of false positives.[5] Define

$$W_j(c; Q) := \phi \,\mathrm{TPR}_Q(c) - \eta \,\mathrm{FPR}_Q(c), \qquad W_{H,j}(Q) := \phi\beta_{H,j} - \eta\alpha_{H,j}.$$

If a set of doctors $\mathcal{R}$ is replaced using a threshold profile $c = (c_j)_{j\in\mathcal{R}}$, aggregate welfare is

$$W(\mathcal{R}, c; Q) := \frac{1}{J}\sum_{j\in\mathcal{R}} W_j(c_j; Q) + \frac{1}{J}\sum_{j\notin\mathcal{R}} W_{H,j}(Q). \tag{3.12}$$

For the deployment law Q, define the replacement set of the matched-FPR benchmark

$$\mathcal{R}^*(Q) := \{j : \beta_{H,j} < g_Q(\alpha_{H,j})\},$$

the population robust replacement set

$$\mathcal{R}_\varepsilon(P_0) := \{j : \beta_{H,j} < g_\varepsilon(\alpha_{H,j}; P_0)\},$$

and the replacement set of the implemented rule

$$\mathcal{R}_\varepsilon^{(n)} := \{j : \hat{q}^\varepsilon_{\max,j} \geq \tau\}.$$

Let $\hat{c} = (\hat{c}_j)_{j\in\mathcal{R}_\varepsilon^{(n)}}$ denote the threshold profile selected by the implemented rule, where each $\hat{c}_j$ is the candidate selected by (3.4).

The matched-FPR welfare comparison is defined at deployment laws $Q \in \mathcal{B}_\varepsilon(P_0)$ for which the following thresholds exist. For each $j \in \mathcal{R}^*(Q)$, let $c_j^*(Q)$ denote a threshold on the deployment ROC satisfying $\mathrm{FPR}_Q(c_j^*(Q)) = \alpha_{H,j}$ and $\mathrm{TPR}_Q(c_j^*(Q)) = g_Q(\alpha_{H,j})$, and write $c^*(Q) := (c_j^*(Q))_{j\in\mathcal{R}^*(Q)}$. Uniform statements about this comparison are restricted to these laws, whereas the comparison with retaining all doctors holds over the whole

[5] If payoffs are specified per case, the class prevalence is absorbed into ϕ and η.

ambiguity set.

Theorem 4 does not use the next condition, whose only role is to sharpen the bound on differences between the replacement sets.

Assumption 5 (Margin condition for a finite group of doctors)**.** *There exist constants $C_M, \gamma_M > 0$ such that, for every $Q \in \mathcal{B}_\varepsilon(P_0)$ and every $t > 0$,*

$$\frac{1}{J}\sum_{j=1}^{J} \mathbb{I}(|g_Q(\alpha_{H,j}) - \beta_{H,j}| \le t) \le C_M t^{\gamma_M}. \tag{3.13}$$

Because J is fixed, this condition requires uniform separation across the entire ambiguity set:

$$\inf_{Q\in\mathcal{B}_\varepsilon(P_0)} \min_{j\le J} |g_Q(\alpha_{H,j}) - \beta_{H,j}| \ge (JC_M)^{-1/\gamma_M} > 0.$$

Indeed, a single doctor at distance $d > 0$ contributes $1/J$ when $t = d$, and a doctor at distance zero violates the condition as $t \downarrow 0$. Under the assumption, therefore, no frontier g_Q with Q in the ambiguity set passes through a doctor's point $(\alpha_{H,j}, \beta_{H,j})$. The constants may depend on the fixed group of doctors and on the radius.

Theorem 4 (Welfare relative to retaining all doctors)**.** *Suppose (2.4) and Assumption 4 hold, and doctor j is replaced exactly when $\mathcal{C}_j^\tau$ in (2.15) is nonempty, with a measurable threshold in that set. Let $\phi, \eta \ge 0$ and $\gamma \in (0,1)$. With sampling probability at least $1-\gamma$, simultaneously for every $Q \in \mathcal{B}_\varepsilon(P_0)$,*

$$W(\mathcal{R}_\varepsilon^{(n)}, \widehat{c}; Q) - W(\emptyset, \varnothing; Q) \ge -\frac{|\mathcal{R}_\varepsilon^{(n)}|}{J}\left[\phi e_{\beta,n} + \eta e_{\alpha,n} + (\phi+\eta)t_n(\gamma)\right], \tag{3.14}$$

where the errors and $t_n(\gamma)$ are defined in Theorem 3.

The bound is the negative of the replacement fraction times the per-doctor loss bound in Theorem 3. The possible harm shrinks with the estimation errors in the worst-case rates and the posterior radius.

For the matched-FPR comparison, partition the replacement decisions into common, missed, and extra replacements:

$$\mathcal{I}_n = \mathcal{R}^*(Q) \cap \mathcal{R}_\varepsilon^{(n)}, \quad \mathcal{M}_n = \mathcal{R}^*(Q) \setminus \mathcal{R}_\varepsilon^{(n)}, \quad \mathcal{A}_n = \mathcal{R}_\varepsilon^{(n)} \setminus \mathcal{R}^*(Q).$$

The welfare difference between the benchmark and the implemented rule is then

$$\begin{aligned} & W(\mathcal{R}^*, c^*; Q) - W(\mathcal{R}_\varepsilon^{(n)}, \widehat{c}; Q) \\ = & \frac{1}{J} \sum_{j \in \mathcal{I}_n} [W_j(c_j^*; Q) - W_j(\widehat{c}_j; Q)] \\ & + \frac{1}{J} \sum_{j \in \mathcal{M}_n} [W_j(c_j^*; Q) - W_{H,j}(Q)] \\ & + \frac{1}{J} \sum_{j \in \mathcal{A}_n} [W_{H,j}(Q) - W_j(\widehat{c}_j; Q)]. \end{aligned} \tag{3.15}$$

The first sum is present even when the two replacement sets coincide. Its terms can have either sign, because the matched-FPR benchmark need not maximize $\phi\,\mathrm{TPR} - \eta\,\mathrm{FPR}$ over unrestricted thresholds.

Definition 4 (Matched-FPR welfare comparison)**.** *For the implemented posterior rule, define the signed welfare difference*

$$\mathrm{Regret}(\varepsilon, Q) := W(\mathcal{R}^*(Q), c^*(Q); Q) - W(\mathcal{R}_\varepsilon^{(n)}, \widehat{c}; Q).$$

For a population comparison, suppose that each $j \in \mathcal{R}_\varepsilon(P_0)$ has a threshold $c_{\varepsilon,j}$ with

$$\overline{\alpha}_\varepsilon(c_{\varepsilon,j}; P_0) = \alpha_{H,j}, \qquad \underline{\beta}_\varepsilon(c_{\varepsilon,j}; P_0) = g_\varepsilon(\alpha_{H,j}; P_0).$$

Write $c_\varepsilon = (c_{\varepsilon,j})_{j \in \mathcal{R}_\varepsilon(P_0)}$ and define

$$\mathrm{Regret}_{\mathrm{pop}}(\varepsilon, Q) := W(\mathcal{R}^*(Q), c^*(Q); Q) - W(\mathcal{R}_\varepsilon(P_0), c_\varepsilon; Q). \tag{3.16}$$

In the population comparison the threshold is chosen so that the worst-case FPR equals the true human FPR. The implemented rule instead selects its threshold using the human posterior. Both are compared with the matched-FPR benchmark under the deployment law.

The threshold contribution can be bounded using quantities available when the decision is made. Let $a_{n,j}$ be a median of doctor j's FPR posterior, which for the Dirichlet model is the unique median of the Beta distribution in (2.17). Define

$$\begin{aligned} \widehat{K}_n := \frac{1}{J} \sum_{j \in \mathcal{R}_\varepsilon^{(n)}} \Big[& \phi\{\widehat{g}_\varepsilon(a_{n,j}) - \widehat{\underline{\beta}}_\varepsilon(\widehat{c}_j)\} \\ & + \eta\{\widehat{\overline{\alpha}}_\varepsilon(\widehat{c}_j) - a_{n,j}\} \Big]_+, \qquad [x]_+ := \max\{x, 0\}. \end{aligned} \tag{3.17}$$

Thus $\widehat{K}_n$ measures, in welfare units, how far the estimated worst-case rates at the selected threshold of each replaced doctor fall short of the estimated frontier at the posterior

median FPR $a_{n,j}$. The median is only a reference point in the bound, and the empirical FPR constraint may be slack there.

The two threshold choices in (2.15) have the same replacement set and the same posterior medians. Writing $\widehat{K}_n(c)$ for (3.17) evaluated at a threshold profile $c = (c_j)$, we have

$$\widehat{K}_n(c) = \frac{1}{J} \sum_{j \in \mathcal{R}_\varepsilon^{(n)}} \left[\phi \widehat{g}_\varepsilon(a_{n,j}) - \eta a_{n,j} - \widehat{V}_\varepsilon(c_j)\right]_+,$$

and (2.16) implies

$$\widehat{K}_n(\widehat{c}^W) \leq \widehat{K}_n(\widehat{c}^q). \tag{3.18}$$

Theorems 3 and 4, and Theorem 5(b) below, also hold for $\widehat{c}^W$: their proofs rely on the deployed threshold meeting the posterior requirement and on a doctor being replaced whenever some candidate threshold meets it, and both properties hold for $\widehat{c}^W$. The replacement decisions in Theorem 2 are the same under both choices, although its convergence statement is about the maximal posterior probability rather than the posterior probability at $\widehat{c}_j^W$. Note that (3.18) compares a term of the bound, not welfare itself. Section 5.4 compares the deployment welfare of the two choices in simulations.

Theorem 5 (Welfare relative to the matched-FPR benchmark)**.** *Suppose (2.4) holds, thresholds are selected by (3.4), and the matched-FPR thresholds $c_j^*(Q)$ exist as stipulated above. The candidate set is either the full threshold interval or a finite set containing the all-negative rule $c = 1$, and the posterior maximum is attained. Let $\phi, \eta \geq 0$, suppose that Assumption 5 holds and that all human FPRs lie in $[\alpha_l + d_H, \alpha_u - d_H]$ for a fixed $d_H > 0$, and set*

$$\Delta_\varepsilon(Q) := \sup_{\alpha \in [\alpha_l, \alpha_u]} \{g_Q(\alpha) - g_\varepsilon(\alpha; P_0)\} \geq 0.$$

(a) If the thresholds $c_{\varepsilon,j}$ in Definition 4 exist, then, for every $Q \in \mathcal{B}_\varepsilon(P_0)$ at which the benchmark is defined,

$$\mathrm{Regret}_{\mathrm{pop}}(\varepsilon, Q) \leq \phi \frac{|\mathcal{R}_\varepsilon(P_0)|}{J} \Delta_\varepsilon(Q) + \phi C_M \Delta_\varepsilon(Q)^{1+\gamma_M}. \tag{3.19}$$

(b) Suppose also that Assumptions 3 and 4 hold. For $\gamma \in (0,1)$, let $\bar{\tau} := \min\{\tau, 1-\tau\}$ and

$$t := \sqrt{\frac{\log\{2JC_0/(\gamma\bar{\tau})\}}{\zeta_0 n_{\min}}} \leq \min\{1, d_H\}. \tag{3.20}$$

Let r_n be the frontier error in Theorem 2, including any candidate-grid approximation,

and let $e_{\alpha,n}, e_{\beta,n}$ be the errors in Theorem 3. Define the nonnegative tolerances

$$\begin{aligned} h_n &:= r_n + (1 + L_g)t, \\ u_n &:= \phi e_{\beta,n} + \eta e_{\alpha,n} + (\phi + \eta)t, \\ v_n &:= \phi r_n + \phi e_{\beta,n} + \eta e_{\alpha,n} + (\phi L_g + \eta)t, \end{aligned}$$

and the replacement fraction $\widehat{\pi}_n := |\mathcal{R}_\varepsilon^{(n)}|/J$. With sampling probability at least $1-\gamma$, simultaneously for every $Q \in \mathcal{B}_\varepsilon(P_0)$ at which the matched-FPR benchmark is defined,

$$\begin{aligned} \mathrm{Regret}(\varepsilon, Q) \le \widehat{K}_n + \widehat{\pi}_n\{\phi\Delta_\varepsilon(Q) + v_n\} \\ + \phi C_M\{\Delta_\varepsilon(Q) + h_n\}^{1+\gamma_M} + C_M u_n h_n^{\gamma_M}. \end{aligned} \tag{3.21}$$

The statement also holds for the welfare choice $\widehat{c}^W$ in (2.15), with $\widehat{K}_n$ evaluated at $\widehat{c}^W$ and the same replacement set. On the same event, the fractions of extra and missed replacements obey

$$\frac{|\mathcal{A}_n|}{J} \le C_M h_n^{\gamma_M}, \qquad \frac{|\mathcal{M}_n|}{J} \le C_M\{\Delta_\varepsilon(Q) + h_n\}^{\gamma_M}. \tag{3.22}$$

In part (b), probabilities are over the histories and the validation sample, given the fitted score. The errors r_n, $e_{\alpha,n}$ and $e_{\beta,n}$ are random and unobserved, and they need not be independent of the doctor posteriors. To turn (3.21) into a numerical confidence statement, one would also need a quantitative bound on these errors.

The first two terms of (3.21) bound threshold differences among common replacements; the last two bound missed and extra replacements. Even as estimation errors shrink, the posterior-maximizing threshold need not approach the matched-FPR threshold, so $\widehat{K}_n$ may persist. Example 1 gives a smooth-score case with a positive radius.

Example 1 (Posterior-maximizing and matched-FPR thresholds)**.** *Let $\mathcal{X} = \mathbb{R}$, $P_0(Y = 1) = 1/2$, and $X \mid Y = y \sim \mathrm{Unif}[0,1]$ for both classes. Let $m(x) = (1 + e^{-x})^{-1}$, suppose there is one doctor, whose pair $(\alpha_H, \beta_H) = (0.8, 0.2)$ satisfies (2.4), and let $(\phi, \eta) = (2, 1)$. Write $s = 2\sqrt{\varepsilon} \in (0, 0.1)$ and let $u = 1 - \log\{c/(1-c)\}$ denote the nominal rate at thresholds with $\log\{c/(1-c)\} \in [0,1]$. For $s \le u \le 1 - s$,*

$$\overline{\alpha}_\varepsilon(c; P_0) = u + s, \qquad \underline{\beta}_\varepsilon(c; P_0) = u - s.$$

At deployment $Q = P_0$, the matched-FPR benchmark uses $u = 0.8$, whereas the population comparison of Definition 4 uses $u = 0.8 - s$. Both replace the doctor, yet

$$\mathrm{Regret}_{\mathrm{pop}}(\varepsilon, P_0) = s.$$

On $[0.7, 0.9]$*, the frontier gap is* $\Delta_\varepsilon(P_0) = 2s$*, and Assumption 5 holds with* $\gamma_M = 1$ *and* $C_M = (0.6 - 2s)^{-1}$*.*

Now suppose that the machine frontier under P_0 *is known and that the doctor's history consists of a growing number of iid cases, with the Dirichlet parameter* $a_0 > 0$ *held fixed. Any exact maximizer of the posterior probability (2.12) has nominal rate* $\widehat{u} \to 0.5$ *almost surely, and replacement occurs eventually for every fixed* $\tau < 1$*. Thus the implemented rule has*

$$\text{Regret}(\varepsilon, P_0) \longrightarrow 0.3, \qquad \widehat{K}_n \longrightarrow 0.3 - s.$$

The population regret is of the order of the frontier gap Δ_ε*, whereas the regret of the posterior-maximizing rule contains a threshold term that does not vanish. In this symmetric example its threshold balances the rates at which the two posterior tail probabilities decay, which happens at* $u = 0.5$ *whatever the welfare weights. Appendix A.8 gives the proofs.*

Proposition 4 in Appendix A.7 gives the stochastic order of what remains of (3.21) once the threshold term $\widehat{K}_n$ and the frontier-gap term $\phi\widehat{\pi}_n\Delta_\varepsilon(Q)$ are subtracted.

4 An extension to pointwise budgets

4.1 A bound on each observation's displacement

The Wasserstein set limits how far the covariates can be moved on average, so a large displacement of some observations is allowed if others move little. In this section we instead bound the displacement of every observation, a restriction related to the input perturbations considered by Madry et al. (2017). Such a bound is natural when each recorded covariate vector can be off by at most a known distance (e.g., after a change in measurement equipment). We call this restriction a pointwise budget, with radius ρ, and keep ε for the average budget. A pointwise budget can also be used to assess local changes in the inputs of random forests (RF) and other tree-based scores.

Definition 5 (Pointwise ambiguity set)**.** *Let* $\mathcal{X}$ *be a closed subset of* $\mathbb{R}^d$*. For* $\rho \geq 0$*, write* $K_\rho(x) = \{z \in \mathcal{X} : \|z - x\|_2 \leq \rho\}$ *and define*

$$\mathcal{B}^{\text{pt}}_\rho(P) := \{(T_X, Id_Y)_\# P : T_X \text{ is Borel measurable and } T_X(X,Y) \in K_\rho(X) \quad P\text{-a.s.}\}.$$

The pushforward is the law of $(T_X(X,Y), Y)$*.*

Thus the set contains the laws obtained from P by deterministic, label-preserving maps that move each covariate vector by at most ρ.

Definition 6 (Local extrema of the score)**.** *For a fixed score $m : \mathcal{X} \to [0,1]$, define*

$$\underline{m}_\rho(x) := \inf_{z \in K_\rho(x)} m(z), \tag{4.1}$$

$$\overline{m}_\rho(x) := \sup_{z \in K_\rho(x)} m(z). \tag{4.2}$$

We use these extrema when they are measurable and are attained at measurable selections $z_\rho^-(x), z_\rho^+(x) \in K_\rho(x)$.

These conditions hold when m is continuous and $\mathcal{X}$ is closed, and for a fixed finite ensemble of axis-aligned trees on $\mathbb{R}^d$ with a specified convention at the split boundaries. Lemma 7 verifies both cases.

Definition 7 (Pointwise frontier)**.** *For $0 < p = P(Y = 1) < 1$, define the strict-threshold rates*

$$\tilde{\beta}_\rho(c; P) = P(\underline{m}_\rho(X) > c \mid Y = 1), \qquad \tilde{\alpha}_\rho(c; P) = P(\overline{m}_\rho(X) > c \mid Y = 0),$$

and the deterministic-threshold frontier

$$\tilde{g}_\rho(\alpha; P) = \sup_{c \in [0,1]} \{\tilde{\beta}_\rho(c; P) : \tilde{\alpha}_\rho(c; P) \le \alpha\}, \qquad \alpha \in [0,1]. \tag{4.3}$$

Let $N_y = \sum_{i=1}^n \mathbb{I}(Y_i = y)$. On samples with both classes, its empirical counterpart is

$$\tilde{g}_\rho(\alpha; \widehat{\mathbb{P}}_n) = \sup_{c \in [0,1]} \left\{ \frac{1}{N_1} \sum_{i:Y_i=1} \mathbb{I}(\underline{m}_\rho(X_i) > c) : \frac{1}{N_0} \sum_{i:Y_i=0} \mathbb{I}(\overline{m}_\rho(X_i) > c) \le \alpha \right\}. \tag{4.4}$$

Since $c = 1$ is always feasible, the frontier is well defined. As in (2.2), thresholds are deterministic and we do not interpolate between them.

Proposition 3 (Pointwise worst-case rates and frontier ordering)**.** *Suppose the measurable attaining selections in Definition 6 exist, P has a finite first moment, and $0 < P(Y = 1) < 1$. Then, for every c,*

$$\inf_{Q \in \mathcal{B}_\rho^{\mathrm{pt}}(P)} \mathrm{TPR}_Q(c) = \tilde{\beta}_\rho(c; P), \tag{4.5}$$

$$\sup_{Q \in \mathcal{B}_\rho^{\mathrm{pt}}(P)} \mathrm{FPR}_Q(c) = \tilde{\alpha}_\rho(c; P). \tag{4.6}$$

Moreover, $\tilde{g}_0(\alpha; P) = g(\alpha; P)$, and for $0 \le \rho \le \varepsilon$,

$$g_\varepsilon(\alpha; P) \le \tilde{g}_\rho(\alpha; P) \le g(\alpha; P). \tag{4.7}$$

The ordering follows from set inclusion. At the same numerical radius, $\mathcal{B}_\rho^{\mathrm{pt}}(P) \subseteq \mathcal{B}_\rho(P)$,

where $\mathcal{B}_\rho(P)$ is the Wasserstein set (2.7) with $\varepsilon = \rho$, and the larger set also contains the concentrated displacements that the pointwise budget excludes. The attainment condition ensures that $\tilde{\beta}_\rho$ and $\tilde{\alpha}_\rho$ are probabilities of classifications that some admissible perturbation actually produces, and Appendix A.9 shows by a counterexample that it cannot be dropped.

Exact extrema for the logistic score under a pointwise budget. For the logistic score on $\mathbb{R}^d$ in (2.19), write $\sigma(t) = (1 + e^{-t})^{-1}$. By the Cauchy–Schwarz inequality and the monotonicity of σ,

$$\underline{m}_\rho(x) = \sigma\{z(x) - \rho\|\beta\|_2\}, \qquad \overline{m}_\rho(x) = \sigma\{z(x) + \rho\|\beta\|_2\}. \tag{4.8}$$

Using the index distributions in Remark 1,

$$\tilde{\alpha}_\rho(c) = 1 - H_0(a_c - \rho), \qquad \tilde{\beta}_\rho(c) = 1 - H_1(a_c + \rho). \tag{4.9}$$

When H_0 is continuous and strictly increasing at the relevant quantiles, set $a_\alpha = H_0^{-1}(1-\alpha)$. The FPR constraint binds at $a_c = a_\alpha + \rho$, giving

$$\tilde{g}_\rho(\alpha) = 1 - H_1(a_\alpha + 2\rho), \qquad g(\alpha) - \tilde{g}_\rho(\alpha) = 2H_1'(a_\alpha)\rho + o(\rho). \tag{4.10}$$

The expansion uses the continuity of H_1' at a_α. Compared with the square-root effect of an average budget in Remark 1, the local effect of a pointwise budget is linear in ρ.

One map attains both worst-case rates, at every threshold:

$$T_\rho(x, y) = x + (1 - 2y)\rho\frac{\beta}{\|\beta\|_2}. \tag{4.11}$$

It moves positive cases in the direction that lowers the fitted index and negative cases in the direction that raises it. Consequently, for $\phi, \eta \geq 0$,

$$\inf_{Q \in \mathcal{B}_\rho^{\mathrm{pt}}(P)} \{\phi \mathrm{TPR}_Q(c) - \eta \mathrm{FPR}_Q(c)\} = \phi\tilde{\beta}_\rho(c; P) - \eta\tilde{\alpha}_\rho(c; P). \tag{4.12}$$

The identity also holds for the empirical distribution, which is why the welfare-based threshold has a worst-case interpretation under the pointwise budget.

When the extrema are approximated by a search over a finite set of points of $K_\rho(x)$ that includes x itself, the resulting scores $\underline{m}_{\rho,K}, \overline{m}_{\rho,K}$ satisfy

$$\underline{m}_\rho \leq \underline{m}_{\rho,K} \leq m \leq \overline{m}_{\rho,K} \leq \overline{m}_\rho.$$

Hence, for the same observations and thresholds, the frontier computed from these scores

lies between the exact pointwise frontier and the nominal frontier. A finite search is therefore optimistic, and a larger validation sample alone does not control its optimization error.

For tree-based scores on $\mathcal{X} = \mathbb{R}^d$, we evaluate the score at the observation itself and at K proposals, each equal to the observation plus a direction drawn uniformly from the unit sphere times an independent length drawn from $\mathrm{Unif}[0, \rho]$.[6] The smallest and largest scores found in this way replace the local extrema in the posterior calculation, and the selected threshold is then deployed. The theory in Section 4.2 assumes exact extrema.

4.2 Estimation with continuous and discrete scores

Throughout this subsection the local extrema are computed exactly and, like the score, are held fixed. For a given radius ρ, define

$$Z_0(x) = \overline{m}_\rho(x), \qquad Z_1(x) = \underline{m}_\rho(x), \qquad F_y(c) = P_0(Z_y(X) \le c \mid Y = y).$$

Then $\tilde{\alpha}_\rho = 1 - F_0$ and $\tilde{\beta}_\rho = 1 - F_1$, so the pointwise frontier is a functional of two univariate distribution functions, which we estimate by the class-conditional empirical CDFs $\widehat{F}_y$.

Assumption 6 (Sampling and local regularity for the pointwise frontier)**.** *The validation sample is iid from P_0, independent of the training sample on which we condition, with $0 < p_0 = P_0(Y = 1) < 1$. The fixed exact extrema in Definition 6 are measurable. For a continuous-frontier result on $I = [\alpha_l, \alpha_u] \subset (0, 1)$, assume there is $C = [c_l, c_u] \subset (0, 1)$ and $b > 0$ such that*

$$1 - F_0(c_l) > \alpha_u + b, \qquad 1 - F_0(c_u) < \alpha_l - b,$$

F_0, F_1 are continuously differentiable on a neighborhood of C, and

$$0 < \underline{f} \le f_0(c), \qquad 0 \le f_1(c) \le L_1 < \infty \quad (c \in C),$$

where $f_y = F_y'$.

Theorem 6 (Estimation of the pointwise frontier)**.** *Under the sampling conditions in Assumption 6, write $d_{y,n} = \|\widehat{F}_y - F_y\|_\infty$ and $\widehat{\tilde{g}}_\rho = \tilde{g}_\rho(\cdot; \widehat{\mathbb{P}}_n)$.*

(i) For continuous or discrete score distributions, $d_{0,n} + d_{1,n} = O_p(n^{-1/2})$. On samples with both classes, whenever $\alpha \pm d_{0,n} \in [0, 1]$,

$$\tilde{g}_\rho(\alpha - d_{0,n}; P_0) - d_{1,n} \le \widehat{\tilde{g}}_\rho(\alpha) \le \tilde{g}_\rho(\alpha + d_{0,n}; P_0) + d_{1,n}. \tag{4.13}$$

[6] The proposals are therefore uniform in the radius rather than in the volume of the ball.

(ii) Under the additional density and interval conditions,

$$\sup_{\alpha\in I}|\widehat{\tilde{g}}_\rho(\alpha)-\tilde{g}_\rho(\alpha;P_0)|=O_p(n^{-1/2}).$$

As a function of (F_0,F_1), the frontier map into $\ell^\infty(I)$ is Hadamard differentiable in the uniform norm on C, tangentially to continuous CDF perturbation directions. With $c_\alpha=F_0^{-1}(1-\alpha)$, its derivative is

$$\dot{\tilde{g}}_{(h_0,h_1)}(\alpha)=-h_1(c_\alpha)+\frac{f_1(c_\alpha)}{f_0(c_\alpha)}h_0(c_\alpha). \tag{4.14}$$

(iii) Suppose instead that the two score laws have fixed finite supports. Let $s_1<\cdots<s_K$ list their union, let $c_1<\cdots<c_M$ list the distinct elements of $\{0,s_1,\dots,s_K,1\}$, and set

$$A_k=1-F_0(c_k),\qquad B_k=1-F_1(c_k),\qquad \mathcal{A}=\{A_k:1\le k\le M\}.$$

For any compact $I\subset(0,1)$ with $\operatorname{dist}(I,\mathcal{A})>0$, the same uniform root-n frontier rate holds. Relative to perturbations of the two conditional mass vectors on these fixed supports, the map into $\ell^\infty(I)$ is Hadamard differentiable. Writing $v_y=(v_{y,1},\dots,v_{y,K})$ for a mass-vector direction and $k(\alpha)=\min\{k:A_k\le\alpha\}$, its derivative is

$$\dot{\tilde{g}}_{(v_0,v_1)}(\alpha)=\sum_{\ell:s_\ell>c_{k(\alpha)}}v_{1,\ell}. \tag{4.15}$$

Parts (ii) and (iii) give two ways in which the root-n accuracy of the two CDFs carries over to the frontier. For a continuous score, the lower bound on f_0 keeps the inverse of the worst-case FPR stable. For a finite tree ensemble the frontier is a step function, and the rate holds on FPR intervals that stay away from the finite set $\mathcal{A}$ of levels at which it can jump; at a level in $\mathcal{A}$ even pointwise consistency can fail (Appendix A.9), so finite support alone gives neither uniform convergence nor directional differentiability on every interior FPR interval.

Corollary 1 (Replacement and welfare under a pointwise budget)**.** *Suppose that (2.4) holds and that the measurable attaining selections assumed in Proposition 3 exist, and fix $\rho\ge 0$. Consider the rule with the same posterior and a candidate set containing the all-negative rule $c=1$, with attained frontiers and measurable posterior maximizers, in which the objects defined from the Wasserstein set are replaced by their pointwise counterparts:*

$$(\mathcal{B}_\varepsilon,g_\varepsilon,\overline{\alpha}_\varepsilon,\underline{\beta}_\varepsilon)\longmapsto(\mathcal{B}_\rho^{\mathrm{pt}},\tilde{g}_\rho,\tilde{\alpha}_\rho,\tilde{\beta}_\rho).$$

Define $\tilde{r}_n$, $\tilde{e}_{\alpha,n}$, $\tilde{e}_{\beta,n}$, $\tilde{\Delta}_\rho(Q)$ and $\widehat{\tilde{K}}_n$ by the same substitution in (3.4), Theorems 2–5 and (3.17). In particular, the frontier error includes the candidate-grid approximation. Then:

(i) Under Assumptions 2–4, with separation and Lipschitz regularity now imposed on $\tilde{g}_\rho$, the conclusions of Theorem 2 hold with $\tilde{r}_n$ in place of r_n.

(ii) For a selected threshold whose posterior probability under the pointwise rates is at least τ, the posterior statement of Theorem 3 holds over $\mathcal{B}_\rho^{\mathrm{pt}}(P_0)$ with errors $\tilde{e}_{\alpha,n}, \tilde{e}_{\beta,n}$. Under Assumption 4, the sampling statement of Theorem 3 and the welfare bound of Theorem 4 hold with those errors and the same $t_n(\gamma)$.

(iii) Theorem 5 holds with the substituted objects when its interior conditions hold and its matching thresholds exist, Assumption 5 is imposed over $Q \in \mathcal{B}_\rho^{\mathrm{pt}}(P_0)$, and, for part (b), Assumptions 3 and 4 hold for that frontier. The sample welfare conclusions also apply to the welfare choice in (2.15).

Proof. Proposition 3 gives the exact worst-case rates, so the substituted version of (3.9) holds for every $Q \in \mathcal{B}_\rho^{\mathrm{pt}}(P_0)$. The smallest feasible threshold attains the pointwise frontier, by the right-continuity of the worst-case FPR and the monotonicity of both rates in the threshold. Every threshold that is feasible for the worst-case FPR constraint is also feasible for the FPR constraint under Q and has TPR under Q at least its worst-case TPR; hence $\tilde{g}_\rho \leq g_Q$. The fixed-rectangle argument, the posterior concentration event and the welfare decomposition in Appendices A.6 and A.7 then apply with the stated conditions and substitutions. For the sample welfare bounds it suffices that the deployed threshold meet the posterior requirement and that a doctor be replaced whenever some candidate threshold meets it. □

For the logistic score, differentiating (4.10) gives

$$\tilde{g}'_\rho(\alpha) = \frac{H'_1(a_\alpha + 2\rho)}{H'_0(a_\alpha)}.$$

Bounding H'_1 from above and H'_0 away from zero over the relevant range therefore gives the Lipschitz condition of Assumption 3 for $\tilde{g}_\rho$ on $[\alpha_l, \alpha_u]$. If, in addition, H_0 and H_1 are continuously differentiable near the corresponding shifted index values and the interval and sampling conditions of Assumption 6 hold, Theorem 6(ii) applies. Even though these formulas are exact, the replacement and welfare bounds in Corollary 1 still contain the sampling errors of the posterior and of the rates computed at the empirical reference law.

5 Monte Carlo simulations

5.1 A smooth-score experiment with known transport distance

The main experiment is built so that the predictors recorded for the machine can shift while the clinical information used by the doctors does not. In the reference environment,

let $Z \sim N(0, I_5)$ and $U \sim N(0, \sigma_u^2)$ be independent, and set

$$R = b_0 + Z^\top \beta + U, \qquad p(Z,U) = \{1 + \exp(-R)\}^{-1}, \qquad Y \mid Z, U \sim \text{Bernoulli}(p(Z,U)). \tag{5.1}$$

The machine observes $X = Z$. Doctor j uses the clinical information (Z, U) and decides according to

$$\widehat{Y}_j = \mathbb{I}\{p(Z,U) + \nu_j > c_{H,j}\}, \qquad \nu_j \sim N(0, \sigma_\nu^2), \qquad c_{H,j} = 0.15 + 0.7 V_j, \quad V_j \sim \text{Beta}(2,3).$$

The noise terms and the doctors' thresholds are independent of the reference cases. We set $\beta = (1, 0.5, -0.3, 0.2, -0.1)^\top$, $b_0 = -1.5$, $\sigma_u = 0.15$ and $\sigma_\nu = 0.225$.

At deployment, only the recorded predictors change:

$$X_\delta = Z + (1 - 2Y)\delta e_1, \qquad \delta \geq 0, \tag{5.2}$$

where e_1 is the first coordinate vector. The law of (Z, U, Y, ν_j) and the doctors' decisions are unchanged. One can think of a change in the way one variable is recorded that pushes the two classes in opposite directions, while clinical assessment continues to use the same information. The shift leaves the prevalence and each doctor's FPR and TPR unchanged, so (2.4) holds, but it does not keep the conditional law of Y given the recorded predictors fixed.

Let Q_δ denote the law of (X_δ, Y). Within each class, Q_δ translates the reference law of the features by a vector of length δ. The translation is a coupling with cost δ, so each conditional W_1 distance is at most δ, and it is at least δ because the conditional means differ by a vector of that length. Consequently,

$$\mathcal{W}_\kappa(P_0, Q_\delta) = \delta. \tag{5.3}$$

Hence $Q_\delta \in \mathcal{B}_\varepsilon(P_0)$ if and only if $\delta \leq \varepsilon$. This is a statement about the population law P_0; the rule itself uses a frontier estimated from the validation data.

Each replication draws 10,000 training observations, 5,000 independent validation observations and a history of 500 cases for each of $J = 500$ doctors. The logistic score is fitted on the training data with an ℓ_2 penalty and then held fixed; the inverse regularization strength is $C = 1$. The prior puts $a_0 = 0.01$ on each cell of the confusion table, and the cutoff is $\tau = 0.95$. For each $\varepsilon \in \{0, 0.05, 0.1, 0.2, 0.3\}$, we compute the exact empirical worst-case rates on a common grid of 1,001 thresholds from 0 to 1 and select, for each doctor, the grid point that maximizes (2.18). This threshold is then used at every deployment distance. With $\varepsilon = 0$ the rule is the nominal rule, that is, the Bayesian comparison (2.3).

The deployment distances are $\delta \in \{0, 0.025, 0.05, 0.1, 0.3, 0.5, 1\}$. For the main radius

$\varepsilon = 0.05$, the law $Q_{0.025}$ lies in the interior of the ambiguity set and $Q_{0.05}$ lies on its boundary. There are 100 independent replications. Within each, the training data, validation data, histories and selected thresholds do not vary with δ. Welfare is reported for the prespecified weights $(\phi, \eta) \in \{(1,1), (2,1), (5,1)\}$, with $(2,1)$ as the main case. Neither the welfare weights nor the deployment outcomes are used in the posterior selection of the thresholds.

Conditional error rates are computed by numerical integration under (5.1) rather than from a test sample (Appendix B), and these population rates also define the matched-FPR benchmark. For each policy and each of the seven evaluated distances on the deployment path (5.2), we report welfare relative to retaining all doctors, the share of doctors replaced, the share of harmful replacements (i.e., the share of replaced doctors for whom replacement lowers welfare), and the common, missed and extra replacement terms of (3.15). We also report the threshold term $\widehat{K}_n$ of Theorem 5, computed from the estimated frontier on the grid and the selected thresholds. Monte Carlo standard errors are computed across replications, and differences between rules are paired within replications.

5.2 Replacement, welfare, and forgone gains

With any of the average budgets $\varepsilon \in \{0.05, 0.1, 0.2, 0.3\}$, the robust rule retains all 500 doctors in all 100 replications, whereas the nominal rule replaces 41.51% of them on average (Monte Carlo standard error 0.88 percentage points). Under the robust rule the posterior probabilities stay far below the cutoff. At $\varepsilon = 0.05$, the largest posterior probability of dominance over all doctors and replications is 5.85×10^{-10}.

Table 1 compares the two rules at each deployment distance. At $\delta = 0$, the nominal rule raises mean welfare from 0.6498 to 0.7323, a paired gain of 0.0826 (0.0017), and at the boundary of the ambiguity set, $\delta = 0.05$, its gain is still 0.0628 (0.0013). The mean welfare of the nominal rule is above all-human welfare at every evaluated distance up to $\delta = 0.1$ and below it at $\delta = 0.3$, 0.5 and 1. At $\delta = 1$, retaining everyone is better by 0.3226 (0.0068). With the radii in this grid, the robust rule avoids the losses from large shifts only by retaining every doctor, and so it also gives up the gains near the reference distribution.

Among doctors replaced by the nominal rule, the mean share of harmful replacements is 0.17% at $\delta = 0$, 0.70% at $\delta = 0.05$ and 99.94% at $\delta = 0.3$, and every replacement is harmful at $\delta = 0.5$ and $\delta = 1$. The share counts harmful replacements but not the size of the losses. It is 100% at both of these distances, while the loss averaged over all doctors grows from 0.1228 to 0.3226 between them. Under the other two pairs of welfare weights the paired welfare differences have the same signs (Appendix B.1).

The shortfall from the matched-FPR benchmark can be split as in (3.15). At $\delta = 0$ the nominal rule falls short of the benchmark by 0.1718 on average, of which 0.0286

Table 1: Welfare of the nominal and robust rules by deployment distance, $(\phi, \eta) = (2, 1)$.

δ	Nominal welfare	Robust welfare	Robust minus nominal	Harmful among nominal-replaced (%)
0	0.7323 (0.0020)	0.6498 (0.0009)	-0.0826 (0.0017)	0.17 (0.03)
0.025	0.7225 (0.0019)	0.6498 (0.0009)	-0.0727 (0.0015)	0.31 (0.05)
0.05	0.7126 (0.0017)	0.6498 (0.0009)	-0.0628 (0.0013)	0.70 (0.08)
0.1	0.6926 (0.0014)	0.6498 (0.0009)	-0.0428 (0.0009)	2.82 (0.20)
0.3	0.6105 (0.0013)	0.6498 (0.0009)	0.0392 (0.0010)	99.94 (0.02)
0.5	0.5270 (0.0027)	0.6498 (0.0009)	0.1228 (0.0027)	100.00 (0.00)
1	0.3272 (0.0067)	0.6498 (0.0009)	0.3226 (0.0068)	100.00 (0.00)

Notes: Means over 100 replications, with Monte Carlo standard errors in parentheses; welfare differences are paired within replications. The nominal rule has $\varepsilon = 0$ and the robust rule $\varepsilon = 0.05$. The other prespecified radii, 0.1, 0.2 and 0.3, give the same policy as 0.05 in every replication (no doctor is replaced), so robust welfare equals all-human welfare and the harmful share among doctors replaced by the robust rule is undefined. The laws Q_δ with $\delta \leq 0.05$ lie in the population ambiguity set of radius 0.05. Error rates are population values computed as in Appendix B.

comes from different thresholds among doctors whom both replace and 0.1433 from missed replacements. The bound (3.21) measures the threshold effect differently, through $\widehat{K}_n$, which averages 0.0327 (0.0008) for the nominal rule. The robust rule replaces nobody, so its $\widehat{K}_n$ is zero, and its shortfall of 0.1864 at $\delta = 0.05$ consists entirely of missed replacements.

5.3 Sensitivity to smaller radii

Since none of the positive radii in Table 1 leads to any replacement, we also look at a finer grid of smaller radii, keeping the samples, the decision rule and the evaluation unchanged. Each radius defines a separate policy, and Figure 2 shows the transition from the nominal rule to complete retention.

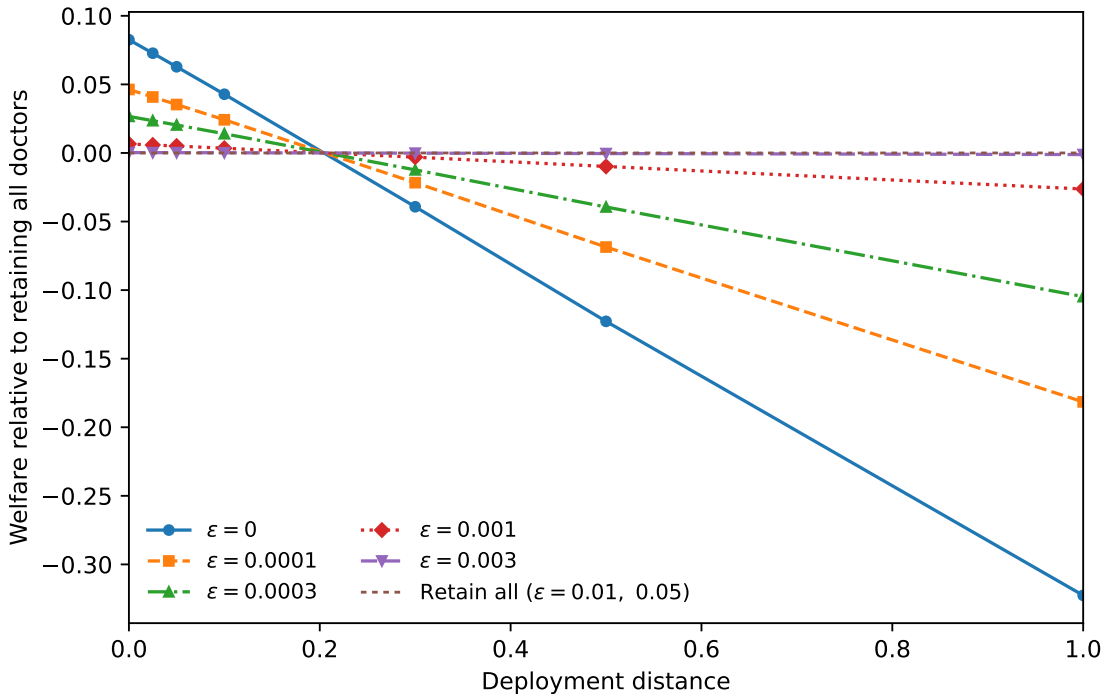


Figure 2: Welfare relative to retaining all doctors, by radius and evaluated deployment distance.

Notes: Mean welfare minus all-human welfare over the 100 replications, $(\phi, \eta) = (2, 1)$. The policies with $\varepsilon = 0.01$ and 0.05 replace nobody and coincide with the zero line. Lines join the evaluated distances. For positive radii below 0.025, $\delta = 0$ is the only evaluated distance at which Q_δ lies in the ambiguity set. Table 5 reports the values with Monte Carlo standard errors.

The share of doctors replaced falls quickly as the radius grows, from 22.79% at $\varepsilon = 0.0001$ to 0.14% at $\varepsilon = 0.003$, and nobody is replaced at 0.01 or 0.05. At $\varepsilon = 0.0001$ the robust rule gains 0.0463 over retaining everyone at $\delta = 0$, against 0.0826 for the nominal rule, and loses 0.0686 at $\delta = 0.5$, against 0.1228. As the radius grows, both the gains at small distances and the losses at large ones shrink toward zero.

For all three pairs of welfare weights, at every evaluated distance up to $\delta = 0.1$, each positive radius in the finer grid that leads to some replacement has lower mean welfare than the nominal rule but higher mean welfare than retaining all doctors. At $\delta = 0.3$, 0.5 and 1 the ranking reverses, and on average these rules lose less than the nominal rule but fall below all-human welfare. At $\delta = 0.5$ and 1, every replacement they make is harmful, in every replication with a replacement.

5.4 Two budgets for the same score

We now keep the 100 fitted logistic scores, validation samples, histories and the grid of 1,001 thresholds fixed and compare the two kinds of budget. The pointwise radii are $\rho \in \{0, 0.05, 0.1, 0.2, 0.3\}$, and the deployment distances and welfare weights are as before. Since (5.2) moves every observation by δ, the law Q_δ belongs to the pointwise set when

$\delta \leq \rho$. At the same radius the pointwise set is contained in the Wasserstein set, which in addition allows the displacement to be concentrated on some observations.

Average and pointwise budgets. No doctor is replaced under any of the average budgets $\varepsilon \in \{0.05, 0.1, 0.2, 0.3\}$. Under the pointwise budget, the rule that maximizes the posterior probability replaces 18.68%, 5.61% and 0.148% of the doctors at $\rho = 0.05$, 0.1 and 0.2, and nobody at $\rho = 0.3$. At $\rho = 0.2$, only 47 of the 100 replications have any replacement. For $(\phi, \eta) = (2, 1)$, the mean gain over retaining all doctors is 0.0275 (0.0009) at $(\rho, \delta) = (0.05, 0.05)$ and 0.0052 (0.0003) at $(0.1, 0.1)$, in both cases at a deployment distribution on the boundary of the population pointwise set. Figure 3 shows the gains over the whole grid of distances, and Appendix B.3 tabulates them.

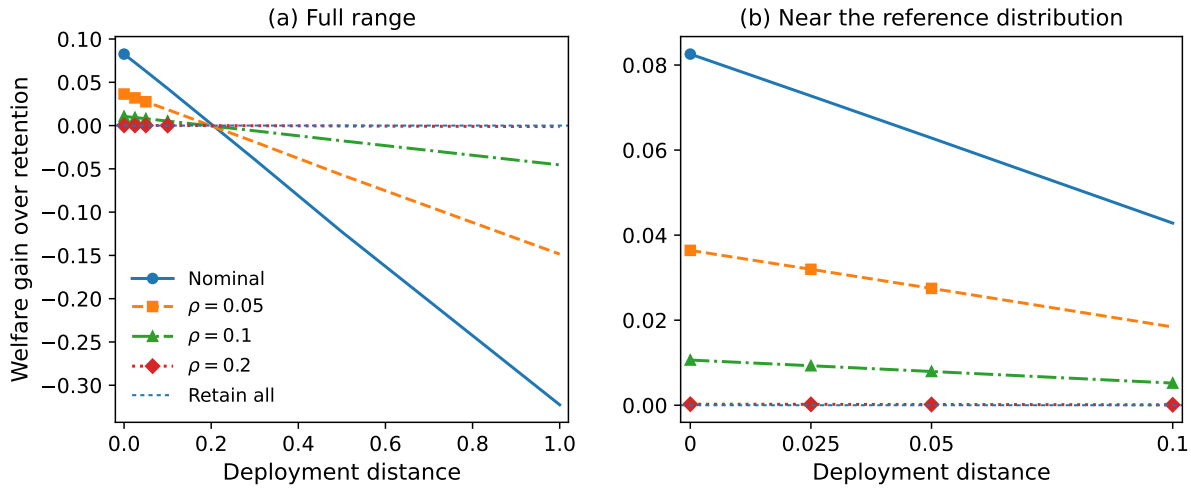


Figure 3: Welfare gain over retaining all doctors under the pointwise budget.

Notes: Mean gain over retaining all doctors for the maximum-probability rule, $(\phi, \eta) = (2, 1)$, over the 100 replications. Panel (a) covers all seven distances and panel (b) enlarges $0 \leq \delta \leq 0.1$; colors and line styles match across panels. Markers indicate distances at which Q_δ lies in the population pointwise set ($\delta \leq \rho$). The curve for $\rho = 0$ is the nominal rule, which is the same under either budget; the pointwise rule with $\rho = 0.3$ and all positive average budgets in this comparison coincide with the retention line. Table 6 reports the means and Monte Carlo standard errors.

Since the posterior probabilities rest on estimated rates, individual replacements can still be harmful, even inside the set. At the two boundary points above, the mean share of harmful replacements is 0.74% and 4.04%. At $(\rho, \delta) = (0.2, 0.1)$ it is 11.52% over the 47 replications with replacements, while the mean share of all doctors whose replacement is harmful is only 0.016%. Five of these 47 replications have a negative aggregate gain. At $\delta = 0.5$ and 1 every replacement is harmful. So partial replacement raises mean welfare at the positive distances evaluated inside the set, and lowers it after larger shifts.

The two threshold choices. Finally, we compare the maximum-probability threshold $\widehat{c}^q$ with the welfare-based threshold $\widehat{c}^W$ of (2.15), using the weights $(2, 1)$. Table 2 shows that the welfare choice trades a lower posterior probability for a higher empirical welfare

lower bound. At $\rho = 0.05$, the mean posterior probability at the selected threshold falls from 0.9774 to 0.9564, the welfare lower bound rises from 0.9003 to 0.9207, and the mean of $\widehat{K}_n$ falls from 0.0131 to 0.0092. In the paired comparison, deployment welfare at $\delta = 0.05$ is higher by 4.15×10^{-3} (Monte Carlo standard error 0.18×10^{-3}), which raises the mean gain over retention to 0.0316.

Table 2: The two threshold choices under the pointwise budget.

		Posterior probability		Welfare lower bound		$10^3\widehat{K}_n$	
ρ	Replaced (%)	$\widehat{c}^q$	$\widehat{c}^W$	$\widehat{c}^q$	$\widehat{c}^W$	$\widehat{c}^q$	$\widehat{c}^W$
0	41.51	0.9824	0.9570	0.9335	0.9605	32.73	21.34
0.05	18.68	0.9774	0.9564	0.9003	0.9207	13.06	9.15
0.1	5.61	0.9732	0.9558	0.8574	0.8736	3.58	2.64
0.2	0.15	0.9661	0.9548	0.7660	0.7764	0.08	0.07
0.3	0.00	–	–	–	–	0.00	0.00

Notes: Both choices use $\tau = 0.95$ and replace the same doctors; the welfare lower bound, $\widehat{K}_n$ and the welfare choice use the weights $(2, 1)$. Posterior probabilities and welfare lower bounds are averaged over the replaced doctors within a replication and then over the replications with replacements (100 at $\rho = 0, 0.05, 0.1$; 47 at 0.2; none at 0.3). Replacement rates and $\widehat{K}_n$ are averaged over all 100 replications and 500 doctors. The row $\rho = 0$ is the nominal rule, which is the same under both budgets, and the positive average budgets in this comparison replace nobody. At $\rho = \delta = 0.05$, the paired difference in deployment welfare between $\widehat{c}^W$ and $\widehat{c}^q$ is 4.15 (Monte Carlo standard error 0.18) in units of 10^{-3}; panel B of Table 6 gives these differences for all radii and distances. Unrounded values are in the data supplement.

The welfare choice has higher mean welfare at every evaluated distance and for every radius with replacements, though not in every replication; at $(\rho, \delta) = (0.2, 1)$, seven replications show small negative differences. The positive average budgets in this comparison leave both choices with empty feasible sets. Where replacement occurs, the two choices replace the same doctors, so their welfare difference comes from the threshold alone. Of the two threshold criteria, only the welfare-based choice requires relative error costs to be specified.

6 An illustration with cross-state income data

The ACSIncome task of Ding et al. (2021), which is built from the American Community Survey (ACS), is to predict whether annual income exceeds $50,000 from demographic and employment variables. Records from California, New York and Texas form the reference environment, and records from five other states are used for evaluation. We fit a logistic regression (LR) and a random forest (RF) with 100 trees on 48,000 reference records, estimate the frontier on 6,000 further records, and use another 6,000 as the histories of a fixed group of 200 synthetic decision makers, about 30 cases each. All policies are selected using only the reference records and then evaluated in the other states. Appendix C.1 describes the allocation of records, the synthetic decisions and the computations.

The nominal policies replace 5 of the 200 decision makers with LR and 10 with RF. With any positive budget in $\{0.1, 0.2, 0.3, 0.5\}$ nobody is replaced, whichever score is used. Even at 0.1, the largest posterior probabilities are 0.0374 for LR and 0.8954 for RF, below $\tau = 0.95$.[7] Table 3 reports welfare on the held-out records of the five evaluation states. Nominal replacement raises welfare in four of them and lowers it in North Dakota, so the policies with positive budgets, which retain everyone, forgo the four gains and avoid the North Dakota loss.

Table 3: Outcomes in the five evaluation states.

		Target ranking AUC			Nominal-policy gain	
State	Marginal W_1	LR	RF	All-human W	LR	RF
WV	1.635	0.8173	0.8404	1.0357	+0.00228	+0.00940
MS	1.624	0.8225	0.8440	1.0703	+0.00614	+0.00767
NM	1.423	0.8396	0.8659	1.0981	+0.00391	+0.01909
ND	1.668	0.7996	0.8335	1.0442	-0.00407	-0.00377
HI	2.033	0.8148	0.8491	1.0119	+0.00981	+0.01173

Notes: Welfare is the average of $2\text{TPR} - \text{FPR}$ over the 200 decision makers, weighted equally; gains are relative to all-human welfare. AUC is the area under the conventional ROC curve in the evaluation state. The nominal LR and RF policies replace 5 and 10 of the 200 decision makers. At the positive budgets evaluated here, nobody is replaced, so the gain is zero in every state. The marginal W_1 distances use the 1,200-record samples described in Appendix C.1. All entries come from a single sample split and a single group of synthetic decision makers and are not averaged over simulations.

The marginal feature distances between the reference and the evaluation states range from 1.423 to 2.033, yet the nominal policies gain or lose very little. These distances describe how the covariates differ, but their relation to a label-preserving budget also depends on the class proportions and on the common reference law. Moreover, the error rates of the synthetic decision makers are not held fixed across states in this exercise, so (2.4) is not imposed, and the results describe only how policies chosen on the reference records perform on the held-out records of the five evaluation states.

In addition to the budget, the RF frontier depends on the size of the search. At $\rho = 0.5$, using the first 1,000 proposals of the search sequence instead of the first 200 lowers the area under the frontier from 0.3937 to 0.2968. Nobody is replaced at any of the six search sizes. The rule based on the exact empirical extrema on this candidate grid also replaces nobody, because its posterior rectangles are contained in those obtained from the search; this conclusion does not require its frontier values, which are not known. In a separate exercise that shifts the age variable used by the machine, both nominal policies have higher welfare than retaining everyone at every shift we consider. Appendix C reports all budgets, search sizes and age shifts.

[7] For LR the budget is an average budget and the worst-case rates are computed exactly; for RF it is a pointwise budget, and the extrema are approximated by a search over $K = 200$ proposals.

7 Conclusion

The replacement rule combines a robust ROC frontier with the human posterior and selects a fixed machine threshold before deployment. Human error rates are assumed stable across environments. The welfare bounds quantify the possible loss relative to retaining the human in terms of estimation error in the worst-case rates and concentration of the human posterior.

Welfare relative to the matched-FPR benchmark also depends on the thresholds used for humans replaced by both policies. Different thresholds can leave a welfare gap even with a known frontier and increasingly informative histories. Choosing by a welfare lower bound within the same posterior feasible set weakly reduces the threshold term in the bound and improves mean deployment welfare in our simulations. This welfare-based choice requires relative error costs, whereas probability maximization does not.

In the same-score comparison, the average budgets $\varepsilon \in \{0.05, 0.1, 0.2, 0.3\}$ retain everyone, while pointwise radii of 0.05 and 0.1 permit partial replacement with positive mean gains over retention at the evaluated set boundaries and losses at the larger evaluated shifts. The finer grid also gives partial replacement at some smaller average budgets. An average budget allows displacement to be concentrated on some observations; a pointwise budget requires a justified bound on each observation's displacement. The choice of budget should reflect the available information about deployment changes.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used ChatGPT (OpenAI) and Claude (Anthropic) to revise the wording, notation and organization of the manuscript. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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A Proofs and additional statistical results

A.1 Transport bounds

We begin with a result on the probability of an event that need not be closed, which is used to handle the strict threshold. For a nonempty set $A \subseteq \mathcal{X}$, write $d(x, A) := \inf_{a \in A} \|x - a\|_2$, and take closures relative to $\mathcal{X}$.

Lemma 1 (Event probabilities at a positive transport radius)**.** *Let $\mathcal{X}$ be a closed subset of $\mathbb{R}^d$, let μ have a finite first moment, and let A be a nonempty Borel subset of $\mathcal{X}$. For $r > 0$,*

$$\sup_{W_1(\nu,\mu) \le r} \nu(A) = \sup_{W_1(\nu,\mu) \le r} \nu(\overline{A}) \tag{A.1}$$

$$= \inf_{\lambda \ge 0} \left\{ \lambda r + \int (1 - \lambda d(x, A))_+ \, \mu(dx) \right\}. \tag{A.2}$$

Proof. Let F be a nonempty closed set. The function $\mathbb{I}_F$ is bounded and upper semicontinuous, the Euclidean cost is nonnegative, continuous and zero only on the diagonal, and $\mathcal{X}$ is Polish. Hence Theorem 1 of Blanchet and Murthy (2019), specialized to closed events as in their Section 2.4, applies. Its inner maximization is

$$\sup_{z \in \mathcal{X}} \{\mathbb{I}_F(z) - \lambda \|x - z\|_2\} = (1 - \lambda d(x, F))_+.$$

This gives (A.2) for F. We now prove (A.1), without assuming that $\mathbb{I}_A$ is upper semicontinuous.

Put $F = \overline{A}$. Fix any ν with $W_1(\nu, \mu) \le r$ and any $t \in (0, 1)$. The mixture $\nu_t = (1 - t)\nu + t\mu$ satisfies

$$W_1(\nu_t, \mu) \le (1 - t) W_1(\nu, \mu) \le (1 - t) r,$$

by mixing a transport plan with the identity plan. Let $\{a_k : k \ge 1\}$ be a countable dense subset of A. For $\delta > 0$, map each $x \in F$ to the first a_k with $\|x - a_k\|_2 < \delta$, and leave points outside F unchanged. Call this map T_δ. It is Borel measurable, and $\|T_\delta(x) - x\|_2 < \delta$ on F. The image law $\widetilde{\nu} = T_{\delta\#}\nu_t$ therefore satisfies

$$W_1(\widetilde{\nu}, \mu) \le \delta + (1 - t) r, \qquad \widetilde{\nu}(A) \ge \nu_t(F) \ge (1 - t)\nu(F).$$

Choose $0 < \delta < tr$, so that $W_1(\widetilde{\nu}, \mu) \le r$. Letting $t \downarrow 0$ and then taking the supremum over ν gives $\sup \nu(A) \ge \sup \nu(F)$, and the reverse inequality holds because $A \subseteq F$. Finally, $d(x, A) = d(x, F)$, so (A.2) for A follows from the formula for the closed set F. □

Proof of Proposition 1. We first prove (2.8). A coupling with finite cost under (2.6) transports mass only within a label class, so its two class masses are p and $1 - p$ under both marginals. Normalizing its two diagonal blocks gives couplings of (P^1, Q^1) and (P^0, Q^0); conversely, any pair of such conditional couplings can be combined with weights p and $1 - p$. Taking infima over the two blocks proves (2.8). If the label marginals differ, every coupling has infinite cost.

Write $S_c = \{m > c\}$ and $N_c = \mathcal{X} \setminus S_c = \{m \le c\}$. Because the other class can be left

unchanged, (2.8) implies

$$\underline{\beta}_\varepsilon(c;P) = \inf_{W_1(\nu,P^1)\le\varepsilon/p} \nu(S_c),$$
$$\overline{\alpha}_\varepsilon(c;P) = \sup_{W_1(\nu,P^0)\le\varepsilon/(1-p)} \nu(S_c).$$

Thus the class probabilities, the denominators of the two conditional rates, are the same for every law in the ambiguity set, and in each conditional problem the transport cost is the Euclidean distance, which is finite.

For the TPR, N_c is closed because m is continuous. By (A.2) applied to N_c, after subtraction from one,

$$\begin{aligned}\underline{\beta}_\varepsilon(c;P) &= 1 - \inf_{\lambda\ge0}\left\{\frac{\lambda\varepsilon}{p} + \int (1-\lambda d(x,N_c))_+ \, P^1(dx)\right\} \\ &= \sup_{\lambda\ge0}\left\{\int \min\{1,\lambda d_c^-(x)\}\, P^1(dx) - \frac{\lambda\varepsilon}{p}\right\}.\end{aligned}$$

For the FPR, S_c is open, and Lemma 1 with $r = \varepsilon/(1-p) > 0$ gives

$$\overline{\alpha}_\varepsilon(c;P) = \inf_{\lambda\ge0}\left\{\frac{\lambda\varepsilon}{1-p} + \int (1-\lambda d_c^+(x))_+ \, P^0(dx)\right\}.$$

Written as expectations under P, these two formulas are (2.10) and (2.11).

To see that λ can be restricted to $[0, 1/\varepsilon]$, multiply the two objectives by p and $1-p$, respectively, and write

$$T(\lambda) = E_P[Y \min\{1,\lambda d_c^-(X)\}] - \lambda\varepsilon, \qquad F(\lambda) = \lambda\varepsilon + E_P[(1-Y)(1-\lambda d_c^+(X))_+].$$

These functions satisfy $T(0) = 0$, $T(\lambda) \le p - \lambda\varepsilon$, $F(0) = 1-p$ and $F(\lambda) \ge \lambda\varepsilon$. Hence no $\lambda > p/\varepsilon$ maximizes T and no $\lambda > (1-p)/\varepsilon$ minimizes F, so both optimizations can be restricted to $[0, 1/\varepsilon]$. The distances to the two nonempty regions are finite, and by bounded convergence both objectives are continuous on this interval, so their extrema are attained. The same bounds hold for the empirical law of any sample in which both classes occur. The statement for constant rules is immediate, since $\{m(X) > c\}$ has probability zero under every law if S_c is empty and probability one if $S_c = \mathcal{X}$. □

Zero radius. The only law at radius zero is the reference law itself. For nonempty S_c, monotone convergence as $\lambda \to \infty$ gives

$$\inf_{\lambda\ge0} \int (1-\lambda d_c^+(x))_+ \, P^0(dx) = P^0\{d_c^+(X) = 0\} = P^0(\overline{S}_c).$$

This need not equal $P^0(S_c)$. For example, let $m(x) = (1+e^{-x})^{-1}$ and $c = 1/2$, and let P^0 be the point mass at zero on $\mathbb{R}$. Because the threshold is strict, the nominal FPR is zero, whereas the dual above, evaluated at zero radius, equals one. At every positive radius the worst-case FPR is one, since all the negative-class mass can be moved a sufficiently small distance to the right of zero. For this reason the zero-radius rates, like those of the constant rules $S_c = \varnothing$ and $S_c = \mathcal{X}$, are computed directly from the definition.

A.2 Frontier attainment and posterior comparison

Lemma 2 (Attainment of the frontier and equivalent posterior criteria)**.** *Let P be a law under which both classes have positive probability, let $\varepsilon \geq 0$, and let m be a measurable score with values in $[0,1]$; thresholds range over $[0,1]$. Let $A(c) = \overline{\alpha}_\varepsilon(c;P)$ and $B(c) = \underline{\beta}_\varepsilon(c;P)$ be the worst-case rates of Definition 1. For every $\alpha \in [0,1]$, the threshold*

$$c_\alpha := \min\{c \in [0,1] : A(c) \leq \alpha\}$$

exists and satisfies $g_\varepsilon(\alpha;P) = B(c_\alpha)$. For any posterior distribution Π_j on $[0,1]^2$,

$$\sup_{c\in[0,1]} \Pi_j\{\alpha_H \geq A(c),\ \beta_H \leq B(c) \mid \mathcal{D}_j\} = \sup_{\alpha\in[0,1]} \Pi_j\{\alpha_H \geq \alpha,\ \beta_H \leq g_\varepsilon(\alpha;P) \mid \mathcal{D}_j\}. \quad \text{(A.3)}$$

The conclusions also hold when P is the empirical law of a sample in which both classes occur, and for a finite grid of thresholds containing $c = 1$, with the frontier computed over that grid.

Proof. Both A and B are nonincreasing. If $c_n \downarrow c$, the events $\{m > c_n\}$ increase to $\{m > c\}$, so by continuity from below under each law and an interchange of the two suprema,

$$\begin{aligned} A(c) &= \sup_{Q\in\mathcal{B}_\varepsilon(P)} \sup_n Q(m(X) > c_n \mid Y = 0) \\ &= \sup_n \sup_{Q\in\mathcal{B}_\varepsilon(P)} Q(m(X) > c_n \mid Y = 0) = \lim_n A(c_n). \end{aligned}$$

Thus A is right-continuous. Since $A(1) = 0$, the set of thresholds with $A(c) \leq \alpha$ is nonempty, and it is an upper interval because A is nonincreasing. By right-continuity it contains its infimum; if the infimum is 1, this holds because $A(1) = 0$. Hence c_α exists. Every feasible c satisfies $c \geq c_\alpha$, so $B(c) \leq B(c_\alpha)$ by the monotonicity of B alone, and therefore $g_\varepsilon(\alpha;P) = B(c_\alpha)$.

Any threshold c is feasible at $\alpha = A(c)$, so $g_\varepsilon(A(c);P) \geq B(c)$, and the event $\{\alpha_H \geq A(c),\ \beta_H \leq B(c)\}$ is contained in the event obtained by replacing $B(c)$ with $g_\varepsilon(A(c);P)$. Hence the left side of (A.3) is no larger than the right side. Conversely, for every α the threshold c_α satisfies $A(c_\alpha) \leq \alpha$ and $B(c_\alpha) = g_\varepsilon(\alpha;P)$, so that

$$\{\alpha_H \geq \alpha,\ \beta_H \leq g_\varepsilon(\alpha;P)\} \subseteq \{\alpha_H \geq A(c_\alpha),\ \beta_H \leq B(c_\alpha)\}.$$

Taking probabilities and suprema gives the reverse inequality. The same argument applies when P is an empirical law. On a finite grid, the set of feasible candidates is finite and nonempty, so it has a smallest element, and the remaining statements follow from monotonicity. □

If the supremum on the right side of (A.3) is attained at some α, the two inclusions show that the threshold c_α attains the supremum on the left side. In (A.3) one threshold is fixed before the posterior probability is taken, unlike in the event $\{\beta_H \leq g_\varepsilon(\alpha_H;P)\}$, where the threshold may vary with the pair (α_H, β_H).

Proof of Proposition 2. Fix the reference law and the posterior. Since $P \in \mathcal{B}_\varepsilon(P)$, we have $g_\varepsilon(\alpha;P) \leq g(\alpha;P)$. For each *fixed* α, therefore,

$$\{\alpha_H \geq \alpha,\ \beta_H \leq g_\varepsilon(\alpha;P)\} \subseteq \{\alpha_H \geq \alpha,\ \beta_H \leq g(\alpha;P)\}.$$

Taking posterior probabilities at this same α and then maximizing over α gives $\widehat{q}^{\varepsilon}_{\max,j} \leq \widehat{q}^{0}_{\max,j}$. Since both rules use the same cutoff τ, the inclusion of the replacement sets follows. □

A.3 Logistic distances and empirical computation

Lemma 3 (Logistic distances, complexity, and finite-sample evaluation)**.** *Let* $m(x) = \{1 + \exp[-(b + x^\top \beta)]\}^{-1}$ *on* $\mathbb{R}^d$*, with* $\beta \neq 0$*, and restrict* c *to a compact subinterval of* $(0, 1)$*. Then* (2.19) *holds. For fixed* $\varepsilon > 0$ *and* $p_0 \in (0, 1)$*, the classes* $\mathcal{F}^{\pm}$ *are bounded, pointwise measurable VC-subgraph classes and satisfy Assumption 1(ii).*

For an empirical law with $n_1, n_0 > 0$*, put* $n = n_1 + n_0$*. For any nonnegative finite distances* $d = (d_1, \ldots, d_{n_y})$*, define*

$$V(d, \varepsilon) = \max \left\{ \frac{1}{n} \sum_{i=1}^{n_y} u_i : 0 \leq u_i \leq 1, \quad \frac{1}{n} \sum_{i=1}^{n_y} d_i u_i \leq \varepsilon \right\}.$$

For $\varepsilon > 0$*, the empirical worst-case rates are*

$$\underline{\widehat{\beta}}_{\varepsilon}(c) = 1 - \frac{V((d_c^-(X_i))_{Y_i=1}, \varepsilon)}{n_1/n}, \qquad \widehat{\overline{\alpha}}_{\varepsilon}(c) = \frac{V((d_c^+(X_i))_{Y_i=0}, \varepsilon)}{n_0/n}.$$

The maximum that defines V *is attained by setting* $u_i = 1$ *in increasing order of* d_i *until the budget is exhausted or all available mass has been moved, with a fractional value for the last observation if necessary.*

Proof. We prove the three claims in turn.

Distances. The positive region is $\{z(x) > k_c\}$. If $z(x) > k_c$, the projection of x onto the hyperplane $\{z = k_c\}$ is $x - (z(x) - k_c)\beta/\|\beta\|_2^2$, at distance $(z(x) - k_c)/\|\beta\|_2$ from x, and by the Cauchy–Schwarz inequality no shorter displacement reaches $\{z \leq k_c\}$. Points already in that set are at distance zero. The distance to $\{z > k_c\}$ is obtained in the same way, by the reverse projection followed, when needed, by an arbitrarily small additional displacement into this open set. This proves both formulas in (2.19).

Complexity. Let $a = \lambda/\|\beta\|_2$ and $\ell(v) = \min\{1, \max\{0, v\}\}$. Before division by p_0 and $1 - p_0$, the functions in $\mathcal{F}^+$ and $\mathcal{F}^-$ are

$$y\ell\{az(x) - ak_c\}, \qquad (1 - y)\ell\{1 + az(x) - ak_c\}.$$

Each has the form $\ell(az + d)$ on one label class, with $d = -ak_c$ or $d = 1 - ak_c$. For $0 \leq u < 1$, the condition $u < \ell(az + d)$ is equivalent to $u < az + d$; for $u < 0$ it always holds, and for $u \geq 1$ it never does. Hence, on $0 \leq u < 1$, each subgraph coincides with a half-plane in the coordinates (z, u), so the class has finite VC dimension. Multiplying by the indicator of a label class sets the function to zero on the other class, and dividing by p_0 or $1 - p_0$ rescales the coordinate u; both operations preserve the finiteness of the VC dimension. For either class $\mathcal{F}$, whose envelope is the constant $M = 1/p_0$ or $M = 1/(1-p_0)$, Theorem 2.6.7 of van der Vaart and Wellner (1996, p. 141), with $r = 2$, gives

$$N(uM, \mathcal{F}, L_2(H)) \leq K_0 V(\mathcal{F})(16e)^{V(\mathcal{F})} u^{-2\{V(\mathcal{F})-1\}}, \qquad 0 < u < 1.$$

Here $V(\mathcal{F})$ is the VC index of the subgraphs, which is one more than their VC dimension, and K_0 is a universal constant. The bound holds for every probability law H and implies Assumption 1(ii). The functions are continuous in (λ, c) at every (x, y) and the parameter set is compact, so the functions indexed by a countable dense set of parameters approximate every member of the class pointwise; hence the classes are pointwise measurable.

Finite-sample evaluation. The linear program that defines V is feasible and bounded. By linear programming duality, after the multipliers of the constraints $u_i \leq 1$ are eliminated,

$$V(d, \varepsilon) = \min_{\lambda \geq 0} \left\{ \lambda\varepsilon + \frac{1}{n} \sum_{i=1}^{n_y} (1 - \lambda d_i)_+ \right\}.$$

Substituting $\min\{1, v\} = 1 - (1 - v)_+$ into the TPR dual, and using the FPR dual as it stands, gives the two formulas for the empirical worst-case rates. In the primal problem, moving the mass of observation i costs d_i per unit. Shifting expenditure from an observation with a larger positive d_i to one with a smaller d_i cannot decrease the mass moved, so, as in the fractional knapsack problem, some optimum takes the observations in increasing order of d_i, starting with those at distance zero. The dual objective is piecewise linear in λ, with knots at $1/d_i$ for positive d_i, and by Proposition 1 it can be minimized over $[0, 1/\varepsilon]$, so it suffices to evaluate it at 0, at $1/\varepsilon$ and at the knots in between, as described in Section 2.4. By (A.1), the same value is obtained for the open event S_c. □

A.4 Uniform estimation of the robust frontier

Proof of Theorem 1. Write $A(c) = \overline{\alpha}_\varepsilon(c; P_0)$ and $B(c) = \underline{\beta}_\varepsilon(c; P_0)$, with hats for the empirical versions. The suprema over c in the error terms below are taken over $\mathcal{C}$, and we condition throughout on the independent training sample.

Step 1: uniform convergence of the dual objectives. By Proposition 1, the population and empirical dual objectives are optimized over $\Lambda = [0, 1/\varepsilon]$. Under the entropy bound in Assumption 1(ii), with constant envelope M, the uniform entropy integral is bounded by

$$\int_0^1 \sqrt{1 + v \log(K/u)}\, du < \infty.$$

The maximal inequality of van der Vaart and Wellner (2011, equation (1.1)) therefore gives

$$\sup_{f \in \mathcal{F}^+ \cup \mathcal{F}^-} |(\widehat{P}_n - P_0) f| = O_p(n^{-1/2}).$$

Assumption 1(ii) provides the measurability and the bounded envelope that this inequality requires, and the entropy integral remains finite for the union of the two classes.

Moreover, $\widehat{p} - p_0 = O_p(n^{-1/2})$. On the event that $\widehat{p}$ is bounded away from zero and one, whose probability tends to one, replacing the denominators p_0 and $1 - p_0$ by $\widehat{p}$ and $1 - \widehat{p}$ changes each objective by $O_p(n^{-1/2})$ uniformly in (λ, c). This includes the term in $\lambda\varepsilon$: since $\lambda\varepsilon \leq 1$ on Λ, the numerator of the TPR objective, $\widehat{P}_n[Y \min\{1, \lambda d_c^-\}] - \lambda\varepsilon$, is at most one in absolute value, and that of the FPR objective is at most two. Hence both empirical dual objectives are within $O_p(n^{-1/2})$ of their population counterparts, uniformly on $\Lambda \times \mathcal{C}$. A sample in which one class is absent has exponentially small probability; on

that event the estimators may be given any bounded value, which does not affect the conclusion.

Step 2: the worst-case rates. The maps $f \mapsto \sup_\lambda f$ and $f \mapsto \inf_\lambda f$ are 1-Lipschitz in the supremum norm. Hence, with

$$\delta_{\alpha,n} = \sup_c |\widehat{A}(c) - A(c)|, \qquad \delta_{\beta,n} = \sup_c |\widehat{B}(c) - B(c)|,$$

Step 1 gives $\delta_{\alpha,n} + \delta_{\beta,n} = O_p(n^{-1/2})$, which is part (i).

Step 3: the inverse of the worst-case FPR. Suppose $\delta_{\alpha,n} < b_C$. Let c_a denote the population inverse at level a, and for $\alpha \in [\alpha_l, \alpha_u]$ let $\widehat{c}_\alpha$ be the smallest threshold that is feasible in the sample. By the strict endpoint inequalities in Assumption 1(iv), these thresholds lie in $\mathcal{C}$. The same inequalities imply that no $c < c_l$ is feasible, and every $c > c_u$ has a worst-case TPR no larger than that of the feasible threshold c_u. Hence restricting attention to $\mathcal{C}$ changes neither frontier on $[\alpha_l, \alpha_u]$. By Lemma 2, the empirical frontier is attained even if the empirical worst-case rates have jumps.

If $c < c_{\alpha+\delta_{\alpha,n}}$, then $A(c) > \alpha + \delta_{\alpha,n}$ and hence $\widehat{A}(c) > \alpha$. On the other hand, $c_{\alpha-\delta_{\alpha,n}}$ is feasible in the sample. Therefore

$$c_{\alpha+\delta_{\alpha,n}} \le \widehat{c}_\alpha \le c_{\alpha-\delta_{\alpha,n}}. \tag{A.4}$$

Since $-A' \ge \underline{L}_\alpha$ on $\mathcal{C}$, the map $a \mapsto c_a$ is Lipschitz with constant $1/\underline{L}_\alpha$, and so

$$\sup_{\alpha\in[\alpha_l,\alpha_u]} |\widehat{c}_\alpha - c_\alpha| \le \delta_{\alpha,n}/\underline{L}_\alpha.$$

Step 4: the frontier. The inclusions between the population and empirical sets of feasible thresholds also give

$$g_\varepsilon(\alpha - \delta_{\alpha,n}; P_0) - \delta_{\beta,n} \le \widehat{g}_\varepsilon(\alpha) \le g_\varepsilon(\alpha + \delta_{\alpha,n}; P_0) + \delta_{\beta,n}. \tag{A.5}$$

The population inverse and B are Lipschitz on the region covered by Assumption 1(iv), so the population frontier is Lipschitz there with constant at most $L_\beta/\underline{L}_\alpha$. Hence

$$\sup_{\alpha\in[\alpha_l,\alpha_u]} |\widehat{g}_\varepsilon(\alpha) - g_\varepsilon(\alpha; P_0)| \le \delta_{\beta,n} + \frac{L_\beta}{\underline{L}_\alpha}\delta_{\alpha,n}.$$

Together with part (i) and $\Pr(\delta_{\alpha,n} < b_C) \to 1$, this proves part (ii). □

Theorem 1 assumes that the worst-case rates are computed exactly and that all thresholds are searched. Otherwise, the uniform numerical error of the rates is added to the bounds and, on a finite grid, so is the error of approximating the frontier on the grid; see Section 3.2.

A.5 Directional differentiability and the limit distribution of the estimated frontier

Theorem 1 bounds the estimation error of the frontier. A limit distribution for the frontier as a function requires, in addition, a first-order expansion that is uniform in the FPR

level, and pointwise differentiability of an optimized value does not deliver this (Firpo et al., 2023). We give a sufficient condition in terms of the sets of optimal multipliers and verify it below for the logistic score under the conditions of Remark 3.

Let $K = \Lambda \times \mathcal{C}$ and

$$f^+_{\lambda,c}(x,y) = y \min\{1, \lambda d^-_c(x)\}, \qquad f^-_{\lambda,c}(x,y) = (1-y)(1-\lambda d^+_c(x))_+.$$

We represent a reference law by the moments

$$\mathcal{M}(P) = \big(PY,\ (Pf^+_{\lambda,c})_K,\ (Pf^-_{\lambda,c})_K\big) \in \mathbb{D} := \mathbb{R} \times \ell^\infty(K)^2, \tag{A.6}$$

where $\mathbb{D}$ is normed by the maximum of the three coordinate norms. We regard the frontier as a function of the array $\mathcal{M}(P)$ and call this function the frontier map; it is defined on the arrays $\mathcal{M}(P)$ of laws P that satisfy the conditions of Proposition 1, with $0 < PY < 1$. Directions of differentiation belong to $\mathbb{T} = \mathbb{R} \times C(K)^2$, where $C(K)$ is the space of continuous functions on K. We call a family P_t of such laws admissible if $\{\mathcal{M}(P_t) - \mathcal{M}(P_0)\}/t \to h \in \mathbb{T}$ in this norm as $t \downarrow 0$; the difference quotients themselves need not belong to $\mathbb{T}$. The sets of optimal multipliers under P_0 are

$$\Gamma_+(c) = \arg\max_{\lambda\in\Lambda}\{P_0 f^+_{\lambda,c} - \lambda\varepsilon\}, \qquad \Gamma_-(c) = \arg\min_{\lambda\in\Lambda}\{P_0 f^-_{\lambda,c} + \lambda\varepsilon\}.$$

Theorem 7 (Functional differentiation on a regular threshold region)**.** *Suppose that Assumption 1 holds, that $(\lambda, c) \mapsto f^\pm_{\lambda,c}$ is continuous into $L_2(P_0)$ on K, and that $c \mapsto \Gamma_\pm(c)$ is continuous in the Hausdorff distance between nonempty compact subsets of Λ. Let $A_0(c) = \overline{\alpha}_\varepsilon(c; P_0)$ and $B_0(c) = \underline{\beta}_\varepsilon(c; P_0)$. Then the frontier, as a function of the moments (A.6), is Hadamard directionally differentiable at $\mathcal{M}(P_0)$ tangentially to $\mathbb{T}$, with values in $\ell^\infty([\alpha_l, \alpha_u])$. For $h = (h_p, h_+, h_-)$, the derivatives of the two worst-case rates are*

$$\dot{B}_h(c) = \frac{\max_{\lambda\in\Gamma_+(c)} h_+(\lambda, c) - B_0(c)h_p}{p_0}, \tag{A.7}$$

$$\dot{A}_h(c) = \frac{\min_{\lambda\in\Gamma_-(c)} h_-(\lambda, c) + A_0(c)h_p}{1-p_0}. \tag{A.8}$$

They are continuous in c. The derivatives of the inverse and of the frontier are, uniformly in $\alpha \in [\alpha_l, \alpha_u]$,

$$\dot{c}_h(\alpha) = -\frac{\dot{A}_h(c_\alpha)}{A_0'(c_\alpha)}, \qquad \dot{g}_h(\alpha) = \dot{B}_h(c_\alpha) - \frac{B_0'(c_\alpha)}{A_0'(c_\alpha)}\dot{A}_h(c_\alpha). \tag{A.9}$$

In particular, for every admissible family P_t,

$$\left\| \frac{g_\varepsilon(\cdot; P_t) - g_\varepsilon(\cdot; P_0)}{t} - \dot{g}_h \right\|_\infty \longrightarrow 0.$$

If each set $\Gamma_\pm(c)$ is a singleton, the derivative is linear and the map is fully Hadamard differentiable tangentially to $\mathbb{T}$.

The condition on $\Gamma_\pm$ allows the optimal multipliers to be nonunique, provided the sets vary continuously with c. It yields the separation (A.12) used in the proof of Lemma 4 below.

Remark 3 (Primitive conditions for the logistic score)**.** *For the logistic score on $\mathbb{R}^d$ with $\beta \neq 0$, let $S = z(X)/\|\beta\|_2$ be the normalized index of Remark 1 and let $a(c) = \operatorname{logit}(c)/\|\beta\|_2$. Suppose that $S \mid Y = y$ has a continuous, strictly positive density on $\mathbb{R}$ and a finite first moment, and that, on a compact interval $\mathcal{C}$ of thresholds,*

$$\frac{\varepsilon}{p_0} < \mathbb{E}[(S - a(c))_+ \mid Y = 1], \qquad \frac{\varepsilon}{1 - p_0} < \mathbb{E}[(a(c) - S)_+ \mid Y = 0] \quad \text{for every } c \in \mathcal{C}. \tag{A.10}$$

Lemma 5 shows that the optimal multipliers are then unique and continuous in c, that the functions $f^{\pm}_{\lambda,c}$ are continuous in $L_2(P_0)$, and that both worst-case rates are continuously differentiable on $\mathcal{C}$ with strictly negative derivatives bounded away from zero. Once an FPR interval is chosen as in Assumption 1(iv), Theorem 7 applies.

Under the conditions of Theorem 7, the empirical process indexed by Y and the two classes $\{f^{\pm}_{\lambda,c}\}$ converges weakly in $\mathbb{D}$ to a tight Gaussian process $\mathbb{G}$ with paths in $\mathbb{T}$ and covariance function $P_0(fg) - (P_0 f)(P_0 g)$. The delta method for directionally differentiable maps then gives

$$\sqrt{n}\{\widehat{g}_\varepsilon - g_\varepsilon(\cdot; P_0)\} \rightsquigarrow \dot{g}_{\mathbb{G}} \quad \text{in } \ell^\infty([\alpha_l, \alpha_u]). \tag{A.11}$$

Step 4 of the proof of Theorem 7 verifies the conditions of Fang and Santos (2019, Theorem 2.1). As in Theorem 1, the limit is conditional on the independent training sample, with the fitted score and the positive radius held fixed as the validation sample grows. It applies to the estimator that searches over all thresholds and computes the rates exactly. A grid-based or numerically approximated estimator shares this limit if its uniform difference from the exact estimator is $o_p(n^{-1/2})$.

If the optimal multipliers are unique, the limit in (A.11) is Gaussian. With a nonlinear directional derivative it need not be, and (A.11) alone does not justify the standard bootstrap (Fang and Santos, 2019, Section 3). By the chain rule, (A.11) also yields the limit distribution of smooth finite-dimensional functionals of the frontier. We do not pursue the estimation of the law of $\dot{g}_{\mathbb{G}}$ or the construction of simultaneous confidence bands.

The proof of Theorem 7 uses a lemma on the derivative of an optimized value that holds uniformly in a parameter. For nonempty compact sets $C, D \subset \Lambda$, the Hausdorff distance is

$$d_{\mathrm{Haus}}(C, D) = \max\left\{\sup_{x \in C} \inf_{y \in D} |x - y|, \sup_{y \in D} \inf_{x \in C} |x - y|\right\}.$$

Lemma 4 (Uniform expansion of an optimized value)**.** *Let F be continuous on the product of compact metric spaces $\Lambda \times \mathcal{C}$, and let $V(c) = \max_{\lambda \in \Lambda} F(\lambda, c)$. Suppose that $c \mapsto \Gamma(c) = \arg\max_\lambda F(\lambda, c)$ is continuous in the Hausdorff distance. If $H_t \to H$ uniformly, with H continuous on the product and each H_t bounded, then*

$$\sup_{c \in \mathcal{C}} \left| \frac{\sup_\lambda\{F(\lambda, c) + tH_t(\lambda, c)\} - V(c)}{t} - \max_{\lambda \in \Gamma(c)} H(\lambda, c) \right| \longrightarrow 0 \quad (t \downarrow 0).$$

The limit $c \mapsto \max_{\lambda \in \Gamma(c)} H(\lambda, c)$ is continuous in c and Lipschitz in H with respect to the uniform norm. The analogous statement holds for minima.

Proof. The function V is continuous because F is uniformly continuous. Fix $\eta > 0$. Points with $\operatorname{dist}(\lambda, \Gamma(c)) \geq \eta$ are uniformly separated from the optimum:

$$\inf_{c \in \mathcal{C},\, \lambda \in \Lambda : \operatorname{dist}(\lambda, \Gamma(c)) \geq \eta} \{V(c) - F(\lambda, c)\} > 0 \tag{A.12}$$

whenever the set over which the infimum is taken is nonempty. If this failed, there would be a sequence (λ_k, c_k) in that set with $V(c_k) - F(\lambda_k, c_k) \to 0$. By compactness, a subsequence converges to some (λ, c), and by the continuity of F, of V and of $(\lambda, c) \mapsto \operatorname{dist}(\lambda, \Gamma(c))$, the limit satisfies $F(\lambda, c) = V(c)$ and $\operatorname{dist}(\lambda, \Gamma(c)) \geq \eta$, which is impossible.

For every c, choose $\lambda_t(c) \in \Lambda$ at which $F(\cdot, c) + tH_t(\cdot, c)$ is within t^2 of its supremum; since the bounds below are deterministic, this choice need not be measurable in c. Because $\|H_t\|_\infty$ is bounded for small t,

$$0 \leq V(c) - F(\lambda_t(c), c) \leq 2t\|H_t\|_\infty + t^2.$$

Equation (A.12) then gives $\sup_c \operatorname{dist}(\lambda_t(c), \Gamma(c)) \to 0$. Evaluating $F + tH_t$ at a maximizer of H on $\Gamma(c)$ shows that the difference quotient is at least $\max_{\Gamma(c)} H - \|H_t - H\|_\infty$, and evaluating it at $\lambda_t(c)$ shows that the difference quotient is at most $H_t(\lambda_t(c), c) + t$. By the uniform continuity of H and the convergence of $\sup_c \operatorname{dist}(\lambda_t(c), \Gamma(c))$ to zero, this upper bound is at most $\max_{\Gamma(c)} H + o(1)$, uniformly in c, which proves the expansion. The continuity of $c \mapsto \max_{\Gamma(c)} H$ follows from the Hausdorff continuity of Γ and the uniform continuity of H, and the Lipschitz property holds because $|\max_{\Gamma(c)} H - \max_{\Gamma(c)} \widetilde{H}| \leq \|H - \widetilde{H}\|_\infty$ for any two continuous functions H and $\widetilde{H}$ on $\Lambda \times \mathcal{C}$. For minima, apply the result to $-F$, $-H_t$ and $-H$. □

Proof of Theorem 7. Let P_t be an admissible family, so that $\mathcal{M}(P_t) = \mathcal{M}(P_0) + th_t$ with $h_t \to h = (h_p, h_+, h_-)$ in $\mathbb{D}$ and $h \in \mathbb{T}$. By Proposition 1, the multipliers can be restricted to $\Lambda = [0, 1/\varepsilon]$ for every P_t. By the Cauchy–Schwarz inequality, the $L_2(P_0)$ continuity assumed in the theorem implies that $P_0 f^{\pm}_{\lambda,c}$ is continuous on K.

Step 1: expansion of the worst-case rates. Write

$$\Psi_0^+(\lambda, c) = \frac{P_0 f^+_{\lambda,c} - \lambda\varepsilon}{p_0}, \qquad \Psi_0^-(\lambda, c) = \frac{P_0 f^-_{\lambda,c} + \lambda\varepsilon}{1 - p_0}.$$

Let $\Psi_t^{\pm}$ denote the same expressions with P_0 and p_0 replaced by P_t and P_tY. Writing each difference $\Psi_t^{\pm} - \Psi_0^{\pm}$ over a common denominator, and using the fact that P_tY is bounded away from zero and one for small t and that h_t is bounded, we obtain, uniformly on K,

$$\frac{\Psi_t^+ - \Psi_0^+}{t} \longrightarrow D_h^+(\lambda, c) = \frac{h_+(\lambda, c) - \Psi_0^+(\lambda, c) h_p}{p_0},$$
$$\frac{\Psi_t^- - \Psi_0^-}{t} \longrightarrow D_h^-(\lambda, c) = \frac{h_-(\lambda, c) + \Psi_0^-(\lambda, c) h_p}{1 - p_0}.$$

Both limits are continuous on K. Applying Lemma 4 to the maximum and to the minimum therefore gives

$$\|B_t - B_0 - t\dot{B}_h\|_{\mathcal{C}} = o(t), \qquad \|A_t - A_0 - t\dot{A}_h\|_{\mathcal{C}} = o(t), \tag{A.13}$$

where $A_t = \overline{\alpha}_\varepsilon(\cdot; P_t)$ and $B_t = \underline{\beta}_\varepsilon(\cdot; P_t)$. Since Ψ_0^+ equals $B_0(c)$ on $\Gamma_+(c)$ and Ψ_0^- equals $A_0(c)$ on $\Gamma_-(c)$, the derivatives are those in (A.7)–(A.8), and by Lemma 4 they are continuous in c.

Step 2: the inverse. Let $c_{\alpha,t} = \min\{c \in \mathcal{C} : A_t(c) \le \alpha\}$. Each A_t is nonincreasing and right-continuous, as in Lemma 2. By the strict endpoint inequalities in Assumption 1(iv) and the argument that leads to (A.4),

$$\sup_{\alpha} |c_{\alpha,t} - c_\alpha| = O(t),$$

and all these thresholds lie in the interior of $\mathcal{C}$ for small t. Let $R_t = A_t - A_0 - t\dot{A}_h$, so that $\|R_t\|_\infty = o(t)$ by (A.13). Since $A_0 + t\dot{A}_h$ is continuous, every jump of A_t in the interior of $\mathcal{C}$ is at most $2\|R_t\|_\infty = o(t)$. By the minimality of $c_{\alpha,t}$ and right-continuity,

$$A_t(c_{\alpha,t}) \le \alpha \le A_t(c_{\alpha,t}-).$$

Consequently,

$$\sup_{\alpha} |A_t(c_{\alpha,t}) - \alpha| = o(t). \tag{A.14}$$

Since $A_0(c_\alpha) = \alpha$, the uniform continuity of A_0' and $\dot{A}_h$ gives

$$A_t(c_{\alpha,t}) - \alpha = A_0'(c_\alpha)(c_{\alpha,t} - c_\alpha) + t\dot{A}_h(c_\alpha) + o(t)$$

uniformly in α. Dividing by t, using (A.14), and dividing by $A_0'(c_\alpha)$, whose absolute value is at least $\underline{L}_\alpha$, proves the formula for $\dot{c}_h$ in (A.9), uniformly in α. Here we have used the fact that, by (A.13) and the continuity of $\dot{A}_h$,

$$\sup_{\alpha} \left| A_t(c_{\alpha,t}) - A_0(c_{\alpha,t}) - t\dot{A}_h(c_\alpha) \right| = o(t).$$

Step 3: the frontier. By Lemma 2, $g_\varepsilon(\alpha; P_t) = B_t(c_{\alpha,t})$. Hence (A.13), the bound on $c_{\alpha,t} - c_\alpha$, and the continuity of B_0' and $\dot{B}_h$ give

$$g_\varepsilon(\alpha; P_t) - g_\varepsilon(\alpha; P_0) = t\dot{B}_h(c_\alpha) + B_0'(c_\alpha)(c_{\alpha,t} - c_\alpha) + o(t)$$

uniformly in α, and substituting $\dot{c}_h$ gives the expansion of the frontier stated in the theorem. The derivative map $h \mapsto \dot{g}_h$ is continuous, in fact Lipschitz, because the maximum and the minimum over a fixed compact set are Lipschitz and the ratios $1/A_0'$ and B_0'/A_0' are bounded on $\mathcal{C}$. Since these expansions hold for every admissible family P_t, the frontier map is Hadamard directionally differentiable at $\mathcal{M}(P_0)$ tangentially to $\mathbb{T}$, in the norm of (A.6); the directions are the limits, in that norm, of the difference quotients of the moment arrays. When a direction is generated by a signed measure H and belongs to $\mathbb{T}$, its coordinates are $h_p = HY$ and $h_\pm(\lambda, c) = Hf^\pm_{\lambda,c}$. If the sets $\Gamma_\pm(c)$ are singletons, the derivatives in (A.7)–(A.9) are linear in h.

Step 4: the weak limit (A.11)**.** The classes $\{f^+_{\lambda,c}\}$ and $\{f^-_{\lambda,c}\}$ differ from $\mathcal{F}^\pm$ only by the constant factors p_0 and $1-p_0$, so they have bounded envelopes and satisfy a polynomial covering bound as in Assumption 1(ii); the same holds after the function Y is added. The finite-dimensional distributions of the centered and scaled empirical averages converge by the multivariate central limit theorem. For tightness, consider differences $f - g$ of two functions in the same class with $\|f - g\|_{L_2(P_0)} \le \delta$. Approximating each function pointwise by members of the countable subclass and applying bounded convergence shows that the supremum of the empirical process over these differences is bounded by its supremum over

countably many differences with $L_2(P_0)$ norm less than 2δ. This countable class satisfies a polynomial covering bound and has envelope at most two, and both it and the class of its squares are measurable. After the envelope is rescaled, the local maximal inequality of van der Vaart and Wellner (2011, Theorem 2.1) bounds the limit superior, as $n \to \infty$, of the expected supremum by a constant times

$$\int_0^{2\delta} \sqrt{1 + v \log(C/u)}\, du,$$

where $C \geq 1$ is a fixed covering constant. This integral tends to zero as $\delta \downarrow 0$, and the remaining term in the inequality tends to zero as $n \to \infty$ for each fixed δ. Suprema over the countable classes are measurable, and outer probabilities can be used for the original processes. Because K is compact and $(\lambda, c) \mapsto f^{\pm}_{\lambda,c}$ is continuous into $L_2(P_0)$, the index set is totally bounded, so this asymptotic equicontinuity implies tightness in $\mathbb{D}$, and the limiting Gaussian process has a version with continuous paths on K. Thus

$$\sqrt{n}\{\mathcal{M}(\widehat{\mathbb{P}}_n) - \mathcal{M}(P_0)\} \rightsquigarrow \mathbb{G}, \qquad \Pr(\mathbb{G} \in \mathbb{T}) = 1.$$

A sample in which a class is absent may be assigned any fixed bounded frontier, since by Assumption 1(iii) the probability of such a sample tends to zero; for every other sample, $\mathcal{M}(\widehat{\mathbb{P}}_n)$ belongs to the domain of the frontier map. The conditions of Fang and Santos (2019, Theorem 2.1) therefore hold: $\mathbb{D}$ and $\ell^\infty([\alpha_l, \alpha_u])$ are normed spaces, the frontier map is Hadamard directionally differentiable at $\mathcal{M}(P_0)$ tangentially to $\mathbb{T}$, and the limit $\mathbb{G}$ is tight and concentrated on $\mathbb{T}$. This proves (A.11). The limit of a continuously differentiable finite-dimensional functional of the frontier then follows from the chain rule. □

Lemma 5 (Verification for the logistic score)**.** *Under the conditions of Remark 3, the two continuity conditions of Theorem 7 hold, and on $\mathcal{C}$ both worst-case rates are continuously differentiable with strictly negative derivatives bounded away from zero.*

Proof. Let F_y and f_y be the distribution function and the density of S given $Y = y$, write $a = a(c)$, and put $r_1 = \varepsilon/p_0$ and $r_0 = \varepsilon/(1-p_0)$. Because the densities are strictly positive and (A.10) holds, there are unique finite $w_1(a), w_0(a) > 0$ that solve

$$\int_0^{w_1(a)} u f_1(a+u)\, du = r_1, \qquad \int_0^{w_0(a)} u f_0(a-u)\, du = r_0. \tag{A.15}$$

Indeed, each integral is continuous and strictly increasing in its upper limit, equals zero when the upper limit is zero, and tends to the corresponding expectation in (A.10) as the upper limit tends to infinity. Since the integrals are jointly continuous in a and the upper limit, and each root is bracketed by two strict inequalities, w_y is continuous on the compact set of values of a.

For $\lambda > 0$, the derivative with respect to λ of the dual objective for the TPR, conditional on $Y = 1$, is

$$\mathbb{E}[(S-a)\mathbf{1}\{a < S < a + 1/\lambda\} \mid Y = 1] - r_1.$$

Here we use the fact that S has no atom at the endpoint $a + 1/\lambda$. The derivative is strictly decreasing in λ, from a positive value to $-r_1$, so its zero is the unique maximizer,

$\lambda_+(c) = 1/w_1(a(c))$. The derivative of the dual objective for the FPR is

$$r_0 - \mathbb{E}[(a - S)\mathbf{1}\{a - 1/\lambda < S < a\} \mid Y = 0],$$

which is strictly increasing and passes through zero; the unique minimizer is therefore $\lambda_-(c) = 1/w_0(a(c))$. By Proposition 1, both lie in Λ, and since they are continuous in c, the Hausdorff continuity condition holds. The functions $f^{\pm}_{\lambda,c}$ are bounded and continuous in (λ, c) at every (x, y), so by dominated convergence they are continuous in $L_2(P_0)$, including at $\lambda = 0$.

Set $z_1(a) = a + w_1(a)$ and $z_0(a) = a - w_0(a)$. Substituting (A.15) into the dual objectives gives

$$B_0(c) = 1 - F_1(z_1(a(c))), \qquad A_0(c) = 1 - F_0(z_0(a(c))).$$

To differentiate these expressions without assuming that the densities are differentiable, write the first equation in (A.15) as $\int_a^{z_1}(v - a)f_1(v)\,dv = r_1$ and the second as $\int_{z_0}^{a}(a - v)f_0(v)\,dv = r_0$. These integrals are continuously differentiable in their endpoints, with nonzero derivatives with respect to z_1 and z_0, and implicit differentiation gives

$$\frac{dB_0}{da} = -\frac{F_1(a + w_1(a)) - F_1(a)}{w_1(a)},$$
$$\frac{dA_0}{da} = -\frac{F_0(a) - F_0(a - w_0(a))}{w_0(a)}.$$

Each ratio is the average of a continuous, strictly positive density over an interval whose endpoints are continuous in a; it is therefore continuous and strictly positive, with a positive minimum and a finite maximum on the compact set of values of a. Finally, $a'(c) = 1/(\|\beta\|_2 c(1 - c))$ is positive and bounded above and away from zero on $\mathcal{C} \subset (0, 1)$, which proves the claim about the derivatives with respect to c. The strict endpoint inequalities in Assumption 1(iv) hold once the FPR interval is chosen strictly inside $A_0(\mathcal{C})$. □

A.6 The human posterior and the selected threshold

Lemma 6 (Dirichlet posterior calculation and concentration)**.** *For a fixed doctor, suppose that the n_j historical pairs $(Y_i, \widehat{Y}_{i,j})$ are independent draws from a common law in the reference environment, with $p_j = \mathbb{P}(Y_i = 1) \in [p_*, 1 - p_*]$ for some $p_* \in (0, 1/2]$, and that the four cell probabilities have a fixed symmetric Dirichlet prior with parameter $a_0 > 0$. Then the posterior rates are independent and have the laws in (2.17), and (2.18) is exactly the posterior probability of the event $\{\alpha_H \geq \widehat{\overline{\alpha}}_\varepsilon(c),\ \beta_H \leq \widehat{\underline{\beta}}_\varepsilon(c)\}$. Moreover, Assumption 4 holds with, for example,*

$$C_j = C(a_0) := \max\{8, \exp(a_0/4)\} + 2, \qquad \zeta_j = \zeta(p_*) := \min\{p_*/16, p_*^2/2\}. \tag{A.16}$$

Proof. The multinomial likelihood and the Dirichlet prior give a Dirichlet posterior for the four cell probabilities. We represent this posterior, which is the one used by Feng et al. (2025), by four independent Gamma variables whose shape parameters are the four posterior cell parameters, each variable being divided by the sum of the four. The sum cancels in the ratios that define the FPR and the TPR, and the two ratios involve disjoint

pairs of the Gamma variables. Hence they are independent and have the Beta laws in (2.17), which proves (2.18).

We next bound the tails of the posterior, conditional on the observed counts. Let $Z \sim \mathrm{Beta}(a, b)$, write $s = a + b$ and $\mu = a/s$, and represent Z as $G_1/(G_1 + G_2)$ with independent Gamma variables of unit scale. For $\mu < x < 1$, the Chernoff bound gives

$$\mathbb{P}(Z \geq x) \leq \inf_{0<v<1/(1-x)} \{1 - v(1-x)\}^{-a}(1 + vx)^{-b}.$$

Taking $v = (x - \mu)/\{x(1-x)\}$ yields $\mathbb{P}(Z \geq x) \leq \exp\{-s\,\mathrm{kl}(\mu, x)\}$, where

$$\mathrm{kl}(\mu, x) = \mu \log(\mu/x) + (1-\mu)\log\{(1-\mu)/(1-x)\}.$$

For $0 < x < \mu$, the same calculation with $v < 0$ bounds $\mathbb{P}(Z \leq x)$. Since $\partial_x \mathrm{kl}(\mu, x) = (x - \mu)/\{x(1-x)\}$ and $x(1-x) \leq 1/4$, integrating from μ to x gives $\mathrm{kl}(\mu, x) \geq 2(x-\mu)^2$ for x on either side of μ. For $u > 0$ with $\mu + u \geq 1$ or $\mu - u \leq 0$, the corresponding tail probability is zero, so in all cases

$$\mathbb{P}(|Z - \mu| > u) \leq 2\exp(-2su^2), \qquad u > 0. \tag{A.17}$$

Since (A.17) is centered at the mean of the Beta law, it remains to relate the posterior means to the true rates.

Let $N_{j0} = \mathrm{FP}_j + \mathrm{TN}_j$, $N_{j1} = \mathrm{TP}_j + \mathrm{FN}_j$ and $m_j = \min(N_{j0}, N_{j1})$. Conditional on the label counts, FP_j and TP_j are binomial under the sampling assumption of the lemma. If $m_j > 0$, let $\widetilde{\theta}_j = (\mathrm{FP}_j/N_{j0}, \mathrm{TP}_j/N_{j1})$ and let μ_j be the vector of posterior means. Then

$$\|\mu_j - \widetilde{\theta}_j\|_\infty \leq b_j := \frac{a_0}{m_j + 2a_0}.$$

The inequality of Hoeffding (1963) for bounded variables gives

$$\mathbb{P}\big(\|\widetilde{\theta}_j - \theta_{H,j}\|_\infty > u \mid N_{j0}, N_{j1}\big) \leq 4\exp(-2m_j u^2). \tag{A.18}$$

If $t \geq 4b_j$, then on the event $\{\|\widetilde{\theta}_j - \theta_{H,j}\|_\infty \leq t/4\}$ the posterior mean is within $t/2$ of $\theta_{H,j}$. Applying (A.17) to the two posterior coordinates then gives

$$T_j(t) \leq 4\exp\{-(m_j + 2a_0)t^2/2\}$$

on this event. By (A.18), the conditional probability of its complement is at most $4\exp(-m_j t^2/8)$. Therefore

$$\mathbb{E}[T_j(t) \mid N_{j0}, N_{j1}] \leq 8\exp(-m_j t^2/8).$$

If $t < 4b_j$, we use $T_j(t) \leq 1$ and

$$\frac{m_j t^2}{8} < \frac{2m_j a_0^2}{(m_j + 2a_0)^2} \leq \frac{a_0}{4}.$$

If one class is absent, then $m_j = 0$ and the bound $T_j(t) \leq 1$ still applies. Hence, for all

label counts and $0 < t \le 1$,

$$\mathbb{E}[T_j(t) \mid N_{j0}, N_{j1}] \le C_a \exp(-m_j t^2/8), \qquad C_a = \max\{8, e^{a_0/4}\}. \tag{A.19}$$

Finally, N_{j1} is binomial with success probability p_j, and Hoeffding's inequality applied to the class counts yields

$$\mathbb{P}(m_j < n_j p_*/2) \le 2\exp(-n_j p_*^2/2).$$

Using (A.19) on the complement of this event and $T_j(t) \le 1$ on the event itself, we obtain

$$\begin{aligned}\mathbb{E}[T_j(t)] &\le C_a \exp(-n_j p_* t^2/16) + 2\exp(-n_j p_*^2/2) \\ &\le (C_a + 2)\exp\{-n_j t^2 \min(p_*/16, p_*^2/2)\},\end{aligned}$$

where the last step uses $t \le 1$. This is the bound in Assumption 4 with the constants in (A.16). □

Proof of Theorem 2. Write $g = g_\varepsilon(\cdot; P_0)$ and work on $\mathcal{E}_n = \{r_n \le \delta_0/4\}$. For a candidate c, we call $\{\theta_H : \alpha_H \ge \widehat{\overline{\alpha}}_\varepsilon(c),\ \beta_H \le \widehat{\underline{\beta}}_\varepsilon(c)\}$ the rectangle of c; its posterior probability is $\widehat{q}_j^{\varepsilon}(c)$. The neighborhood

$$K_j = \{\theta_H : \|\theta_H - \theta_{H,j}\|_\infty \le t_0\}$$

of the true pair has posterior probability $1 - T_j(t_0)$.

For a dominated doctor, let $\alpha^* = \alpha_{H,j} - t_0$, which belongs to $[\alpha_l, \alpha_u]$. By Lemma 2, the empirical frontier over the candidate set is attained, so there is a candidate c^* with

$$\widehat{\overline{\alpha}}_\varepsilon(c^*) \le \alpha^*, \qquad \widehat{\underline{\beta}}_\varepsilon(c^*) = \widehat{g}_\varepsilon(\alpha^*).$$

Every $\theta_H \in K_j$ satisfies $\alpha_H \ge \alpha^* \ge \widehat{\overline{\alpha}}_\varepsilon(c^*)$. In addition,

$$\begin{aligned}\widehat{g}_\varepsilon(\alpha^*) &\ge g(\alpha_{H,j}) - L_g t_0 - r_n \\ &\ge \beta_{H,j} + \delta_0 - L_g t_0 - r_n > \beta_{H,j} + t_0 \ge \beta_H,\end{aligned}$$

because $(1 + L_g)t_0 + r_n \le \delta_0/2$. Thus the rectangle of c^* contains K_j, and since $\widehat{q}^{\varepsilon}_{\max,j}$ is the maximum over all candidates, $\widehat{q}^{\varepsilon}_{\max,j} \ge 1 - T_j(t_0)$.

For a non-dominated doctor, suppose that the rectangle of some candidate c intersects K_j. Then $\widehat{\overline{\alpha}}_\varepsilon(c) \le \alpha_{H,j} + t_0$ and $\widehat{\underline{\beta}}_\varepsilon(c) \ge \beta_{H,j} - t_0$. But c is then feasible at level $\alpha_{H,j} + t_0$, so

$$\begin{aligned}\widehat{\underline{\beta}}_\varepsilon(c) &\le \widehat{g}_\varepsilon(\alpha_{H,j} + t_0) \\ &\le g(\alpha_{H,j}) + L_g t_0 + r_n < \beta_{H,j} - t_0,\end{aligned}$$

a contradiction. Hence the rectangle of every candidate is disjoint from K_j and has posterior probability at most $T_j(t_0)$, so that $\widehat{q}^{\varepsilon}_{\max,j} \le T_j(t_0)$.

On $\mathcal{E}_n$, a dominated doctor is retained only if $T_j(t_0) > 1 - \tau$, and a non-dominated doctor is replaced only if $T_j(t_0) \ge \tau$. Hence, on $\mathcal{E}_n$, the event that at least one decision is wrong is contained in $\bigcup_j \{T_j(t_0) \ge \bar{\tau}\}$. By Markov's inequality,

$$\mathbb{P}(\text{at least one wrong decision}) \le \mathbb{P}(\mathcal{E}_n^c) + \frac{1}{\bar{\tau}} \sum_{j=1}^{J} \mathbb{E}[T_j(t_0)],$$

which, by Assumption 4, proves (3.7). The same tail bound gives $T_j(t_0) \to_p 0$, and with it the limits of $\widehat{q}^{\varepsilon}_{\max,j}$ stated in the theorem. □

Proof of Theorem 3. For any selected candidate, (3.9) follows from the definitions of $e_{\alpha,n}$ and $e_{\beta,n}$ and of the population worst-case rates. Every pair $\theta_H = (\alpha_H, \beta_H)$ in the rectangle of $\widehat{c}_j$ satisfies

$$\widehat{\overline{\alpha}}_{\varepsilon}(\widehat{c}_j) \le \alpha_H, \qquad \widehat{\underline{\beta}}_{\varepsilon}(\widehat{c}_j) \ge \beta_H.$$

By (3.9), the two inequalities in the event displayed in the theorem then hold for this pair, simultaneously for all $Q \in \mathcal{B}_{\varepsilon}(P_0)$. The rectangle of $\widehat{c}_j$, whose posterior probability is at least τ, is therefore contained in that event.

If $T_j(t) < \tau$, this rectangle cannot be disjoint from the t-neighborhood $\{\theta_H : \|\theta_H - \theta_{H,j}\|_\infty \le t\}$ of the true pair, because the posterior probability of their intersection is at least $\tau - T_j(t) > 0$. Taking a pair θ_H in the intersection, we obtain

$$\widehat{\overline{\alpha}}_{\varepsilon}(\widehat{c}_j) \le \alpha_{H,j} + t, \qquad \widehat{\underline{\beta}}_{\varepsilon}(\widehat{c}_j) \ge \beta_{H,j} - t.$$

Substituting these inequalities into (3.9) proves (3.10).

If the square root in (3.11) is less than one, Markov's inequality and a union bound give

$$\mathbb{P}\{\max_j T_j(t_n(\gamma)) > \tau/2\} \le \frac{2JC_0}{\tau} \exp\{-\zeta_0 n_{\min} t_n(\gamma)^2\} \le \gamma.$$

If the square root is at least one, then $t_n(\gamma) = 1$ and $T_j(1) = 0$, because the true pair and every pair in the support of the posterior lie in $[0,1]^2$. On the complement of the event in the last display, $T_j(t_n(\gamma)) < \tau$ for every j, so the rectangle of every selected threshold intersects the $t_n(\gamma)$-neighborhood of the true pair, and the last statement of the theorem follows. Here the probability is over the doctor histories, and $e_{\alpha,n}$ and $e_{\beta,n}$ are the errors, possibly random, of the validation sample at hand. □

A.7 Welfare comparisons

Proof of Theorem 4 and identity (3.15). Consider the event on which the last statement of Theorem 3 holds; its sampling probability is at least $1-\gamma$. On this event, every replaced doctor satisfies

$$\begin{aligned} W_j(\widehat{c}_j; Q) - W_{H,j}(Q) &= \phi[\mathrm{TPR}_Q(\widehat{c}_j) - \beta_{H,j}] - \eta[\mathrm{FPR}_Q(\widehat{c}_j) - \alpha_{H,j}] \\ &\ge -\phi e_{\beta,n} - \eta e_{\alpha,n} - (\phi + \eta) t_n(\gamma) \end{aligned}$$

for every $Q \in \mathcal{B}_{\varepsilon}(P_0)$. Doctors who are not replaced contribute nothing to the difference from the all-human policy, so summing over the replaced doctors proves (3.14) on this event.

For (3.15), which uses no margin assumption, partition the doctors into $\mathcal{I}_n$, $\mathcal{M}_n$, $\mathcal{A}_n$ and those in neither replacement set. Doctors in the first group are replaced under both the benchmark and the implemented rule, those in the second only under the benchmark, those in the third only under the implemented rule, and those in the fourth under neither. The corresponding terms of the welfare difference are exactly the three sums in (3.15) and zero. □

Proof of Theorem 5. Part (a). Let $Q \in \mathcal{B}_\varepsilon(P_0)$. A threshold whose worst-case FPR is at most α also has FPR at most α under Q, and its TPR under Q is at least its worst-case TPR. Thus $g_\varepsilon(\cdot; P_0) \le g_Q$ and $\mathcal{R}_\varepsilon(P_0) \subseteq \mathcal{R}^*(Q)$. For a doctor in both sets,

$$\begin{aligned} W_j(c_j^*; Q) - W_j(c_{\varepsilon,j}; Q) &= \phi\{g_Q(\alpha_{H,j}) - \mathrm{TPR}_Q(c_{\varepsilon,j})\} - \eta\{\alpha_{H,j} - \mathrm{FPR}_Q(c_{\varepsilon,j})\} \\ &\le \phi\{g_Q(\alpha_{H,j}) - g_\varepsilon(\alpha_{H,j}; P_0)\} \le \phi\Delta_\varepsilon(Q). \end{aligned}$$

The first inequality uses $\mathrm{TPR}_Q(c_{\varepsilon,j}) \ge g_\varepsilon(\alpha_{H,j}; P_0)$ and the fact that the FPR term is nonpositive. If $j \in \mathcal{R}^*(Q) \setminus \mathcal{R}_\varepsilon(P_0)$, the forgone gain is exactly $\phi\{g_Q(\alpha_{H,j}) - \beta_{H,j}\}$, and

$$0 < g_Q(\alpha_{H,j}) - \beta_{H,j} \le \Delta_\varepsilon(Q).$$

By Assumption 5, the fraction of such doctors is at most $C_M \Delta_\varepsilon(Q)^{\gamma_M}$, and adding the two contributions gives (3.19). When $\Delta_\varepsilon(Q) = 0$, the two sets coincide and the difference for every doctor in them is nonpositive, so (3.19) holds in this case as well.

Part (b), Step 1: the posterior event. With $T_j(t)$ as in (3.5) and t as in (3.20), define

$$\mathcal{E}(t) := \{\max_{j \le J} T_j(t) \le \bar{\tau}/2\}.$$

By Markov's inequality and a union bound,

$$\Pr\{\mathcal{E}(t)^c\} \le \frac{2JC_0}{\bar{\tau}} \exp(-\zeta_0 n_{\min} t^2) = \gamma.$$

The event $\mathcal{E}(t)$ does not depend on Q. On $\mathcal{E}(t)$, every median $a_{n,j}$ lies in $[\alpha_{H,j} - t, \alpha_{H,j} + t]$, since otherwise at least half of the posterior mass of α_H would lie outside this interval, whereas $T_j(t) \le \bar{\tau}/2 \le 1/4$. Because $t \le d_H$, the medians and the points $\alpha_{H,j} \pm t$ all belong to $[\alpha_l, \alpha_u]$.

Step 2: locating extra and missed replacements. Suppose first that $g_\varepsilon(\alpha_{H,j}; P_0) - \beta_{H,j} > h_n$, and let $\alpha_* := \alpha_{H,j} - t$. Then

$$\widehat{g}_\varepsilon(\alpha_*) \ge g_\varepsilon(\alpha_{H,j}; P_0) - L_g t - r_n > \beta_{H,j} + t.$$

By Lemma 2, whether the candidate set is the full interval or a finite set containing the all-negative rule $c = 1$, some candidate attains $\widehat{g}_\varepsilon(\alpha_*)$; its estimated worst-case FPR is at most $\alpha_{H,j} - t$ and its estimated worst-case TPR exceeds $\beta_{H,j} + t$. The rectangle of this candidate contains the t-neighborhood of $\theta_{H,j}$, so its posterior probability is at least $1 - T_j(t) > \tau$, and doctor j is replaced.

Conversely, if doctor j is replaced, the rectangle of $\widehat{c}_j$ has posterior probability at least $\tau > T_j(t)$. Hence it intersects the t-neighborhood of $\theta_{H,j}$, so that

$$\widehat{\overline{\alpha}}_\varepsilon(\widehat{c}_j) \le \alpha_{H,j} + t, \qquad \widehat{\underline{\beta}}_\varepsilon(\widehat{c}_j) \ge \beta_{H,j} - t. \tag{A.20}$$

Since $\widehat{\underline{\beta}}_\varepsilon(\widehat{c}_j) \le \widehat{g}_\varepsilon(\alpha_{H,j} + t)$, it follows that

$$\beta_{H,j} \le g_\varepsilon(\alpha_{H,j}; P_0) + r_n + (1 + L_g)t = g_\varepsilon(\alpha_{H,j}; P_0) + h_n.$$

For $j \in \mathcal{A}_n$, we have $\beta_{H,j} \geq g_Q(\alpha_{H,j}) \geq g_\varepsilon(\alpha_{H,j}; P_0)$, so

$$0 \leq \beta_{H,j} - g_Q(\alpha_{H,j}) \leq h_n.$$

For $j \in \mathcal{M}_n$, doctor j is not replaced, so the first implication gives $g_\varepsilon(\alpha_{H,j}; P_0) - \beta_{H,j} \leq h_n$, and hence

$$0 < g_Q(\alpha_{H,j}) - \beta_{H,j} \leq \Delta_\varepsilon(Q) + h_n.$$

Assumption 5 now gives (3.22), without appeal to the separation δ_0 of Assumption 2.

Step 3: threshold differences among common replacements. Write $A_{n,j} = \widehat{\overline{\alpha}}_\varepsilon(\widehat{c}_j)$ and $B_{n,j} = \widehat{\underline{\beta}}_\varepsilon(\widehat{c}_j)$. For each $j \in \mathcal{I}_n$, (3.9) and the definition of $c_j^*(Q)$ imply

$$W_j(c_j^*; Q) - W_j(\widehat{c}_j; Q) \leq \phi\{g_Q(\alpha_{H,j}) - B_{n,j}\} + \eta\{A_{n,j} - \alpha_{H,j}\} + \phi e_{\beta,n} + \eta e_{\alpha,n}.$$

On $\mathcal{E}(t)$,

$$g_Q(\alpha_{H,j}) \leq \Delta_\varepsilon(Q) + g_\varepsilon(\alpha_{H,j}; P_0) \leq \Delta_\varepsilon(Q) + \widehat{g}_\varepsilon(a_{n,j}) + r_n + L_g t.$$

Moreover, $\eta(a_{n,j} - \alpha_{H,j}) \leq \eta t$. Substituting these bounds gives

$$\begin{aligned} W_j(c_j^*; Q) - W_j(\widehat{c}_j; Q) \leq \phi\Delta_\varepsilon(Q) + v_n \\ + \phi\{\widehat{g}_\varepsilon(a_{n,j}) - B_{n,j}\} + \eta\{A_{n,j} - a_{n,j}\}. \end{aligned}$$

We bound the sum of the last two terms by its positive part and sum over $j \in \mathcal{I}_n$. Since $\mathcal{I}_n \subseteq \mathcal{R}_\varepsilon^{(n)}$, the contribution of the common replacements is at most

$$\widehat{\pi}_n\{\phi\Delta_\varepsilon(Q) + v_n\} + \widehat{K}_n.$$

Step 4: the remaining contributions. For $j \in \mathcal{M}_n$, the definition of $c_j^*(Q)$ gives

$$W_j(c_j^*; Q) - W_{H,j}(Q) = \phi\{g_Q(\alpha_{H,j}) - \beta_{H,j}\} \leq \phi\{\Delta_\varepsilon(Q) + h_n\}.$$

For $j \in \mathcal{A}_n$, (A.20) and (3.9) give

$$W_{H,j}(Q) - W_j(\widehat{c}_j; Q) \leq \phi(e_{\beta,n} + t) + \eta(e_{\alpha,n} + t) = u_n.$$

Multiplying these bounds by the bounds on $|\mathcal{M}_n|/J$ and $|\mathcal{A}_n|/J$ in (3.22) and adding the contribution of the common replacements, we obtain (3.21) from (3.15). All the inequalities hold on the single event $\mathcal{E}(t)$, for every $Q \in \mathcal{B}_\varepsilon(P_0)$ at which the matched-FPR benchmark is defined. □

Proposition 4 (Frontier accuracy and the welfare remainder)**.** *Suppose that the conditions of Theorem 5 hold with J, $\varepsilon > 0$ and Q fixed, and that $n_{\min} \to \infty$ and*

$$r_n + e_{\alpha,n} + e_{\beta,n} = O_p\left(\sqrt{\frac{\log n}{n}}\right), \qquad \bar{r}_n := \sqrt{\frac{\log n}{n}} + n_{\min}^{-1/2}.$$

Then the positive part of the remainder satisfies

$$\left[\mathrm{Regret}(\varepsilon, Q) - \widehat{K}_n - \phi\widehat{\pi}_n\Delta_\varepsilon(Q)\right]_+ = O_p\left(\bar{r}_n + \{\Delta_\varepsilon(Q) + \bar{r}_n\}^{1+\gamma_M}\right). \tag{A.21}$$

Moreover, in (3.21) *the term due to missed replacements is* $O_p(\{\Delta_\varepsilon(Q) + \bar{r}_n\}^{1+\gamma_M})$ *and the term due to extra replacements is* $O_p(\bar{r}_n^{1+\gamma_M})$.

For fixed ε and Q with $\Delta_\varepsilon(Q) > 0$, the right side of (A.21) does not tend to zero, so in general the bound on the remainder is only $O_p(1)$. The estimation errors r_n, $e_{\alpha,n}$ and $e_{\beta,n}$ depend on the fitted score, the complexity of the function classes and the distance calculations, and through them possibly on the dimension of the covariates.

Proof of Proposition 4. Fix $\gamma \in (0,1)$. Because J, C_0 and ζ_0 are fixed, the quantity t in (3.20) is $O(n_{\min}^{-1/2})$ and satisfies $t \le \min\{1, d_H\}$ for all sufficiently large $n_{\min}$. The assumed rates then imply $h_n, u_n, v_n = O_p(\bar{r}_n)$. Subtracting $\widehat{K}_n + \phi\widehat{\pi}_n\Delta_\varepsilon(Q)$ from (3.21), taking the positive part and using $0 \le \widehat{\pi}_n \le 1$, we find that, on the event of probability at least $1-\gamma$ on which (3.21) holds, the left side of (A.21) is at most

$$v_n + \phi C_M\{\Delta_\varepsilon(Q) + h_n\}^{1+\gamma_M} + C_M u_n h_n^{\gamma_M}.$$

The first term is $O_p(\bar{r}_n)$, the second is $O_p(\{\Delta_\varepsilon(Q) + \bar{r}_n\}^{1+\gamma_M})$, and the third is $O_p(\bar{r}_n^{1+\gamma_M})$. Since γ is arbitrary, (A.21) follows from these bounds and the tightness of $(n/\log n)^{1/2}(r_n + e_{\alpha,n} + e_{\beta,n})$. □

A.8 Calculations for the threshold-choice example

Proof of the claims in Example 1. The frontier at positive radius. Write $b = \log\{c/(1-c)\}$ and $u = 1 - b$. For $b \in [0,1]$, the nominal TPR and FPR both equal u. Each class has probability one half, so the transport budget within a class is $r = 2\varepsilon = s^2/2$. With a uniform density, moving an additional mass x across the threshold costs $x^2/2$, provided that this mass is available on the relevant side. Thus, if $s \le u \le 1 - s$, the worst-case FPR is $u + s$ and the worst-case TPR is $u - s$. The same values follow from the distance formula in Proposition 1, with the mass closest to the threshold moved first. For the open event $\{m > c\}$, the supremum is approached by moving mass to points just beyond the boundary, although it need not be attained. When $u \ge 1 - s$ the worst-case FPR is one, and when $u \le s$ the worst-case TPR is zero. For thresholds with b outside $[0,1]$, likewise, either the worst-case FPR is one or the worst-case TPR is zero.

For $\alpha \in [0.7, 0.9]$ and $s \in (0, 0.1)$, the smallest feasible threshold has nominal rate $u = \alpha - s$, so

$$g_{P_0}(\alpha) = \alpha, \qquad g_\varepsilon(\alpha; P_0) = \alpha - 2s.$$

In this range the worst-case rates are continuously differentiable in c, and the derivative of the worst-case FPR is bounded away from zero. The score is Lipschitz and logistic with a nonzero slope, and the relevant thresholds lie in a compact subinterval of $(0,1)$, so Lemma 3 applies. The doctor is robustly dominated, since $g_\varepsilon(0.8; P_0) - 0.2 = 0.6 - 2s > 0$. For every $Q \in \mathcal{B}_\varepsilon(P_0)$ we have $g_Q(0.8) \ge 0.8 - 2s$, so the indicator in (3.13) is zero for $t < 0.6 - 2s$ and is at most $t/(0.6 - 2s)$ for larger t. Assumption 5 therefore holds with $\gamma_M = 1$ and $C_M = (0.6 - 2s)^{-1}$.

At $Q = P_0$, the welfare of the machine at a threshold with nominal rate u is $2u - u = u$. The matched-FPR threshold therefore gives welfare 0.8, and the threshold c_ε of Definition 4, whose worst-case FPR equals 0.8, gives $0.8 - s$. The two replacement sets coincide, so $\mathrm{Regret}_{\mathrm{pop}}(\varepsilon, P_0) = s$. The doctor's welfare is $2(0.2) - 0.8 = -0.4$, so replacement improves on retention at either threshold.

A tail limit for Beta laws. Suppose that $Z_N \sim \mathrm{Beta}(a_N, b_N)$ with $a_N/N \to a > 0$ and $b_N/N \to b > 0$, and let $\theta = a/(a+b)$. For fixed $x \in (\theta, 1)$,

$$-\frac{1}{N}\log \Pr(Z_N > x) \longrightarrow (a+b)\,\mathrm{kl}(\theta, x), \tag{A.22}$$

where $\mathrm{kl}(\theta, x) = \theta \log(\theta/x) + (1-\theta)\log\{(1-\theta)/(1-x)\}$. To see this, write the probability as the ratio of the integrals of the Beta density kernel over $[x, 1]$ and $[0, 1]$. The logarithm of the integrand, divided by N, is

$$f_N(z) = \frac{a_N - 1}{N}\log z + \frac{b_N - 1}{N}\log(1-z).$$

For large N this function is strictly concave, its maximizer converges to θ, and it is decreasing on $[x, 1]$. The numerator is therefore at most $\exp\{N f_N(x)\}$ and at least $h\exp\{N f_N(x+h)\}$ for $0 < h < 1 - x$; letting $N \to \infty$ and then $h \downarrow 0$ shows that its logarithm divided by N tends to $f(x) = a\log x + b\log(1-x)$. Bounding the denominator above by the value of the integrand at its mode and below by its integral over an interval around θ shows in the same way that the corresponding limit is $f(\theta)$. Subtracting gives (A.22).

The posterior-maximizing threshold. Let the historical observations be iid with $P(Y = 1) = 1/2$, $P(\widehat{Y}_H = 1 \mid Y = 0) = 0.8$ and $P(\widehat{Y}_H = 1 \mid Y = 1) = 0.2$. Such decisions can be generated by a diagnostic signal whose law given the label is the same in every environment, in line with (2.4). By the strong law of large numbers,

$$\frac{\mathrm{TN}_N}{N}, \frac{\mathrm{TP}_N}{N} \longrightarrow 0.1, \qquad \frac{\mathrm{FP}_N}{N}, \frac{\mathrm{FN}_N}{N} \longrightarrow 0.4 \quad \text{almost surely.}$$

Under the posterior, $Z_{0,N} := 1 - \alpha_H$ and $Z_{1,N} := \beta_H$ are independent Beta variables with respective shape parameters

$$(a_0 + \mathrm{TN}_N, a_0 + \mathrm{FP}_N), \qquad (a_0 + \mathrm{TP}_N, a_0 + \mathrm{FN}_N).$$

Let $F_{0,N}$ and $F_{1,N}$ be their CDFs. For $s < u < 1 - s$, the exact posterior probability (2.12) is

$$q_N(u) = F_{0,N}(1 - u - s)F_{1,N}(u - s).$$

This function is continuous and vanishes at the two endpoints, and the posterior probability is zero outside the interval, so a maximizer exists. Put $x_0 = 0.5 - s > 0.2$. At $u = 0.5$,

$$1 - q_N(0.5) \le \Pr(Z_{0,N} > x_0) + \Pr(Z_{1,N} > x_0).$$

For any fixed $d \in (0, x_0 - 0.2)$ and all $u \ge 0.5 + d$,

$$1 - q_N(u) \ge \Pr(Z_{0,N} > x_0 - d),$$

and for $u \le 0.5 - d$ the same lower bound holds with $Z_{1,N}$ in place of $Z_{0,N}$. On the event of probability one on which the limits above hold, (A.22) applies to both variables with $(a, b) = (0.1, 0.4)$. Since $\mathrm{kl}(0.2, x)$ is strictly increasing in x for $x > 0.2$, the lower bounds for $|u - 0.5| \ge d$ are eventually strictly larger than the upper bound at $u = 0.5$, so every maximizer satisfies $|u - 0.5| < d$ for all sufficiently large N. Applying this to a

sequence $d \downarrow 0$ proves that $\widehat{u}_N \to 0.5$ almost surely. Moreover, $q_N(0.5) \to 1$, so the doctor is eventually replaced for every fixed $\tau < 1$.

Because the posterior concentrates, the posterior median $a_{N,1}$ of the FPR also converges to 0.8. For $a_{N,1}$ close to 0.8, $g_\varepsilon(a_{N,1}; P_0) = a_{N,1} - 2s$, while the worst-case FPR and TPR at the selected threshold are $\widehat{u}_N + s$ and $\widehat{u}_N - s$. Consequently,

$$\widehat{K}_N = \left[2\{a_{N,1} - 2s - (\widehat{u}_N - s)\} + (\widehat{u}_N + s - a_{N,1})\right]_+ \longrightarrow 0.3 - s.$$

The welfare of the deployed threshold tends to 0.5 and that of the benchmark is 0.8, so $\mathrm{Regret}(\varepsilon, P_0) \to 0.3$, as claimed. Because the reference frontier is known, the limit reflects only the choice of threshold. □

A.9 The extension to pointwise budgets

We first show that the measurable selections required in Definition 6 exist in the two cases considered in Section 4.1.

Lemma 7 (Measurable local extrema)**.** *The selections in Definition 6 exist in either of the following cases:*

(i) $\mathcal{X}$ *is closed in* $\mathbb{R}^d$ *and* $m : \mathcal{X} \to [0, 1]$ *is continuous;*

(ii) $\mathcal{X} = \mathbb{R}^d$ *and* m *is constant on each cell of a finite axis-aligned rectangular partition, whose open and closed faces are specified.*

A fixed finite ensemble of axis-aligned trees is of the form in (ii), with the partition obtained by intersecting the leaf cells of its trees.

Proof. In case (i), $K_\rho(x)$ is nonempty and compact, so the minimum and the maximum of m over it exist. For any real a, the set $\{x : \underline{m}_\rho(x) \le a\}$ is closed: if $x_n \to x$ and $z_n \in K_\rho(x_n)$ satisfy $m(z_n) \le a$, then the z_n are bounded, and a subsequence converges to some $z \in K_\rho(x)$ with $m(z) \le a$. Hence $\underline{m}_\rho$ is Borel measurable, and the same argument applied to $-m$ shows that $\overline{m}_\rho$ is as well.

A measurable selection can be constructed as follows. For a closed bounded cube D, let $v_D(x)$ be the minimum of m over $K_\rho(x) \cap D$, with $v_D(x) = +\infty$ if this set is empty. The subsequence argument above shows that v_D is Borel measurable. The set of x for which some minimizer of m over $K_\rho(x)$ lies in D is $\{x : v_D(x) = \underline{m}_\rho(x)\}$, which is therefore Borel. Enumerate closed unit cubes that cover $\mathbb{R}^d$ and, for each x, take the first one that contains a minimizer. Divide this cube into its 2^d closed subcubes of half the side length, take the first of these that contains a minimizer, and continue in this way. Each of these choices is measurable in x. The selected cubes are nested, each of them meets the compact set of minimizers, and their diameters tend to zero, so their centers converge to a minimizer, and the limit is measurable in x. The same construction applied to $-m$ gives the maximizer.

In case (ii), fix a nonempty cell D and a point $z_D \in D$, and let $\pi_D(x)$ denote the projection of x onto the closure of D, which is a closed rectangle; the map π_D is continuous. The cell D contains a point of $K_\rho(x)$ if and only if

$$\|x - \pi_D(x)\| < \rho, \quad \text{or} \quad \|x - \pi_D(x)\| \le \rho \text{ and } \pi_D(x) \in D.$$

Indeed, the distance from x to a closed convex set is attained only at the projection of x. Under the strict inequality, the points $\pi_D(x) + 2^{-k}(z_D - \pi_D(x))$ lie in D and, for large

k, in $K_\rho(x)$. The set of x from which D can be reached is Borel. For such x, take $\pi_D(x)$ if it belongs to $D \cap K_\rho(x)$, and otherwise the point above with the smallest k for which it belongs to $K_\rho(x)$; this defines a measurable map from x to a point of $D \cap K_\rho(x)$. At least one cell can be reached, because $x \in K_\rho(x)$. Among the finitely many cells that can be reached, choose one with the smallest score, breaking ties by a fixed ordering of the cells, and likewise one with the largest score. The corresponding points are the required measurable selections. □

Proof of Proposition 3. For every map T_X as in Definition 5, we have, P-almost surely,

$$\underline{m}_\rho(X) \le m(T_X(X,Y)) \le \overline{m}_\rho(X).$$

Since the labels and the class probabilities are unchanged, $\mathrm{TPR}_Q(c) \ge \tilde{\beta}_\rho(c;P)$ and $\mathrm{FPR}_Q(c) \le \tilde{\alpha}_\rho(c;P)$ for every $Q \in \mathcal{B}_\rho^{\mathrm{pt}}(P)$. The map that sends each positive case to $z_\rho^-(X)$ and each negative case to $z_\rho^+(X)$ is measurable and defines a law in $\mathcal{B}_\rho^{\mathrm{pt}}(P)$. Because the selections attain the extrema, this law attains both bounds at every threshold, also for observations with $\underline{m}_\rho(X) = c$ or $\overline{m}_\rho(X) = c$. This proves (4.5)–(4.6).

At $\rho = 0$ we have $K_0(x) = \{x\}$, so $\underline{m}_0 = \overline{m}_0 = m$ and $\tilde{g}_0(\cdot;P) = g(\cdot;P)$. For any $Q \in \mathcal{B}_\rho^{\mathrm{pt}}(P)$, the map T_X that defines Q gives a coupling of P and Q with cost at most ρ. Hence, if $\rho \le \varepsilon$,

$$\{P\} \subseteq \mathcal{B}_\rho^{\mathrm{pt}}(P) \subseteq \mathcal{B}_\varepsilon(P).$$

It follows that, at every threshold, the worst-case rates over $\mathcal{B}_\rho^{\mathrm{pt}}(P)$ lie between the nominal rates and the worst-case rates over $\mathcal{B}_\varepsilon(P)$, and taking the supremum over the thresholds that are feasible in each case gives (4.7). □

The requirement that the extrema be attained cannot be dropped for arbitrary Borel scores. On $\mathcal{X} = [0,1]$, let $m(0) = 1$ and $m(z) = z$ for $z > 0$, and take $x = 0$, $\rho = 1$ and $c = 0$. Then $m(z) > 0$ for every $z \in K_\rho(0)$, although $\underline{m}_\rho(0) = 0$. Thus $\mathbb{I}\{\underline{m}_\rho(0) > c\} = 0$, whereas $\mathbb{I}\{m(z) > c\} = 1$ for every $z \in K_\rho(0)$. Lemma 7 excludes such cases; in particular, the scores of finite tree ensembles attain their extrema although they are discontinuous.

Consider a finite set of proposals in $K_\rho(x)$ that contains x. The smallest score over this set lies between $\underline{m}_\rho(x)$ and $m(x)$, and the largest between $m(x)$ and $\overline{m}_\rho(x)$. Hence the class frequencies computed from these scores for the same observations lie between the pointwise and the nominal rates at every threshold, and so the frontier formed from them over the same candidate thresholds lies between the pointwise and the nominal frontiers, as stated in Section 4.1.

Proof of Theorem 6. We condition throughout on the independent training sample.

Part (i). Write $\pi_y = P_0(Y = y) > 0$ and $\widehat{\pi}_y = N_y/n$. The classes

$$\{\mathbb{I}(Y = y)\mathbb{I}(Z_y(X) \le c) : c \in [0,1]\}, \qquad y = 0, 1,$$

consist of indicators of nested sets and have bounded envelopes and finite VC dimension. They are pointwise measurable, whether or not F_y has atoms, because the indicator at c is the pointwise limit of the indicators at rational thresholds decreasing to c. Their covering numbers are polynomial, so the uniform entropy integral is finite, and the maximal inequality of van der Vaart and Wellner (2011, equation (1.1)) gives a uniform $O_p(n^{-1/2})$

bound for the centered empirical averages. The class frequencies satisfy $\widehat{\pi}_y - \pi_y = O_p(n^{-1/2})$. On the event $\widehat{\pi}_y > 0$, the identity

$$\widehat{F}_y(c) - F_y(c) = \frac{(\widehat{\mathbb{P}}_n - P_0)[\mathbb{I}(Y = y)\mathbb{I}(Z_y(X) \le c)] - F_y(c)(\widehat{\pi}_y - \pi_y)}{\widehat{\pi}_y}$$

gives $d_{y,n} = O_p(n^{-1/2})$. In a sample in which a class is absent, $\widehat{F}_y$ may be set equal to an arbitrary fixed CDF for the purpose of these rates; such a sample has probability $p_0^n + (1-p_0)^n$, and in the computations the corresponding conditional rate is reported as missing.

The population and empirical rates $1 - F_y$ and $1 - \widehat{F}_y$ are right-continuous and nonincreasing in c. Since $c = 1$ is feasible, a smallest feasible threshold exists, and it maximizes the TPR among the feasible thresholds. A threshold that is feasible in the population at level $\alpha - d_{0,n}$ is feasible in the sample at level α, and a threshold that is feasible in the sample at level α is feasible in the population at level $\alpha + d_{0,n}$; moreover, the population and empirical TPRs differ by at most $d_{1,n}$. These two inclusions prove (4.13).

Part (ii). By the endpoint inequalities in Assumption 6 and $f_0 \ge \underline{f}$, the population inverse thresholds lie in C, and the empirical ones do so with probability tending to one. For every α whose inverse threshold c_α lies in C,

$$\tilde{g}_\rho(\alpha; P_0) = 1 - F_1(c_\alpha), \qquad \left|\frac{d}{d\alpha}\tilde{g}_\rho(\alpha; P_0)\right| = \frac{f_1(c_\alpha)}{f_0(c_\alpha)} \le L_1/\underline{f}.$$

Combining this Lipschitz bound with (4.13) gives the uniform rate in part (ii).

For differentiability, let $F_{y,t} = F_y + t h_{y,t}$ be CDFs with $\|h_{y,t} - h_y\|_{\infty,C} \to 0$ as $t \downarrow 0$, where h_y is continuous on C. For small t, the inverses at all $\alpha \in I$ lie in C, and the feasibility inclusions used for (4.13), together with $f_0 \ge \underline{f}$, give

$$\sup_{\alpha \in I} |c_{\alpha,t} - c_\alpha| = O(t), \qquad c_{\alpha,t} = \inf\{c : F_{0,t}(c) \ge 1 - \alpha\}.$$

The perturbed CDF $F_{0,t}$ may have jumps, but $F_0 + t h_0$ is continuous on C and differs from $F_{0,t}$ by $o(t)$ uniformly on C, so the jumps of $F_{0,t}$ at the points $c_{\alpha,t}$, $\alpha \in I$, are $o(t)$ uniformly in α. Since

$$F_{0,t}(c_{\alpha,t}-) \le 1 - \alpha \le F_{0,t}(c_{\alpha,t}),$$

it follows that $F_{0,t}(c_{\alpha,t}) - (1-\alpha) = o(t)$ uniformly in α. A Taylor expansion of F_0, uniform on C, and the uniform continuity of h_0 then give

$$0 = f_0(c_\alpha)(c_{\alpha,t} - c_\alpha) + t h_0(c_\alpha) + o(t).$$

Thus $(c_{\alpha,t} - c_\alpha)/t \to -h_0(c_\alpha)/f_0(c_\alpha)$ uniformly in α, and expanding $1 - F_{1,t}(c_{\alpha,t})$ in the same way proves (4.14). The derivative is linear and continuous in (h_0, h_1), with norm at most $1 + L_1/\underline{f}$. Hence the map from (F_0, F_1) to the frontier is Hadamard differentiable with respect to the uniform norm on C, tangentially to the continuous directions. Differentiability as a function of P follows by composition with the map from P to (F_0, F_1), in which each F_y is normalized by the class probability as in the quotient displayed in part (i).

Part (iii). For every $c \in [0,1]$ there is a candidate in $\{0, s_1, \ldots, s_K, 1\}$ that gives the same indicators $\mathbb{I}\{Z_y(X) > c\}$ on the supports. Let $d = \operatorname{dist}(I, \mathcal{A}) > 0$, and let $\widehat{A}_k$ and $\widehat{B}_k$ denote the empirical counterparts of A_k and B_k. If $\max_k |\widehat{A}_k - A_k| < d/2$, then for every $\alpha \in I$ each candidate is feasible in the sample exactly when it is feasible in the population. The first feasible index $k(\alpha)$ is then the same in the sample and in the population, and since the TPR is nonincreasing in the threshold,

$$\widehat{\tilde{g}}_\rho(\alpha) - \tilde{g}_\rho(\alpha; P_0) = \widehat{B}_{k(\alpha)} - B_{k(\alpha)}.$$

The supremum over $\alpha \in I$ of the absolute value of the right side is at most $d_{1,n}$, and by part (i) the event $\{\max_k |\widehat{A}_k - A_k| < d/2\}$ has probability tending to one; this proves the rate. The same argument applies to any sufficiently small perturbation of the two conditional mass vectors on the fixed supports. Since each B_k is a sum of positive-class masses and hence linear in them, the last identity gives (4.15) uniformly on I. The derivative is taken only with respect to the masses on the fixed supports, so perturbations that introduce new score values are not covered. □

The condition $\operatorname{dist}(I, \mathcal{A}) > 0$ in part (iii) cannot be dropped, even with a positive radius, a finite tree and exact extrema. Consider the one-dimensional tree score $m(x) = 1/4$ for $x \le 0$ and $m(x) = 3/4$ for $x > 0$, with $\rho = 1/4$, and suppose that in each class X takes the values -1 and 1, with

$$P_0(X = 1 \mid Y = 0) = a, \qquad P_0(X = 1 \mid Y = 1) = b, \quad 0 < a, b < 1.$$

No ball $K_\rho(x)$ centered at a support point crosses the split, so $\underline{m}_\rho = \overline{m}_\rho = m$ at both support points. The available pairs of rates are $(1,1)$, (a,b) and $(0,0)$, and

$$\tilde{g}_\rho(\alpha; P_0) = \begin{cases} 0, & 0 \le \alpha < a, \\ b, & a \le \alpha < 1, \\ 1, & \alpha = 1. \end{cases}$$

Let $\widehat{a}$ denote the empirical counterpart of a. If $a = 1/2$, the event $\{\widehat{a} > a\}$ has probability tending to $1/2$ under iid validation sampling with $0 < p_0 < 1$, and on this event $\widehat{\tilde{g}}_\rho(a) = 0$, whereas $\tilde{g}_\rho(a; P_0) = b$. Thus the estimated frontier is not even pointwise consistent at $\alpha = a$. Nor is the frontier directionally differentiable at a: changing the mass of the negative class at 1 from a to $a + t$, for any $t > 0$, changes the frontier at a from b to zero, although the two conditional CDFs change by at most t.

In the continuous case, it is likewise not enough that F_0 be strictly increasing. Let the CDFs of the extremal scores on $[0,1]$ be $F_0(c) = 1/2 + 4(c - 1/2)^3$ and $F_1(c) = c$. For any fixed $\rho > 0$ they arise from the score $m(x) = \operatorname{logistic}(x)$: if V_y has CDF F_y, let $X \mid Y = 0$ have the law of $\operatorname{logit}(V_0) - \rho$ and $X \mid Y = 1$ the law of $\operatorname{logit}(V_1) + \rho$. Both CDFs are continuous and strictly increasing, but $f_0(1/2) = 0$. The CDF $F_{0,t}(c) = (1-t)F_0(c) + t\mathbb{I}\{c \ge 1/4\}$ differs from F_0 by at most t, yet at $\alpha = 1/2$ its inverse differs from that of F_0 by $-[t/\{8(1-t)\}]^{1/3}$ for sufficiently small t, and the frontier changes by the same amount with the opposite sign. At such a point, a uniform error of order $n^{-1/2}$ in the estimated CDFs therefore does not imply an error of order $n^{-1/2}$ in the frontier. The lower bound on f_0 in part (ii) rules this out.

A.10 Replacement using confidence bounds

If simultaneous confidence bounds on the worst-case rates are available, they can be combined with the fixed-threshold posterior rule of Definition 3. Let $U_{\alpha,n}(c), L_{\beta,n}(c) \in [0,1]$ be an upper confidence bound for the worst-case FPR and a lower confidence bound for the worst-case TPR at each of the candidates $c \in \mathcal{C}_n$ that are searched, and suppose that they are at least as conservative as the estimates themselves:

$$U_{\alpha,n}(c) \geq \widehat{\overline{\alpha}}_\varepsilon(c), \qquad L_{\beta,n}(c) \leq \widehat{\underline{\beta}}_\varepsilon(c). \tag{A.23}$$

Any given one-sided bounds can be modified to satisfy (A.23), by taking the maximum or the minimum with the estimates, and this preserves their coverage. Define the fixed-threshold probability

$$\widehat{q}_j^L(c) = \Pi_j\{\alpha_H \geq U_{\alpha,n}(c),\ \beta_H \leq L_{\beta,n}(c) \mid \mathcal{D}_j\}. \tag{A.24}$$

The rule based on these bounds replaces doctor j if a maximizer $\widehat{c}_j^L$ of (A.24) over $\mathcal{C}_n$ satisfies $\widehat{q}_j^L(\widehat{c}_j^L) \geq \tau$, and it deploys that threshold. We call it the protected rule and the event in (A.24) the protected rectangle of c; the rule of Definition 3 with the estimated rates is called the plug-in rule. As for the plug-in rule, a finite grid and a fixed tie-breaking rule ensure that the maximizer exists and is measurable.

Proposition 5 (Conditional guarantee for replacement using confidence bounds)**.** *Suppose that* (2.4) *and* (A.23) *hold and that, for some* $\gamma \in (0,1)$*, the simultaneous coverage event*

$$\mathcal{E}_n^L = \{\overline{\alpha}_\varepsilon(c; P_0) \leq U_{\alpha,n}(c), \quad \underline{\beta}_\varepsilon(c; P_0) \geq L_{\beta,n}(c) \text{ for every } c \in \mathcal{C}_n\}$$

has probability at least $1-\gamma$*. Then:*

(i) the set of doctors replaced by the protected rule is contained in the set replaced by the plug-in rule with the same candidates and the same cutoff τ*;*

(ii) on $\mathcal{E}_n^L$*, for every doctor replaced by the protected rule, the posterior probability that, simultaneously for all* $Q \in \mathcal{B}_\varepsilon(P_0)$*, the threshold* $\widehat{c}_j^L$ *has no higher FPR and no lower TPR than the doctor is at least* τ*;*

(iii) under Assumptions 2–4, the protected rule is consistent if

$$\sup_{c \in \mathcal{C}_n} \big\{ |U_{\alpha,n}(c) - \overline{\alpha}_\varepsilon(c; P_0)| + |L_{\beta,n}(c) - \underline{\beta}_\varepsilon(c; P_0)| \big\} = o_p(1),$$

and, for each dominated doctor, the candidate sets contain a feasible population threshold attaining $g_\varepsilon(\alpha_{H,j} - t_0; P_0)$*, with* t_0 *as in Theorem 2, or candidates whose worst-case rates converge to those at that threshold.*

We do not construct such bounds here, and the limit distribution (A.11), which holds only on the regular threshold region of Theorem 7, does not by itself provide them. (Here $1-\gamma$ is a frequentist coverage level and τ a posterior cutoff, and neither is an unconditional probability that the machine dominates the doctor.)

Proof of Proposition 5. By (A.23), the protected rectangle of every candidate c is contained in the rectangle of c used by the plug-in rule, in every sample. Hence $\widehat{q}_j^L(c) \leq \widehat{q}_j^\varepsilon(c)$ for every c, and maximizing over the same candidate set proves (i).

On $\mathcal{E}_n^L$, every pair (α_H, β_H) in the protected rectangle of $\widehat{c}_j^L$ satisfies

$$\mathrm{FPR}_Q(\widehat{c}_j^L) \le \overline{\alpha}_\varepsilon(\widehat{c}_j^L; P_0) \le U_{\alpha,n}(\widehat{c}_j^L) \le \alpha_H,$$

$$\mathrm{TPR}_Q(\widehat{c}_j^L) \ge \underline{\beta}_\varepsilon(\widehat{c}_j^L; P_0) \ge L_{\beta,n}(\widehat{c}_j^L) \ge \beta_H$$

for every $Q \in \mathcal{B}_\varepsilon(P_0)$. For a doctor replaced by the protected rule, the posterior probability of this rectangle is at least τ, which proves (ii).

For (iii), let $\delta_n^L = o_p(1)$ be a bound, uniform in c, for the two differences in the display of part (iii). For a dominated doctor, let c_j° denote the feasible population threshold in (iii), attaining $g_\varepsilon(\alpha_{H,j} - t_0; P_0)$. Lemma 2 ensures that such a threshold exists. Its worst-case FPR is at most $\alpha_{H,j} - t_0$, and separation and Lipschitz regularity give a worst-case TPR of at least $\beta_{H,j} + \delta_0 - L_g t_0 > \beta_{H,j} + t_0$. The candidate condition in (iii) supplies this threshold or an approximation to its rates. When δ_n^L and the approximation error are small enough, the protected rectangle of that candidate contains a fixed neighborhood of $\theta_{H,j}$. By Assumption 4 and Markov's inequality, the posterior probability of this neighborhood tends to one in probability, and hence so does the posterior probability of the protected rectangle.

For a non-dominated doctor, suppose that the protected rectangle of some candidate c intersects $\{\|\theta_H - \theta_{H,j}\|_\infty \le t\}$. Then

$$\overline{\alpha}_\varepsilon(c; P_0) \le \alpha_{H,j} + t + \delta_n^L, \qquad \underline{\beta}_\varepsilon(c; P_0) \ge \beta_{H,j} - t - \delta_n^L.$$

Thus $\beta_{H,j} \le g_\varepsilon(\alpha_{H,j} + t + \delta_n^L; P_0) + t + \delta_n^L$. Fix $0 < t < \min\{d_H/2, \delta_0/[4(1 + L_g)]\}$. Since $\delta_n^L = o_p(1)$, both $\alpha_{H,j} + t + \delta_n^L \in [\alpha_l, \alpha_u]$ and $(1 + L_g)(t + \delta_n^L) < \delta_0$ hold with probability tending to one. By the Lipschitz property in Assumption 3, the bound on $\beta_{H,j}$ obtained above then gives $\beta_{H,j} < g_\varepsilon(\alpha_{H,j}; P_0) + \delta_0$, which contradicts Assumption 2 for a non-dominated doctor. Hence, with probability tending to one, the protected rectangle of every candidate lies in the complement of this neighborhood, whose posterior probability $T_j(t)$ tends to zero in probability, so that $\max_{c \in \mathcal{C}_n} \widehat{q}_j^L(c) \to_p 0$. Since J is fixed, the two cases together prove that the rule is consistent simultaneously for all doctors. □

B Population evaluation in the smooth-score experiment

In the reference environment the index in (5.1) satisfies $R \sim N(b_0, v)$, with $v = \|\beta\|_2^2 + \sigma_u^2$. Write $\pi(r) = (1 + e^{-r})^{-1}$ and $p_0 = E[\pi(R)]$, and let Φ denote the standard normal CDF. Given $R = r$, doctor j makes a positive decision with probability

$$H_j(r) = \Phi\left(\frac{\pi(r) - c_{H,j}}{\sigma_\nu}\right).$$

The doctor's rates, which are the same at every δ, are therefore

$$\alpha_{H,j} = \frac{E[(1 - \pi(R))H_j(R)]}{1 - p_0}, \qquad \beta_{H,j} = \frac{E[\pi(R)H_j(R)]}{p_0}. \tag{B.1}$$

We condition on the fitted score $m(x) = \{1 + \exp[-(a + x^\top w)]\}^{-1}$, with $w \neq 0$, and

write $M = a + Z^\top w$. Since (M, R) is jointly normal,

$$M \mid R = r \sim N(\mu(r), s_M^2), \qquad \mu(r) = a + \frac{w^\top \beta}{v}(r - b_0), \qquad s_M^2 = \|w\|_2^2 - \frac{(w^\top \beta)^2}{v} > 0.$$

The inequality $s_M^2 > 0$ follows from $\sigma_u > 0$ and the Cauchy–Schwarz inequality. For a threshold $c \in (0, 1)$, put $t = \log\{c/(1-c)\}$. Under Q_δ the index of the machine is $M + (1 - 2Y)\delta w_1$, so

$$\begin{aligned} \mathrm{FPR}_{Q_\delta}(c) &= \frac{E[(1 - \pi(R))\Phi\{(\mu(R) + \delta w_1 - t)/s_M\}]}{1 - p_0}, \\ \mathrm{TPR}_{Q_\delta}(c) &= \frac{E[\pi(R)\Phi\{(\mu(R) - \delta w_1 - t)/s_M\}]}{p_0}. \end{aligned} \tag{B.2}$$

At the endpoints $c = 0$ and $c = 1$ the rates are evaluated directly. All the expectations in (B.1)–(B.2) are one-dimensional integrals over the law of R.

Let $A_0(t)$ be the probability, under the reference law and given $Y = 0$, that M exceeds t, and let $t_{0,j}$ solve $A_0(t_{0,j}) = \alpha_{H,j}$. Because $s_M > 0$ and $0 < \pi(R) < 1$, the conditional densities of M given the label are positive on the real line, so this solution exists and is unique. Under Q_δ, the threshold on the index scale that matches doctor j's FPR is $t_{0,j} + \delta w_1$, and its TPR is the reference probability, given $Y = 1$, that M exceeds $t_{0,j} + 2\delta w_1$. The benchmark uses this threshold. Each evaluated policy deploys its own selected threshold: the posterior-probability maximizer or, in the comparison of Section 5.4, the welfare maximizer within the posterior feasible set.

The integrals are computed by Gauss–Hermite quadrature with 1,024 nodes. As a check, we recompute them with 512 nodes at all evaluated human rates, selected machine thresholds and matched-FPR thresholds and require the two results to differ by at most 5×10^{-7}; we also require the matched-FPR probability residual to be at most 10^{-10}. These checks indicate numerical convergence but do not bound the integration error. The numbers of welfare gains and of benchmark dominance comparisons that lie within the numerical tolerance of zero are recorded with the results. In a replication with no replacements, the share of harmful replacements among replaced doctors is undefined (we report it as missing), while the share among all doctors is zero.

For the simulations, the data supplement reports results for the maximum-probability rule at every combination of radius, deployment distance and welfare weights, including all 147 combinations for the finer grid of radii; paired differences from the nominal rule and between the two threshold choices; the share of harmful replacements among replaced doctors and among all doctors; the sizes of the losses, including losses averaged over all doctors; and the number of replications behind each conditional mean.

B.1 Alternative welfare weights

The posterior rule does not use (ϕ, η), so the replacement decisions and thresholds are the same under all three pairs of welfare weights. Table 4 reports the paired welfare difference between the robust and the nominal rule at each prespecified deployment distance, using all 100 replications. Since the four positive radii of Table 1 all lead to complete retention, each entry applies to every one of them.

Table 4: Robust-minus-nominal welfare under the prespecified weights.

δ	$(\phi, \eta) = (1, 1)$	$(2, 1)$	$(5, 1)$
0	-0.0507 (0.0010)	-0.0826 (0.0017)	-0.1781 (0.0039)
0.025	-0.0445 (0.0009)	-0.0727 (0.0015)	-0.1575 (0.0034)
0.05	-0.0382 (0.0008)	-0.0628 (0.0013)	-0.1368 (0.0030)
0.1	-0.0254 (0.0005)	-0.0428 (0.0009)	-0.0951 (0.0023)
0.3	0.0275 (0.0006)	0.0392 (0.0010)	0.0745 (0.0022)
0.5	0.0822 (0.0017)	0.1228 (0.0027)	0.2446 (0.0055)
1	0.2168 (0.0045)	0.3226 (0.0068)	0.6398 (0.0135)

Notes: Means over 100 independent replications, with standard errors in parentheses (the standard deviation of the within-replication difference divided by $\sqrt{100}$). The entries are the same for every positive radius in $\{0.05, 0.1, 0.2, 0.3\}$. Positive entries favor the robust rule, which retains every doctor. The column for $(2, 1)$ repeats the corresponding column of Table 1.

B.2 Results for the finer grid of radii

Table 5 reports the values plotted in Figure 2, together with the replacement rates and Monte Carlo standard errors.

Table 5: Replacement and welfare relative to retaining all doctors, by radius.

		Welfare minus all-human welfare, $(\phi, \eta) = (2, 1)$						
ε	Replaced (%)	$\delta = 0$	0.025	0.05	0.1	0.3	0.5	1
0	41.51 (0.88)	0.08257 (0.00173)	0.07274 (0.00153)	0.06284 (0.00133)	0.04283 (0.00093)	-0.03923 (0.00098)	-0.12279 (0.00266)	-0.32258 (0.00676)
0.0001	22.79 (0.70)	0.04628 (0.00138)	0.04083 (0.00122)	0.03532 (0.00105)	0.02419 (0.00073)	-0.02166 (0.00074)	-0.06860 (0.00214)	-0.18150 (0.00550)
0.0003	12.98 (0.49)	0.02666 (0.00099)	0.02354 (0.00087)	0.02039 (0.00076)	0.01401 (0.00052)	-0.01233 (0.00051)	-0.03940 (0.00151)	-0.10477 (0.00394)
0.001	3.18 (0.17)	0.00659 (0.00035)	0.00582 (0.00031)	0.00505 (0.00027)	0.00347 (0.00018)	-0.00307 (0.00018)	-0.00983 (0.00054)	-0.02627 (0.00143)
0.003	0.14 (0.02)	0.00028 (0.00004)	0.00025 (0.00003)	0.00022 (0.00003)	0.00015 (0.00002)	-0.00013 (0.00002)	-0.00043 (0.00006)	-0.00115 (0.00016)
0.01	0.00 (0.00)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)
0.05	0.00 (0.00)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)

Notes: Means over the 100 replications of Table 1, with Monte Carlo standard errors in parentheses; welfare differences from the all-human policy are paired within replications. At $\varepsilon = 0.003$, at least one doctor is replaced in 44 replications, but the means are taken over all 100. For radii from 0.0001 to 0.01, $\delta = 0$ is the only evaluated distance at which Q_δ lies in the corresponding population ambiguity set.

B.3 Budgets and threshold choices for the same score

Tables 6 and 7 report welfare and the share of harmful replacements at all seven deployment distances and five pointwise radii. In this comparison the positive average budgets replace nobody under either threshold choice, and at zero radius the two kinds of budget give the same policy.

Table 6: Welfare under the pointwise budget for the two threshold choices.

$\rho\backslash\delta$	0	0.025	0.05	0.1	0.3	0.5	1
A. Maximum posterior probability: gain over retaining everyone							
0	0.08257†	0.07274	0.06284	0.04283	-0.03923	-0.12279	-0.32258
	(0.00173)	(0.00153)	(0.00133)	(0.00093)	(0.00098)	(0.00266)	(0.00676)
0.05	0.03642†	0.03196†	0.02748†	0.01841	-0.01890	-0.05701	-0.14851
	(0.00123)	(0.00108)	(0.00093)	(0.00063)	(0.00071)	(0.00200)	(0.00509)
0.1	0.01063†	0.00929†	0.00794†	0.00520†	-0.00606	-0.01760	-0.04543
	(0.00053)	(0.00047)	(0.00040)	(0.00026)	(0.00032)	(0.00090)	(0.00230)
0.2	0.00025†	0.00021†	0.00018†	0.00010†	-0.00019	-0.00049	-0.00122
	(0.00003)	(0.00003)	(0.00002)	(0.00002)	(0.00003)	(0.00007)	(0.00017)
0.3	0.00000†	0.00000†	0.00000†	0.00000†	0.00000†	0.00000	0.00000
	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)
B. Welfare choice minus maximum-probability choice							
0	0.01196†	0.01193	0.01191	0.01183	0.01123	0.01023	0.00646
	(0.00037)	(0.00037)	(0.00036)	(0.00036)	(0.00033)	(0.00029)	(0.00017)
0.05	0.00416†	0.00415†	0.00415†	0.00413	0.00397	0.00366	0.00243
	(0.00018)	(0.00018)	(0.00018)	(0.00018)	(0.00016)	(0.00015)	(0.00009)
0.1	0.00101†	0.00101†	0.00101†	0.00101†	0.00098	0.00092	0.00064
	(0.00006)	(0.00006)	(0.00006)	(0.00006)	(0.00006)	(0.00005)	(0.00003)
0.2	0.00002†	0.00002†	0.00002†	0.00002†	0.00002	0.00002	0.00001
	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)
0.3	0.00000†	0.00000†	0.00000†	0.00000†	0.00000†	0.00000	0.00000
	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)

Notes: Means over the 100 replications, including those without replacements, with Monte Carlo standard errors in parentheses; differences are paired within replications. The welfare weights are $(2, 1)$, and a dagger marks a point in the population pointwise set ($\delta \leq \rho$). The positive average budgets in this comparison give zero gain over retaining everyone and zero difference between the two threshold choices.

Table 7: Share of harmful replacements under the pointwise budget for the two threshold choices.

$\rho\backslash\delta$	0	0.025	0.05	0.1	0.3	0.5	1
A. Maximum posterior probability							
0	0.17†	0.31	0.70	2.82	99.94	100.00	100.00
	(0.03)	(0.05)	(0.08)	(0.20)	(0.02)	(0.00)	(0.00)
0.05	0.24†	0.40†	0.74†	2.99	99.98	100.00	100.00
	(0.06)	(0.09)	(0.13)	(0.27)	(0.02)	(0.00)	(0.00)
0.1	0.13†	0.25†	0.65†	4.04†	100.00	100.00	100.00
	(0.06)	(0.08)	(0.14)	(0.59)	(0.00)	(0.00)	(0.00)
0.2	0.00†	0.00†	3.37†	11.52†	100.00	100.00	100.00
	(0.00)	(0.00)	(2.28)	(4.37)	(0.00)	(0.00)	(0.00)
0.3	–	–	–	–	–	–	–
	–	–	–	–	–	–	–
B. Welfare choice							
0	0.03†	0.06	0.17	0.86	98.71	100.00	100.00
	(0.01)	(0.02)	(0.03)	(0.09)	(0.13)	(0.00)	(0.00)
0.05	0.10†	0.12†	0.31†	1.11	99.66	100.00	100.00
	(0.03)	(0.04)	(0.07)	(0.14)	(0.06)	(0.00)	(0.00)
0.1	0.04†	0.07†	0.16†	1.60†	99.92	100.00	100.00
	(0.04)	(0.05)	(0.08)	(0.31)	(0.04)	(0.00)	(0.00)
0.2	0.00†	0.00†	2.13†	8.69†	100.00	100.00	100.00
	(0.00)	(0.00)	(2.13)	(3.77)	(0.00)	(0.00)	(0.00)
0.3	–	–	–	–	–	–	–
	–	–	–	–	–	–	–

Notes: Mean share of harmful replacements among replaced doctors, in percent, with Monte Carlo standard errors in parentheses; welfare weights and daggers as in Table 6. Means are over all 100 replications at radii 0, 0.05 and 0.1 and over the 47 replications with replacements at 0.2; at 0.3 nobody is replaced and the share is undefined.

C Details of the income-data illustration

C.1 Sample construction and policy evaluation

We use the 2018 one-year ACS records with the ten original numeric features of the ACSIncome task and its eligibility restrictions: age above 16, income above \$100, positive hours worked and a person weight of at least one. All retained records receive equal weight. We draw a fixed sample of 20,000 eligible records from each of the three reference states; of these 60,000 records, 48,000 are used to fit the scores, 6,000 to estimate the machine frontier and the remaining 6,000 as histories for the decision makers. Median imputation and standardization are fitted on the training sample only, and the two scores are a logistic regression and a random forest with 100 trees.

A single group of $J = 200$ synthetic decision makers is used throughout, with thresholds $c_{H,j}$ drawn from $0.15 + 0.7\,\mathrm{Beta}(2, 3)$. To generate a decision, we clip the fitted logistic proxy to $[10^{-5}, 1 - 10^{-5}]$, add U to its logit, truncate the result to $[-50, 50]$ and apply the logistic function, which gives p_{priv}; the decision is $\mathbb{I}\{p_{\mathrm{priv}} + \nu > c_{H,j}\}$, with $U \sim N(0, 0.15^2)$ and $\nu \sim N(0, 0.225^2)$. The counts in the reference histories give the independent Beta posteriors (2.17) with $a_0 = 0.01$, and a fixed machine threshold is selected by Definition 3, with $\tau = 0.95$, using only the reference records. The policy is then evaluated in the other states.

For the retrospective comparison of conditional rates, the records are assigned at random to the 200 decision makers, evenly within each observed label class, so that every decision maker has cases from both classes; both scores are evaluated on the same cases and human decisions. All eligible records from each evaluation state, up to 20,000 per state, are reserved for evaluation.

Budgets, search and evaluation. The candidate set contains 400 equally spaced thresholds from 0.001 to 0.999, together with zero and one. For the logistic regression we compute the empirical worst-case rates over the Wasserstein set exactly; for the random forest we use the finite search of Section 4.1 under the pointwise budget, with $K = 200$. The average budget ε for the logistic regression and the pointwise budget ρ for the forest both take the values $0, 0.1, 0.2, 0.3, 0.5$, a grid fixed without reference to the feature distances in Table 3. To see how the forest results depend on the size of the search, at $\rho = 0.5$ we also use the first K proposals of a single sequence of 1,000 proposals, for $K \in \{20, 50, 100, 200, 500, 1000\}$, so that the searches are nested. Each policy deploys the threshold used in its posterior calculation.

The feature distances in Table 3 compare equally weighted samples of 1,200 records from the pooled validation set and from each evaluation state, using the unscaled Euclidean cost in the coordinates standardized on the training sample. Since categorical features enter as numeric codes, perturbing them measures the sensitivity of the fitted score rather than a feasible change in personal characteristics.

In every evaluation, the welfare of a synthetic decision maker is twice the TPR of the assigned policy minus its FPR, and these values are averaged over decision makers with equal weights; rates computed from the pooled cases are reported separately. Harm is measured on the finite evaluation sample, and the conditional harm measures are undefined when nobody is replaced. The ranking AUCs in Table 3 and in Panel C of Table 8 are conventional areas under the ROC curve. The frontier areas in Panels A and B of Table 8 are taken under the frontier over the candidate thresholds, rather than under

the ROC curve of a single least-favorable distribution.

The age-shift exercise uses 5,000 further California records that are not part of the reference sample. We add $\delta \in \{0, 0.5, 1, 1.5, 2, 2.5, 3\}$ to the standardized age variable used by the machine and keep the labels and the unshifted human decisions fixed. The nominal policies and the policies with budget 0.5 are those selected from the reference records. The shift is measured relative to these California records, not to the pooled reference law.

C.2 Results by budget, search size and age shift

Table 8 collects the results by budget, search size and age shift. The data supplement contains the results for every combination of policy and environment, the confusion counts of each decision maker, welfare computed from the pooled cases and the conditional harm measures.

Table 8: Frontier areas, posterior probabilities and age-shift results in the ACS illustration.

A. Budget grid: frontier areas and maximum posterior probabilities

Budget	LR area	RF area	LR maximum q	RF maximum q
0	0.8526	0.8790	0.9850	0.9907
0.1	0.2631	0.7386	0.0374	0.8954
0.2	0.0851	0.6337	1.03×10^{-4}	0.7032
0.3	0.0155	0.5230	2.79×10^{-9}	0.4298
0.5	0.0000	0.3937	0	0.1522

B. Nested forest searches at $\rho = 0.5$

K	Frontier area	Maximum q	Number replaced
20	0.5816	0.5834	0
50	0.5004	0.3855	0
100	0.4433	0.2485	0
200	0.3937	0.1522	0
500	0.3356	0.0786	0
1000	0.2968	0.0454	0

C. Age shift in the additional California records: nominal policies

δ	LR gain over all-human	RF gain over all-human	RF ranking AUC
0	+0.00750	+0.01316	0.8814
0.5	+0.00845	+0.01227	0.8737
1	+0.01212	+0.01360	0.8687
1.5	+0.01100	+0.01424	0.8633
2	+0.00860	+0.01258	0.8609
2.5	+0.00883	+0.01055	0.8569
3	+0.01048	+0.01022	0.8508

Notes: In Panel A, LR uses the exact empirical worst-case rates under the average budget and RF a search with $K = 200$ proposals under the pointwise budget. Areas are computed as described in Appendix C.1. Panel B keeps the model, the validation records and the sequence of proposals fixed. Panel C uses the same 5,000 records and human decisions throughout; all-human welfare is 1.13206, with $(\phi, \eta) = (2, 1)$, and the ranking AUC of LR is 0.85690 at every shift. The two policies with budget 0.5 retain everyone and so have zero gain at every shift. All results come from a single sample split, and no standard errors are reported.